# Climate Control System



**Principles of Operation**

For additional refrigerant system and component information, refer to Section 412-01.

**The Refrigerant Cycle**

During stabilized conditions (A/C system shutdown), the refrigerant pressures are equal throughout the system. When the A/C compressor is in operation, it increases pressure on the refrigerant vapor, raising its temperature. The high-pressure and high-temperature vapor is then released into the top of the A/C condenser core.

The A/C condenser, being close to ambient temperature, causes the refrigerant vapor to condense into a liquid when heat is removed from the refrigerant by ambient air passing over the fins and tubing. The now liquid refrigerant, still at high pressure, exits from the bottom of the A/C condenser and enters the inlet side of the A/C receiver/drier. The receiver/drier is designed to remove moisture from the refrigerant.

The outlet of the receiver/drier is connected to the TXV . The TXV provides the orifice which is the restriction in the refrigerant system and separates the high and low pressure sides of the A/C system. As the liquid refrigerant passes across this restriction, its pressure and boiling point are reduced.

The liquid refrigerant is now at its lowest pressure and temperature. As it passes through the A/C evaporator, it absorbs heat from the airflow passing over the plate/fin sections of the A/C evaporator. This addition of heat causes the refrigerant to boil (convert to gas). The now cooler air can no longer support the same humidity level of the warmer air and this excess moisture condenses on the exterior of the evaporator coils and fins and drains outside the vehicle.

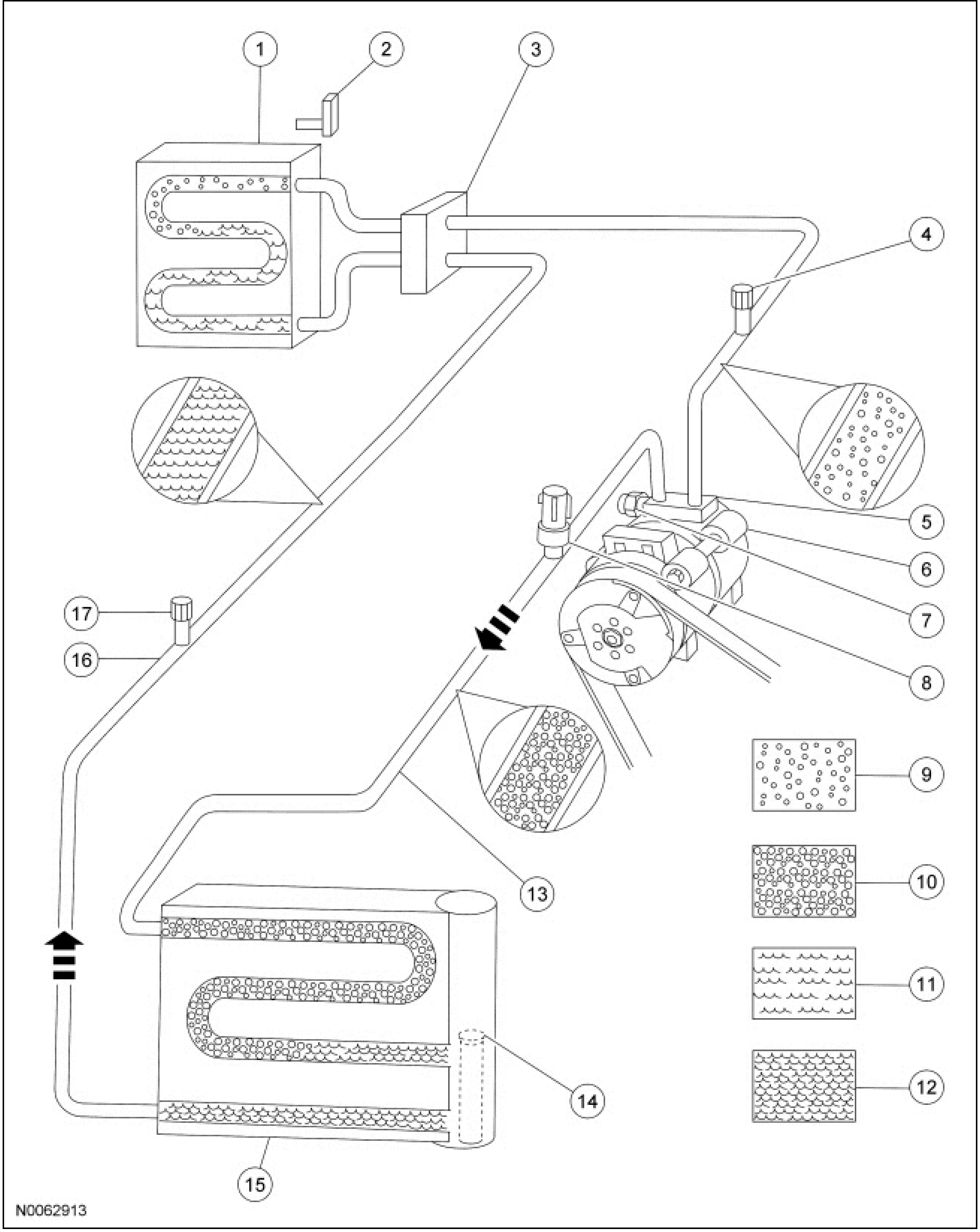
The refrigerant cycle is now repeated with the A/C compressor again increasing the pressure and temperature of the refrigerant.

The PCM controls the A/C clutch relay. The evaporator temperature sensor monitors the temperature of the air that has passed through the evaporator core and sends a signal to the PCM. If the temperature of the evaporator core discharge air is low enough to cause the condensed water vapor to freeze, the A/C clutch is disengaged by the PCM.

The line pressure is monitored so that A/C compressor operation is interrupted if the system pressure becomes too high or too low.

The A/C compressor relief valve opens and vents refrigerant to relieve unusually high system pressure.

**TXV Type Refrigerant System**




| Item | Description |
|---|---|
| 1 | A/C evaporator core |
| 2 | A/C evaporator core outlet temperature thermistor |
| 3 | TXV |
| 4 | A/C charge valve port (low side) |
| 5 | A/C suction line |
| 6 | A/C compressor |

| Item | Description |
|---|---|
| 7 | A/C pressure relief valve |
| 8 | A/C pressure transducer |
| 9 | Low-pressure vapor |
| 10 | High-pressure vapor |
| 11 | Low-pressure liquid |
| 12 | High-pressure liquid |
| 13 | Compressor discharge line |
| 14 | A/C desiccant cartridge |
| 15 | A/C condenser core with A/C receiver/drier |
| 16 | Condenser-to-evaporator line |
| 17 | A/C charge valve port (high side) |

**Inspection and Verification**

1. Verify the customer's concern by operating the climate control system to duplicate the condition.
2. Inspect to determine if one of the following mechanical or electrical concerns apply:

**Visual Inspection Chart**

| Mechanical | Electrical |
|---|---|
| • Loose, missing or damaged A/C compressor drive belt<br>• Loose or disconnected A/C clutch<br>• Broken or binding door/actuator<br>• Broken or leaking refrigerant lines<br>• Obstructed in-vehicle temperature and humidity sensor<br>• Disconnected in-vehicle temperature aspirator hose | • BJB fuses:<br>▪ 30 (10A)<br>▪ 51 (40A)<br>▪ 52 (5A)<br>▪ 77 (10A)<br>• BCM fuse 46 (10A)<br>• Blower motor inoperative<br>• A/C compressor inoperative<br>• Circuitry open/shorted<br>• Disconnected electrical connectors<br>• Cooling fan inoperative |

3. If the inspection reveals obvious concerns that can be readily identified, repair as necessary.
4. **NOTE:** *Make sure to use the latest scan tool software release.*

   If the cause is not visually evident, connect the scan tool to the DLC .
5. **NOTE:** *The VCM LED prove-out confirms power and ground from the DLC are provided to the VCM .*

   If the scan tool does not communicate with the VCM :
   - check the VCM connection to the vehicle.
   - check the scan tool connection to the VCM .
   - refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the VCM .

6. If the scan tool does not communicate with the vehicle:
    - verify the ignition key is in the ON position.
    - verify the scan tool operation with a known good vehicle.
    - refer to Section 418-00 to diagnose no response from the PCM, HVAC module.

7. Carry out the network test.
    - If the scan tool responds with no communication from one or more modules, refer to Section 418-00.
    - If the network test passes, retrieve and record the CMDTCs .

8. Clear the continuous DTCs and carry out the self-test diagnostics from the PCM or HVAC module.

9. **NOTE:** *Some PCM DTCs can inhibit A/C operation. If any PCM DTCs are retrieved, diagnose those first. Refer to the PCM DTC Chart.*

    If the HVAC DTCs retrieved are related to the concern, go to the HVAC Module DTC Chart. If the PCM DTCs retrieved are related to the concern, go to the PCM DTC Chart. For all other DTCs, refer to Section 419-10.

10. If no DTCs related to the concern are retrieved, GO to Symptom Chart — Climate Control Systems or GO to Symptom Chart — NVH.

### DTC Charts

#### PCM DTC Chart

| DTC | Description | Action to Take |
|---|---|---|
| P0532 | A/C Refrigerant Pressure Sensor A Circuit Low | GO to Pinpoint Test A. |
| P0533 | A/C Refrigerant Pressure Sensor A Circuit High | GO to Pinpoint Test A. |
| P0645 | A/C Clutch Relay Control Circuit | GO to Pinpoint Test B. |
| P1464 | A/C Demand Out Of Self Test Range | If the HVAC selector was not powered off, POWER the HVAC off, CLEAR the DTCs and REPEAT the self-test. If the DTC does not return, IGNORE the DTC and continue diagnosing other DTCs or symptoms. If the DTC returns, GO to Pinpoint Test Q. |
| All Other DTCs | — | REFER to Section 303-14. |

#### HVAC Module DTC Chart

**NOTE:** *This module utilizes a 5-character DTC followed by a 2-character failure type code. The failure type code provides information about specific fault conditions such as opens or shorts to ground. CMDTCs have an additional 2-character DTC status code suffix to assist in determining DTC history.*

**NOTE:** *Network DTCs (U-codes) are often a result of intermittent concerns such as damaged wiring or low battery voltage occurrences. Additionally, vehicle repair procedures such as module reprogramming often set network DTCs. Replacing a module to resolve a network DTC is unlikely to resolve the concern. To prevent repeat network DTC concerns, inspect all network wiring, especially connectors. Test the vehicle battery, refer to Section 414-01.*

**NOTE:** *Some PCM DTCs may inhibit A/C operation. If any PCM DTCs are retrieved, diagnose those first. Refer to the PCM DTC Chart.*

| DTC | Description | Action to Take |
|---|---|---|
| B1034:11 | Left Front Seat Heater Element: Circuit Short to Ground | REFER to Section 501-10. |
| B1034:13 | Left Front Seat Heater Element: Circuit Open | REFER to Section 501-10. |
| B1034:15 | Left Front Seat Heater Element: Circuit Short to Battery or Open | REFER to Section 501-10. |
| B1036:11 | Right Front Seat Heater Element: Circuit Short to Ground | REFER to Section 501-10. |
| B1036:13 | Right Front Seat Heater Element: Circuit Open | REFER to Section 501-10. |
| B1036:15 | Right Front Seat Heater Element: Circuit Short to Battery or Open | REFER to Section 501-10. |
| B1038:11 | Left Front Seat Heater Sensor: Circuit Short to Ground | REFER to Section 501-10. |
| B1038:12 | Left Front Seat Heater Sensor: Circuit Short to Battery | REFER to Section 501-10. |
| B103A:11 | Right Front Seat Heater Sensor: Circuit Short to Ground | REFER to Section 501-10. |
| B103A:12 | Right Front Seat Heater Sensor: Circuit Short to Battery | REFER to Section 501-10. |
| B1081:00 | Left Temperature Damper Motor: No Sub Type Information | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test R.<br><br>For EMTC with 4.2 inch display which has 7 blower motor speed positions, GO to Pinpoint Test S.<br><br>For Dual Zone EATC , GO to Pinpoint Test U. |
| B1081:07 | Left Temperature Damper Motor: Mechanical Failure | INSPECT for binding/damaged linkage or door. REPAIR as necessary. If no condition is found, INSTALL a new driver side temperature blend door actuator. REFER to Section 412-01 |
| B1081:11 | Left Temperature Damper Motor: Circuit Short to Ground | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test R.<br><br>For EMTC with 4.2 inch display which has 7 blower motor speed positions, GO to Pinpoint Test S.<br><br>For Dual Zone EATC , GO to Pinpoint Test U. |
| B1081:12 | Left Temperature Damper Motor: Circuit Short to Battery | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test R.<br><br>For EMTC with 4.2 inch display which has 7 blower motor speed positions, GO to Pinpoint Test S.<br><br>For Dual Zone EATC , GO to Pinpoint Test U. |

| DTC | Description | Action to Take |
|---|---|---|
| B1081:13 | Left Temperature Damper Motor: Circuit Open | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test R.<br><br>For EMTC with 4.2 inch display which has 7 blower motor speed positions, GO to Pinpoint Test S.<br><br>For Dual Zone EATC , GO to Pinpoint Test U. |
| B1082:00 | Right Temperature Damper Motor: No Sub Type Information | GO to Pinpoint Test V. |
| B1082:07 | Right Temperature Damper Motor: Mechanical Failure | INSPECT for binding/damaged linkage or door. REPAIR as necessary. If no condition is found, INSTALL a new passenger side temperature blend door actuator. REFER to Section 412-01 |
| B1082:11 | Right Temperature Damper Motor: Circuit Short to Ground | GO to Pinpoint Test V. |
| B1082:12 | Right Temperature Damper Motor: Circuit Short to Battery | GO to Pinpoint Test V. |
| B1082:13 | Right Temperature Damper Motor: Circuit Open | GO to Pinpoint Test V. |
| B1083:00 | Recirculation Damper Motor: No Sub Type Information | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test J.<br><br>For other systems, GO to Pinpoint Test K. |
| B1083:07 | Recirculation Damper Motor: Mechanical Failure | INSPECT for binding/damaged linkage or door. REPAIR as necessary. If no condition is found, INSTALL a new air inlet door actuator. REFER to Section 412-01 |
| B1083:11 | Recirculation Damper Motor: Circuit Short to Ground | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test J.<br><br>For other systems, GO to Pinpoint Test K. |
| B1083:12 | Recirculation Damper Motor: Circuit Short to Battery | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test J.<br><br>For other systems, GO to Pinpoint Test K. |
| B1083:13 | Recirculation Damper Motor: Circuit Open | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test J.<br><br>For other systems, GO to Pinpoint Test K. |
| B1086:00 | Air Distribution Damper Motor: No Sub Type Information | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test L.<br><br>For other systems, GO to Pinpoint Test M. |
| B1086:07 | Air Distribution Damper Motor: Mechanical Failure | INSPECT for binding/damaged linkage or door. REPAIR as necessary. If no condition is found, INSTALL a new defrost/panel/floor door actuator. REFER to Section 412-01 |
| B1086:11 | Air Distribution Damper Motor: Circuit Short to Ground | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test L.<br><br>For other systems, GO to Pinpoint Test M. |
| B1086:12 | Air Distribution Damper Motor: Circuit Short to Battery | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test L.<br><br>For other systems, GO to Pinpoint Test M. |
| B1086:13 | Air Distribution Damper Motor: Circuit Open | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test L. |

| DTC | Description | Action to Take |
|---|---|---|
| | | For other systems, GO to Pinpoint Test M. |
| B10AF:11 | Blower Fan Relay: Circuit Short to Ground | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test X.<br><br>For other systems, GO to Pinpoint Test Z. |
| B10AF:12 | Blower Fan Relay: Circuit Short to Battery | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test W.<br><br>For other systems, GO to Pinpoint Test Y. |
| B10AF:13 | Blower Fan Relay: Circuit Open | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test W.<br><br>For other systems, GO to Pinpoint Test Y. |
| B10B8:00 | Push Buttons: No Sub Type Information | A button held too long can set DTC. ATTEMPT to clean buttons. CLEAR the DTCs. REPEAT the self-test. If code returns, INSTALL a new HVAC module. REFER to Section 412-01 |
| B10B9:12 | Blower Control: Circuit Short to Battery | GO to Pinpoint Test Z. |
| B10B9:14 | Blower Control: Circuit Short to Ground or Open | If the blower motor is inoperative, GO to Pinpoint Test Y.<br><br>If the blower motor is always on HIGH, GO to Pinpoint Test Z. |
| B11E5:11 | Left HVAC Damper Position Sensor: Circuit Short to Ground | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test R.<br><br>For EMTC with 4.2 inch display with 7 blower motor speed positions, GO to Pinpoint Test S.<br><br>For Dual Zone EATC , GO to Pinpoint Test U. |
| B11E5:15 | Left HVAC Damper Position Sensor: Circuit Short to Battery or Open | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test R.<br><br>For EMTC with 4.2 inch display with 7 blower motor speed positions, GO to Pinpoint Test S.<br><br>For Dual Zone EATC , GO to Pinpoint Test U. |
| B11E6:11 | Right HVAC Damper Position Sensor: Circuit Short to Ground | GO to Pinpoint Test V. |
| B11E6:15 | Right HVAC Damper Position Sensor: Circuit Short to Battery or Open | GO to Pinpoint Test V. |
| B11E7:11 | Air Distribution Damper Position Sensor: Circuit Short to Ground | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test L.<br><br>For other systems, GO to Pinpoint Test M. |
| B11E7:15 | Air Distribution Damper Position Sensor: Circuit Short to Battery or Open | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test L.<br><br>For other systems, GO to Pinpoint Test M. |
| B11F0:11 | Air Intake Damper Position Sensor: Circuit Short to Ground | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test J.<br><br>For other systems, GO to Pinpoint Test K. |

| DTC | Description | Action to Take |
|---|---|---|
| B11F0:15 | Air Intake Damper Position Sensor: Circuit Short to Battery or Open | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test J.<br><br>For other systems, GO to Pinpoint Test K. |
| B1A61:11 | Cabin Temperature Sensor: Circuit Short to Ground | GO to Pinpoint Test D. |
| B1A61:15 | Cabin Temperature Sensor: Circuit Short to Battery or Open | GO to Pinpoint Test D. |
| B1A63:11 | Right Solar Sensor: Circuit Short to Ground | GO to Pinpoint Test F. |
| B1A63:15 | Right Solar Sensor: Circuit Short to Battery or Open | GO to Pinpoint Test F. |
| B1A64:11 | Left Solar Sensor: Circuit Short to Ground | GO to Pinpoint Test F. |
| B1A64:15 | Left Solar Sensor: Circuit Short to Battery or Open | GO to Pinpoint Test F. |
| B1A69:11 | Humidity Sensor: Circuit Short to Ground | GO to Pinpoint Test E. |
| B1A69:15 | Humidity Sensor: Circuit Short to Battery or Open | GO to Pinpoint Test E. |
| B1B71:11 | Evaporator Temperature Sensor: Circuit Short to Ground | GO to Pinpoint Test G. |
| B1B71:15 | Evaporator Temperature Sensor: Circuit Short to Battery or Open | GO to Pinpoint Test G. |
| B1C83:11 | Rear Defog Relay: Circuit Short to Ground | REFER to Section 501-09. |
| B1C83:13 | Rear Defog Relay: Circuit Open | REFER to Section 501-09. |
| C1B14:11 | Sensor Supply Voltage A: Circuit Short to Ground | GO to Pinpoint Test C. |
| C1B14:12 | Sensor Supply Voltage A: Circuit Short to Battery | GO to Pinpoint Test C. |
| U0140:00 | Lost Communication With Body Control Module: No Sub Type Information | GO to Pinpoint Test AA. |
| U0256:00 | Lost Communication With FCIM "A": No Sub Type Information | GO to Pinpoint Test AC. |
| U0401:09 | Invalid Data Received From ECM/PCM A: Component Failure | This DTC can set due to a fault in the PCM. CARRY OUT a self-test of the PCM and DIAGNOSE any PCM DTCs present, REFER to Section 303-14. |
| U0401:68 | Invalid Data Received From ECM/PCM A: Event Information | This DTC can set due to a fault in the PCM. CARRY OUT a self-test of the PCM and DIAGNOSE any PCM DTCs present, REFER to Section 303-14. |
| U0401:81 | Invalid Data Received From ECM/PCM A: Invalid Serial Data Received | This DTC can set due to a fault in the PCM. CARRY OUT a self-test of the PCM and DIAGNOSE any PCM DTCs present, REFER to Section 303-14. |
| U0401:82 | Invalid Data Received From ECM/PCM A: Alive/Sequence | This DTC can set due to a fault in the PCM. CARRY OUT a self-test of the PCM and DIAGNOSE any PCM DTCs present, REFER to Section 303-14. |

| DTC | Description | Action to Take |
|---|---|---|
| | Counter Incorrect/Not Updated | |
| U0422:68 | Invalid Data Received From Body Control Module: Event Information | This DTC can set due to a fault in the BCM . CARRY OUT a self-test of the BCM and DIAGNOSE any BCM DTCs present, REFER to Section 419-10. |
| U0422:81 | Invalid Data Received From Body Control Module: Invalid Serial Data Received | This DTC can set due to a fault in the BCM . CARRY OUT a self-test of the BCM and DIAGNOSE any BCM DTCs present, REFER to Section 419-10. |
| U0423:82 | Invalid Data Received from Instrument Panel Cluster Control Module: Alive/Sequence Counter Incorrect/Not Updated | This DTC can set due to a fault in the IPC . CARRY OUT a self-test of the IPC and DIAGNOSE any IPC DTCs present, REFER to Section 413-01. |
| U0554:82 | Invalid Data Received From Accessory Protocol Interface Module: Alive / Sequence Counter Incorrect / Not Updated | This DTC can set due to a fault in the APIM . CARRY OUT a self-test of the APIM and DIAGNOSE any APIM DTCs present, Refer to the appropriate section in Group 415 for the procedure. |
| U1000:00 | Solid State Driver Protection Actived - Driver Disabled: | The HVAC module has temporarily disabled an output because an excessive current draw exists (such as a short to ground). The HVAC module cannot enable the output until the cause of the short is corrected. ADDRESS all other Diagnostic Trouble Codes (DTCs) first. After the cause of the concern is corrected, CLEAR the Diagnostic Trouble Codes (DTCs). REPEAT the self-test. |
| U2100:00 | Initial Configuration Not Complete: No Sub Type Information | CARRY OUT PMI on the HVAC module. REFER to Section 418-01. REPEAT the self-test. If PMI is successful, the DTC will not be present. If the DTC returns, INSTALL a new HVAC module. REFER to Section 412-01. |
| U3000:49 | Control Module: Internal Electronic Failure | ADDRESS all other DTCs first. CLEAR the DTCs. REPEAT the self-test. If DTC U3000:49 is retrieved again, INSTALL a new HVAC module. REFER to Section 412-01. |
| U3002:62 | Vehicle Identification Number: Signal Compare Failure | CARRY OUT PMI on the HVAC module. REFER to Section 418-01. REPEAT the self-test. If PMI is successful, the DTC will not be present. If the DTC returns, INSTALL a new HVAC module. REFER to Section 412-01. |
| U3003:16 | Battery Voltage: Circuit Voltage Below Threshold | GO to Pinpoint Test H. |
| U3003:17 | Battery Voltage: Circuit Voltage Above Threshold | GO to Pinpoint Test H. |
| All other DTCs | — | REFER to the Master DTC Chart in Section 419-10. |

**Symptom Chart — Climate Control Systems**

Symptom Chart — Climate Control Systems

| Condition | Possible Causes | Action |
|---|---|---|
| **NOTE:** *Some PCM DTCs may inhibit A/C operation. If any PCM DTCs are retrieved, diagnose those first. Refer to the PCM DTC Chart.* | | |
| • No communication with the HVAC module | • HVAC module | • REFER to Section 418-00. |

| | | |
|---|---|---|
| | • Communication network | |
| • Unable to duplicate the customer concern and no DTCs present | • HVAC system and/or related components | • GO to Pinpoint Test I. |
| • Reduced outlet airflow | • Wiring, terminals or connectors<br>• A/C compressor clutch air gap<br>• A/C evaporator temperature sensor<br>• A/C clutch relay<br>• Blower motor control<br>• Blower motor resistor<br>• PCM<br>• HVAC module<br>• FCIM | • If the A/C compressor does not cycle, GO to Pinpoint Test P.<br>• For EMTC base with 4 blower motor speed positions, if the A/C compressor cycles normally, GO to Pinpoint Test X.<br>• For other systems, if the A/C compressor cycles normally, GO to Pinpoint Test Z. |
| • The air inlet door is inoperative | • Wiring, terminals or connectors<br>• Air inlet mode door actuator<br>• Stuck or bound linkage or door<br>• HVAC module<br>• FCIM | • ForEMTC base with 4 blower motor speed positions, GO to Pinpoint Test J.<br>• For other systems, GO to Pinpoint Test K. |
| • Incorrect/erratic direction of airflow from outlets | • Wiring, terminals or connectors<br>• Defrost/panel/floor mode door actuator<br>• Stuck or bound linkage or door<br>• HVAC module<br>• FCIM | • For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test L.<br>• For other systems, GO to Pinpoint Test M. |

| | | |
|---|---|---|
| • Insufficient, erratic or no heat | • Low engine coolant level<br>• Plugged or partially plugged heater core<br>• Temperature blend door(s) binding or stuck<br>• Temperature blend door actuator(s) | • GO to Pinpoint Test N. |
| • Insufficient A/C cooling | • Improper refrigerant level<br>• Temperature blend door(s) binding or stuck<br>• Temperature blend door actuator(s) | • CARRY OUT the refrigerant system tests. Refrigerant System Tests If OK, DIAGNOSE for a temperature blend door actuator not operating correctly.<br>• For inadequate temperature out of the EMTC base with 4 blower motor speed positions, GO to Pinpoint Test R.<br>• For inadequate temperature out of the EMTC with 4.2 inch display with 7 blower motor speed positions, GO to Pinpoint Test S.<br>• For inadequate temperature out of the EATC , dual zone driver side, GO to Pinpoint Test U.<br>• For inadequate temperature out of the EATC , dual zone passenger side, GO to Pinpoint Test V. |
| • The air conditioning (A/C) is inoperative | • Fuse(s)<br>• Wiring, terminals or connectors<br>• A/C system discharged/low charge<br>• PCM<br>• HVAC module<br>• FCIM<br>• A/C clutch relay<br>• A/C compressor clutch air gap<br>• A/C pressure transducer | • GO to Pinpoint Test O. |

| | | |
|---|---|---|
| | • Evaporator temperature sensor<br>• A/C compressor clutch field coil | |
| • The air conditioning (A/C) is always on — A/C compressor does not cycle | • Wiring, terminals or connectors<br>• PCM<br>• Evaporator temperature sensor<br>• A/C clutch relay<br>• A/C compressor clutch air gap | • GO to Pinpoint Test P. |
| • The air conditioning (A/C) is always on — A/C mode always commanded ON | • Wiring, terminals or connectors<br>• PCM<br>• FCIM<br>• HVAC module | • GO to Pinpoint Test Q. |
| • Temperature control is inoperative/does not operate correctly — EMTC base with 4 blower motor speed positions | • Wiring, terminals or connectors<br>• HVAC module<br>• Stuck or bound linkage or door<br>• Temperature blend door actuator | • GO to Pinpoint Test R. |
| • Temperature control is inoperative/does not operate correctly — EMTC with 4.2 inch display with 7 blower motor speed positions | • Wiring, terminals or connectors<br>• HVAC module<br>• Stuck or bound linkage or door<br>• Temperature blend door actuator | • GO to Pinpoint Test S. |
| • Temperature control is inoperative/does not operate correctly — EATC , dual zone driver side | • Wiring, terminals or connectors<br>• HVAC module<br>• Stuck or bound linkage or door | • GO to Pinpoint Test U. |

| | | |
|---|---|---|
| | • Temperature blend door actuator<br>• FCIM | |
| • Temperature control is inoperative/does not operate correctly — EATC , dual zone passenger side | • Wiring, terminals or connectors<br>• HVAC module<br>• Stuck or bound linkage or door<br>• Temperature blend door actuator<br>• FCIM | • GO to Pinpoint Test V. |
| • The blower motor is inoperative — EMTC base with 4 blower motor speed positions | • Fuse(s)<br>• Wiring, terminals or connectors<br>• Blower motor relay<br>• Blower motor<br>• HVAC module | • GO to Pinpoint Test W. |
| • The blower motor does not operate correctly — EMTC base with 4 blower motor speed positions | • Wiring, terminals or connectors<br>• Blower motor resistor<br>• HVAC module | • GO to Pinpoint Test X. |
| • The blower motor is inoperative — EATC dual zone and EMTC with 4.2 inch display with 7 blower motor speed positions | • Fuse(s)<br>• Wiring, terminals or connectors<br>• Blower motor relay<br>• Blower motor<br>• HVAC module<br>• Blower motor speed control | • GO to Pinpoint Test Y. |
| • The blower motor does not operate correctly — EATC dual zone and EMTC with 4.2 inch display with 7 blower motor speed positions | • Wiring, terminals or connectors<br>• Blower motor speed control<br>• HVAC module | • GO to Pinpoint Test Z. |

| | | |
|---|---|---|
| | • FCIM | |
| • A/C pressure relief valve discharging | • High system pressure<br>• A/C pressure relief valve | • CHECK the high side system pressure. If the pressure is below the A/C pressure relief valve open pressure, REPLACE the A/C pressure relief valve. If the system pressure is above the A/C pressure relief valve open pressure, REPAIR the system for a restriction. |

**Symptom Chart — NVH**

Symptom Chart — NVH

| Condition | Possible Causes | Action |
|---|---|---|
| **NOTE:** *NVH symptoms will be identified using the diagnostic tools that are available. For a list of these tools, an explanation of their uses and a glossary of common terms, refer to Section 100-04. Since it is possible any one of multiple systems may be the cause of a symptom, it may be necessary to use a process of elimination type of diagnostic approach to pinpoint the responsible system. If this is not the causal system for the symptom, refer back to Section 100-04 for the next likely system and continue diagnosis.* | | |
| • Noisy A/C compressor | • A/C compressor clutch air gap out of specification | • CHECK and ADJUST the A/C compressor clutch gap if necessary. REFER to Air Conditioning (A/C) Clutch Air Gap Adjustment in this section. TEST the system for normal operation.<br>• If the A/C compressor clutch gap is OK, INSTALL an A/C compressor clutch. REFER to Section 412-01. TEST the system for normal operation. |
| | • A/C compressor pulley bearing worn | • INSPECT the A/C compressor pulley bearing for roughness. If bearing roughness is found, INSTALL an A/C compressor pulley. REFER to Section 412-01. TEST the system for normal operation. |
| | • A/C compressor bearing worn | • INSPECT the A/C compressor bearing for roughness. If bearing roughness is found, INSTALL an A/C compressor. REFER to Section 412-01. TEST the system for normal operation. |

**Pinpoint Tests**

**Pinpoint Test A: DTC P0532 or P0533**

Refer to Wiring Diagrams Cell 54 , Manual Climate Control System for schematic and connector information.

Refer to Wiring Diagrams Cell 55 , Automatic Climate Control System for schematic and connector information.

**Normal Operation**

The A/C pressure transducer receives a ground from the PCM. A 5-volt reference voltage is supplied to the A/C pressure transducer from the PCM. The A/C pressure transducer then sends a voltage to the PCM to indicate the A/C pressure.

## DTC Fault Trigger Conditions

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| P0532 | A/C Refrigerant Pressure Sensor A Circuit Low: No Sub Type Information | The A/C pressure transducer inputs a feedback voltage to the PCM. This DTC sets if the feedback voltage is less than 0.26 volts for at least 2 seconds and the ambient air temperature is greater than 0°C (32°F). |
| P0533 | A/C Refrigerant Pressure Sensor A Circuit High: No Sub Type Information | The A/C pressure transducer inputs a feedback voltage to the PCM. This DTC sets if the feedback voltage is greater than 4.95 volts for at least 2 seconds and the ambient air temperature is greater than 0°C (32°F). |

**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- A/C pressure transducer
- PCM

**PINPOINT TEST A : DTC P0532 OR P0533**

**A1 COMPARE THE A/C PRESSURE SENSOR (ACP_PRESS) PCM PARAMETER IDENTIFICATION (PID) WITH THE MANIFOLD GAUGE SET READINGS**

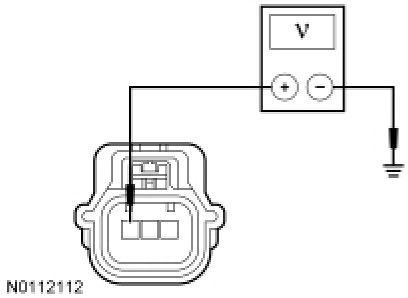
- Allow the A/C system to stabilize to the outside ambient temperature.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: PCM DataLogger .
- With the R-134a manifold gauge set connected, compare the pressure readings of the manifold gauge set and the ACP_PRESS PCM PID.

**Are the pressure values of the manifold gauge set and the Parameter Identification (PID) within +/-15 psi?**

| | |
|---|---|
| **Yes** | IGNORE the Diagnostic Trouble Codes (DTCs). REFER to the Symptom Chart in this section. |
| **No** | GO to A2. |

**A2 CHECK THE PCM REFERENCE VOLTAGE CIRCUIT**

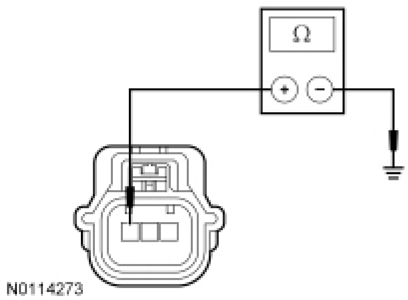
- Ignition OFF.
- Disconnect: A/C Pressure Transducer C1260 .
- Ignition ON.
- Measure the voltage between ground and A/C pressure transducer C1260 Pin 3, circuit LE424 (YE/GN), harness side.

- 


**Is the voltage between 4.7 and 5.1 volts?**

| | |
|---|---|
| **Yes** | GO to A5. |
| **No** | If the voltage is below 4.7 volts, GO to A3.<br>If the voltage is greater than 5.1 volts, REPAIR the circuit for a short to voltage. |

**A3 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR A SHORT TO GROUND**

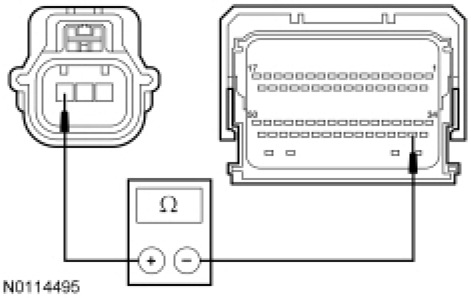
- Ignition OFF.
- Disconnect: PCM C1381B (5.0L, 6.2L) .
- Disconnect: PCM C175B (3.7L) .
- Disconnect: PCM C1551B (3.5L) .
- Measure the resistance between ground and A/C pressure transducer C1260 Pin 3, circuit LE424 (YE/GN), harness side.
- 


**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to A4. |
| **No** | REPAIR the circuit. |

**A4 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR AN OPEN**

- **For 3.7L, 5.0L and 6.2L**

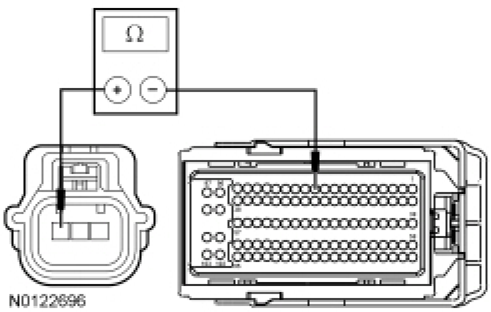



N0114495

Measure the resistance between A/C pressure transducer C1260 Pin 3, circuit LE424 (YE/GN), harness side and:

- For 3.7L, PCM C175B Pin 52, circuit LE424 (YE/GN), harness side.
- For 5.0L and 6.2L, PCM C1381B Pin 52, circuit LE424 (YE/GN), harness side.

- **For 3.5L**




N0122696

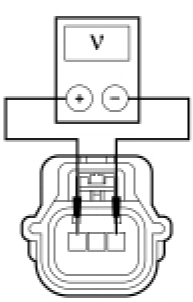
For 3.5L, measure the resistance between A/C pressure transducer C1260 Pin 3, circuit LE424 (YE/GN), harness side and PCM C1551B Pin 11, circuit LE424 (YE/GN), harness side.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to A14. |
| **No** | REPAIR the circuit. |

## A5 CHECK THE PCM SENSOR GROUND

- Measure the voltage between A/C pressure transducer C1260 Pin 1, circuit RE407 (YE/VT), harness side and A/C pressure transducer C1260 Pin 3, circuit LE424 (YE/GN), harness side.

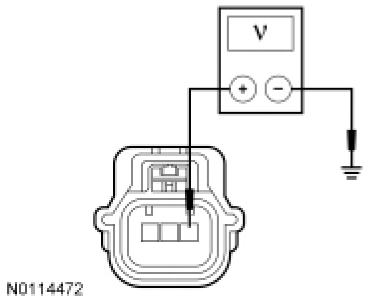
- 


N0112113

**Is the voltage greater than 4.7 volts?**

| | |
|---|---|
| **Yes** | If diagnosing DTC P0532, GO to A8. If diagnosing DTC P0533, GO to A11. |
| **No** | GO to A6. |

## A6 CHECK THE SIGNAL RETURN CIRCUIT FOR VOLTAGE

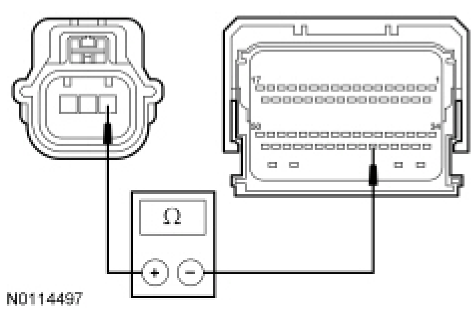
- Measure the voltage between ground and A/C pressure transducer C1260 Pin 1, circuit RE407 (YE/VT), harness side.
- 


**Is any voltage present?**

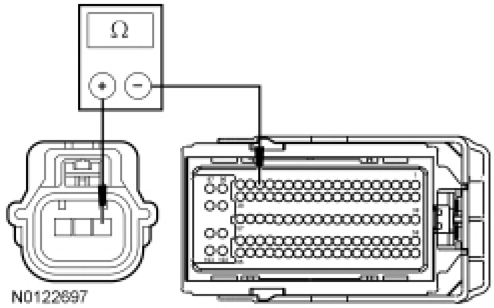
| | |
|---|---|
| **Yes** | If the voltage is 5.1 volts or less, REPAIR circuit RE407 (YE/VT) for a short to circuit LE424 (YE/GN).<br>If the voltage is greater than 5.1 volts, REPAIR circuit RE407 (YE/VT) for a short to voltage. |
| **No** | GO to A7. |

## A7 CHECK THE SIGNAL RETURN CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: PCM C1381B (5.0L, 6.2L) .
- Disconnect: PCM C175B (3.7L) .
- Disconnect: PCM C1551B (3.5L) .
- **For 3.7L, 5.0L and 6.2L**


  Measure the resistance between A/C pressure transducer C1260 Pin 1, circuit RE407 (YE/VT), harness side and:
  - For 3.7L, PCM C175B Pin 56, circuit RE407 (YE/VT), harness side.
  - For 5.0L and 6.2L, PCM C1381B Pin 56, circuit RE407 (YE/VT), harness side.
- **For 3.5L**



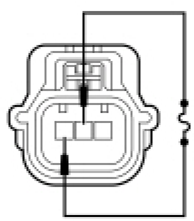

For 3.5L, measure the resistance between A/C pressure transducer C1260 Pin 1, circuit RE407 (YE/VT), harness side and PCM C1551B Pin 17, circuit RE407 (YE/VT), harness side.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to A14. |
| **No** | REPAIR the circuit. |

**A8 CHECK THE A/C PRESSURE SENSOR (ACP_V) PCM PID WITH THE A/C PRESSURE TRANSDUCER SHORTED**

- Enter the following diagnostic mode on the scan tool: PCM DataLogger .
- While observing the ACP_V PCM PID, connect a fused jumper between A/C pressure transducer C1260 Pin 3, circuit LE424 (YE/GN), harness side and A/C pressure transducer C1260 Pin 2, circuit VH433 (VT/OG), harness side.



- 

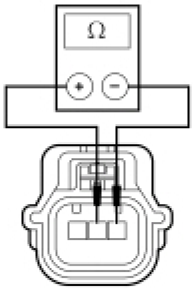
**Does the ACP_V PCM PID voltage read between 4.7 and 5.1 volts?**

| | |
|---|---|
| **Yes** | INSTALL a new A/C pressure transducer. |
| **No** | GO to A9. |

**A9 CHECK THE A/C PRESSURE TRANSDUCER FEEDBACK CIRCUIT FOR A SHORT TO THE SIGNAL RETURN CIRCUIT**

- Ignition OFF.
- Disconnect: PCM C1381B (5.0L, 6.2L) .
- Disconnect: PCM C175B (3.7L) .
- Disconnect: PCM C1551B (3.5L) .

- Measure the resistance between A/C pressure transducer C1260 Pin 2, circuit VH433 (VT/OG), harness side and A/C pressure transducer C1260 Pin 1, circuit RE407 (YE/VT), harness side.

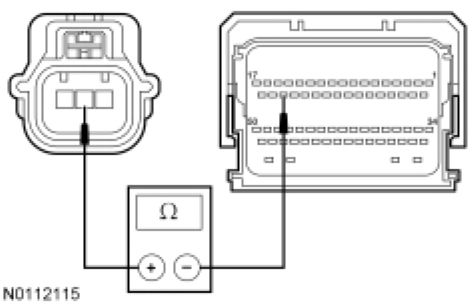
- 
N0112118

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to A10. |
| **No** | REPAIR circuit VH433 (VT/OG) for a short to circuit RE407 (YE/VT). |

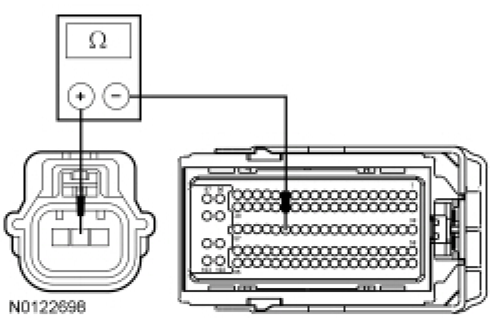
**A10 CHECK THE A/C PRESSURE TRANSDUCER FEEDBACK CIRCUIT FOR AN OPEN**

- **For 3.7L, 5.0L and 6.2L**


N0112115

Measure the resistance between A/C pressure transducer C1260 Pin 2, circuit VH433 (VT/OG), harness side and:
  - For 3.7L, PCM C175B Pin 31, circuit VH433 (VT/OG), harness side.
  - For 5.0L and 6.2L, PCM C1381B Pin 31, circuit VH433 (VT/OG), harness side.

- **For 3.5L**


N0122698

For 3.5L, measure the resistance between A/C pressure transducer C1260 Pin 2, circuit VH433 (VT/OG), harness side and PCM C1551B Pin 52, circuit VH433 (VT/OG), harness side.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to A14. |
| **No** | REPAIR the circuit. |

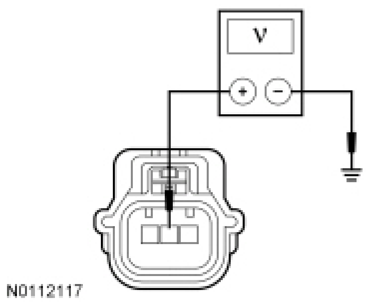
**A11 CHECK THE A/C PRESSURE SENSOR (ACP_V) PCM PID WITH THE A/C PRESSURE TRANSDUCER DISCONNECTED**

- Enter the following diagnostic mode on the scan tool: PCM DataLogger .
- Observe the ACP_V PCM PID voltage.

**Does the ACP_V PCM PID voltage read 0 volts?**

| | |
|---|---|
| **Yes** | INSTALL a new A/C pressure transducer. |
| **No** | GO to A12. |

**A12 CHECK THE A/C PRESSURE TRANSDUCER FEEDBACK CIRCUIT FOR VOLTAGE**

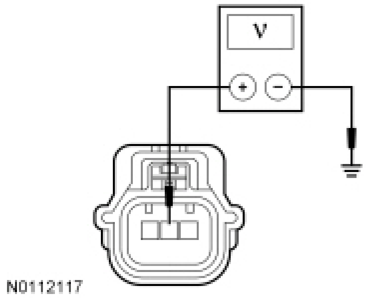
- Measure the voltage between ground and A/C pressure transducer C1260 Pin 2, circuit VH433 (VT/OG), harness side.
- 


**Is any voltage present?**

| | |
|---|---|
| **Yes** | GO to A13. |
| **No** | GO to A14. |

**A13 CHECK THE A/C PRESSURE TRANSDUCER FEEDBACK CIRCUIT FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: PCM C1381B (5.0L, 6.2L) .
- Disconnect: PCM C175B (3.7L) .
- Disconnect: PCM C1551B (3.5L) .
- Ignition ON.
- Measure the voltage between ground and A/C pressure transducer C1260 Pin 2, circuit VH433 (VT/OG), harness side.

- 


**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit. |
| **No** | REPAIR circuit VH433 (VT/OG) for a short to circuit LE424 (YE/GN). |

**A14 CHECK FOR CORRECT PCM OPERATION**

- Ignition OFF.
- Disconnect and inspect all PCM connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all PCM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new PCM. REFER to Section 303-14. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test B: DTC P0645**

Refer to Wiring Diagrams Cell 54 , Manual Climate Control System for schematic and connector information.

Refer to Wiring Diagrams Cell 55 , Automatic Climate Control System for schematic and connector information.

**Normal Operation**

Voltage is provided to the A/C clutch relay coil from Battery Junction Box (BJB) fuse 77 (10A). When A/C is requested and A/C line pressures allow, a ground is provided to the A/C clutch relay coil from the PCM, energizing the A/C clutch relay.

## DTC Fault Trigger Conditions

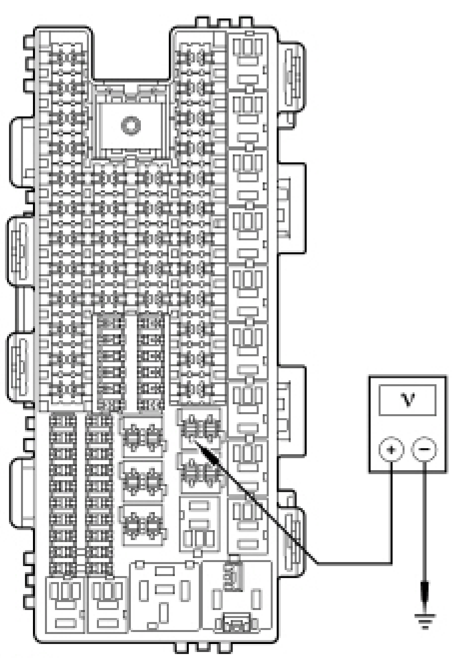
| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| P0645 | A/C Clutch Relay Control Circuit: No Sub Type Information | The DTC sets when the PCM grounds the relay circuit and more voltage than expected is detected on the relay circuit. The DTC also sets when the relay circuit is off and no voltage is detected on the relay circuit. |

**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- A/C clutch relay
- PCM

**PINPOINT TEST B : DTC P0645**

**B1 CHECK THE VOLTAGE TO THE A/C CLUTCH RELAY**

- Ignition OFF.
- Disconnect: A/C Clutch Relay .
- Ignition ON.
- Measure the voltage between ground and the A/C clutch relay socket, circuit CBB77 (WH).
- 


  N0069088

**Is the voltage greater than 11 volts?**

| | |
|---|---|
| **Yes** | GO to B2. |
| **No** | VERIFY BJB fuse 77 (10A) is OK. If OK, REPAIR circuit CBB77 (WH) for an open. TEST the system for normal operation. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. |

## B2 CHECK THE A/C CLUTCH RELAY

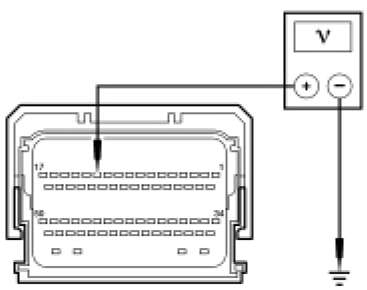
- Carry out the A/C clutch relay component test.
  Refer to Wiring Diagrams Cell 149 for component testing.

**Did the relay pass the component test?**

| | |
|---|---|
| **Yes** | GO to B3. |
| **No** | INSTALL a new A/C clutch relay. TEST the system for normal operation. |

## B3 CHECK THE A/C CLUTCH RELAY COIL SWITCHED GROUND CIRCUIT FOR A SHORT TO VOLTAGE

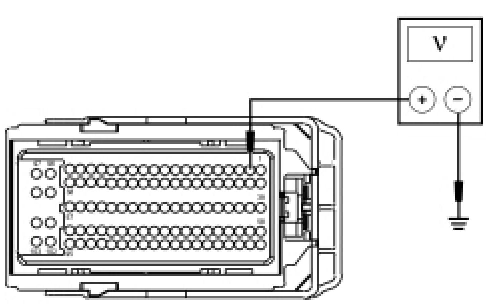
- Ignition OFF.
- Disconnect: PCM C1381B (5.0L, 6.2L) .
- Disconnect: PCM C175B (3.7L) .
- Disconnect: PCM C1551B (3.5L) .
- Ignition ON.
- **For 3.7L, 5.0L and 6.2L**




N0086589

Measure the voltage between ground and:
  - For 3.7L, PCM C175B Pin 12, circuit CH302 (WH/BN), harness side.
  - For 5.0L and 6.2L, PCM C1381B Pin 12, circuit CH302 (WH/BN), harness side.

- **For 3.5L**




N0122699

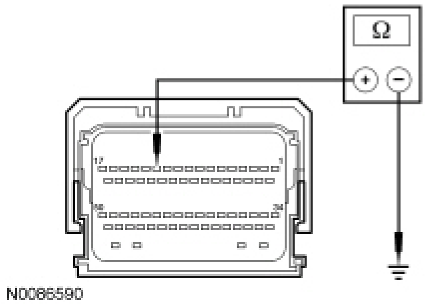
For 3.5L, measure the voltage between ground and PCM C1551B Pin 2, circuit CH302 (WH/BN), harness side.

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit. |
| **No** | GO to B4. |

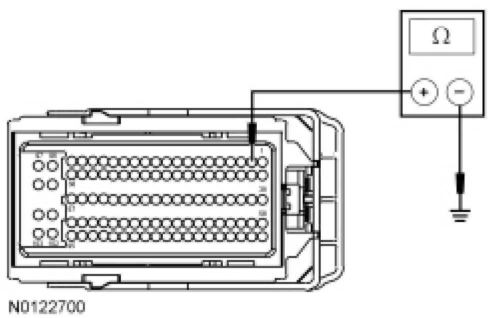
**B4 CHECK THE A/C CLUTCH RELAY COIL SWITCHED GROUND CIRCUIT FOR A SHORT TO GROUND**

- Ignition OFF.
- **For 3.7L, 5.0L and 6.2L**




Measure the resistance between ground and:
  - For 3.7L, PCM C175B Pin 12, circuit CH302 (WH/BN), harness side.
  - For 5.0L and 6.2L, PCM C1381B Pin 12, circuit CH302 (WH/BN), harness side.

- **For 3.5L**



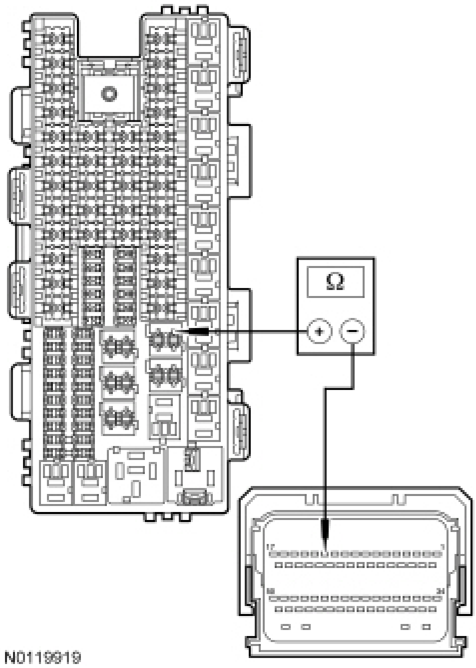

For 3.5L, measure the resistance between ground and PCM C1551B Pin 2, circuit CH302 (WH/BN), harness side.

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to B5. |
| **No** | REPAIR the circuit. |

**B5 CHECK THE A/C CLUTCH RELAY COIL SWITCHED GROUND CIRCUIT FOR AN OPEN**

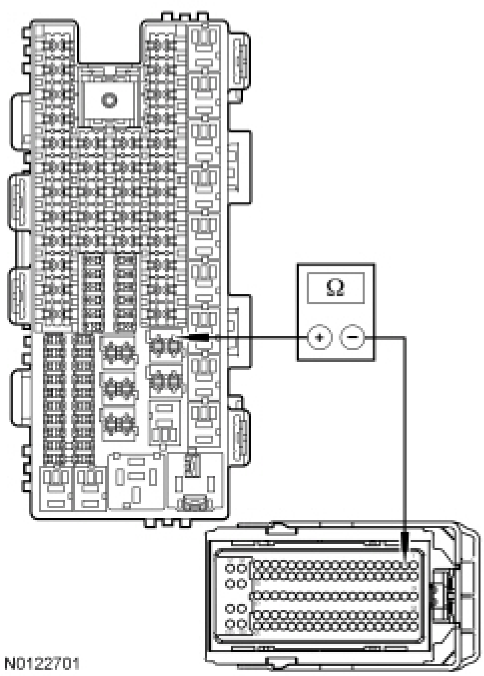
- **For 3.7L, 5.0L and 6.2L**




Measure the resistance between A/C clutch relay socket, circuit CH302 (WH/BN) and:

- For 3.7L, PCM C175B Pin 12, circuit CH302 (WH/BN), harness side.
- For 5.0L and 6.2L, PCM C1381B Pin 12, circuit CH302 (WH/BN), harness side.

- **For 3.5L**




For 3.5L, measure the resistance between A/C clutch relay socket, circuit CH302 (WH/BN) and PCM C1551B Pin 2, circuit CH302 (WH/BN), harness side.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to B6. |
| **No** | REPAIR the circuit. |

**B6 CHECK FOR CORRECT PCM OPERATION**

- Ignition OFF.
- Disconnect and inspect all PCM connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all PCM connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new PCM. REFER to Section 303-14. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test C: DTC C1B14:11 or C1B14:12**

Refer to Wiring Diagrams Cell 54 , Manual Climate Control System for schematic and connector information.

Refer to Wiring Diagrams Cell 55 , Automatic Climate Control System for schematic and connector information.

**Normal Operation**

A 5-volt reference voltage is supplied to the sensors and actuators from the HVAC module.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| C1B14:11 | Sensor Supply Voltage A: Circuit Short to Ground | The HVAC module senses less than 5 volts on the sensor reference voltage circuit, indicating a short to ground. |
| C1B14:12 | Sensor Supply Voltage A: Circuit Short to Battery | The HVAC module senses greater than 5 volts on the sensor reference voltage circuit, indicating a short to voltage. |

**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- HVAC module

**PINPOINT TEST C : DTC C1B14:11 OR C1B14:12**

**C1 CHECK THE SENSOR RESISTANCE**

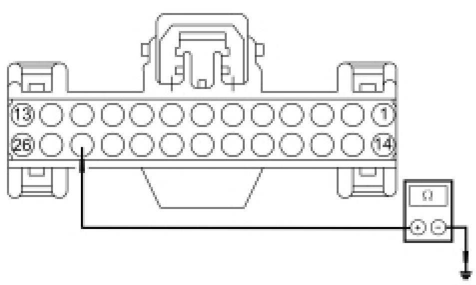
- Check for HVAC DTCs.

**Is DTC C1B14:11 present?**

| | |
|---|---|
| **Yes** | GO to C2. |
| **No** | GO to C5. |

**C2 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR A SHORT TO GROUND**

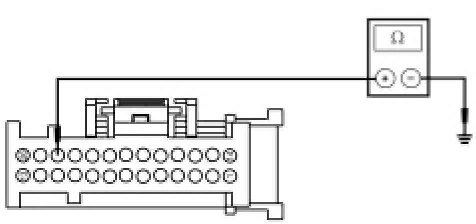
- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: HVAC Module — Base EMTC C294B .
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357B .
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- **For vehicles with Base EMTC**




N0147993

Measure the resistance between ground and HVAC module — Base EMTC C294A Pin 24, circuit LH111 (BN/WH), harness side.

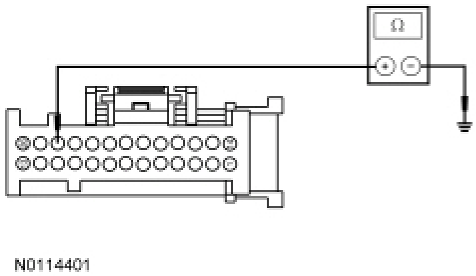
- **For vehicles with Remote Mount EMTC**




N0114401

Measure the resistance between ground and HVAC module — Remote Mount EMTC C2357A Pin 24, circuit LH111 (BN/WH), harness side.

- **For vehicles with EATC**



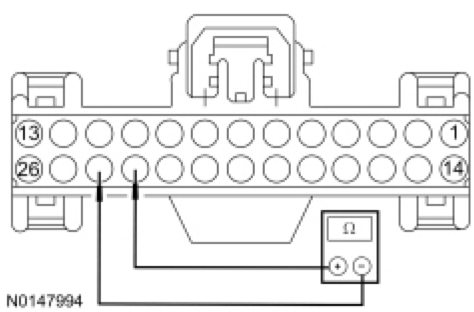

Measure the resistance between ground and HVAC module — EATC C228A Pin 24, circuit LH111 (BN/WH), harness side.

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to C3. |
| **No** | REPAIR the circuit. |

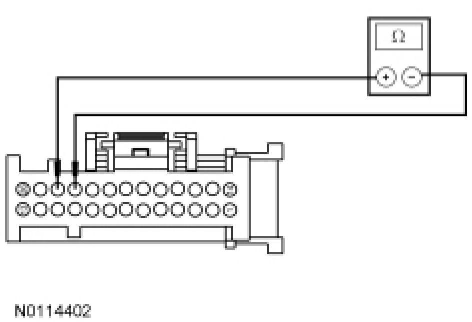
**C3 CHECK THE REFERENCE VOLTAGE CIRCUIT AND THE SIGNAL RETURN CIRCUIT FOR A SHORT TOGETHER**

- **For vehicles with Base EMTC**



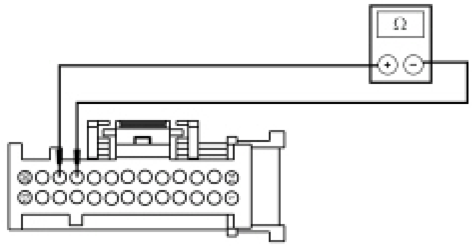

Measure the resistance between HVAC module — Base EMTC C294A Pin 24, circuit LH111 (BN/WH), harness side and HVAC module — EMTC C294A Pin 23, circuit RH111 (GY/BU), harness side.

- **For vehicles with Remote Mount EMTC**




Measure the resistance between HVAC module — Remote Mount EMTC C2357A Pin 24, circuit LH111 (BN/WH), harness side and HVAC module — Remote Mount EMTC C2357A Pin 23, circuit RH111 (GY/BU), harness side.

- **For vehicles with EATC**




N0114402

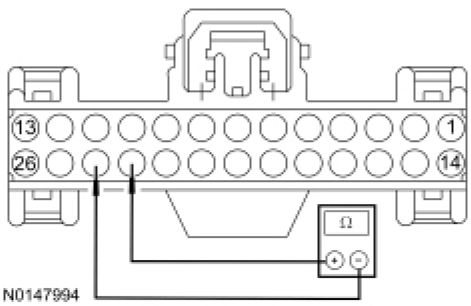
Measure the resistance between HVAC module — EATC C228A Pin 24, circuit LH111 (BN/WH), harness side and HVAC module — EATC C228A Pin 23, circuit RH111 (GY/BU), harness side.

**Is the resistance greater than 200 ohms?**

| | |
|---|---|
| **Yes** | GO to C6. |
| **No** | GO to C4. |

**C4 CHECK THE ACTUATORS**

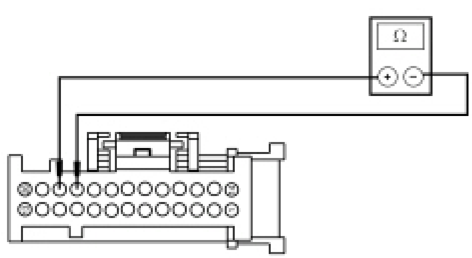
- **For vehicles with BaseEMTC**




N0147994

Measure the resistance between HVAC module — Base EMTC C294A Pin 24, circuit LH111 (BN/WH), harness side and HVAC module — Base EMTC C294A Pin 23, circuit RH111 (GY/BU), harness side.

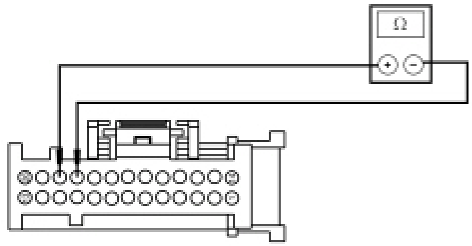
- **For vehicles with Remote Mount EMTC**




N0114402

Measure the resistance between HVAC module — Remote Mount EMTC C2357A Pin 24, circuit LH111 (BN/WH), harness side and HVAC module Remote Mount EMTC C2357A Pin 23, circuit RH111 (GY/BU), harness side.

- **For vehicles with EATC**



N0114402

Measure the resistance between HVAC module — EATC C228A Pin 24, circuit LH111 (BN/WH), harness side and HVAC module C228A Pin 23, circuit RH111 (GY/BU), harness side.

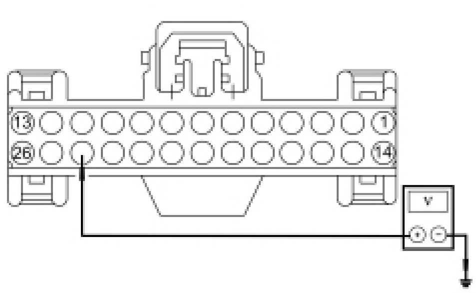
- While measuring the resistance, disconnect the following components one at a time, in order. Stop disconnecting components if the measured resistance rises above 200 ohms.
  - In-vehicle temperature and humidity sensor C2247
  - Air inlet mode door actuator C289
  - Defrost/panel/floor mode door actuator C236
  - Temperature blend door actuator C2381 (EMTC only)
  - Driver side temperature blend door actuator C2091 (EATC only)
  - Passenger side temperature blend door actuator C2092 (EATC only)

**Did the resistance rise above 200 ohms?**

| | |
|---|---|
| **Yes** | INSTALL a new actuator or sensor (the last one to be disconnected). REFER to Section 412-01. When installing an actuator, CONNECT the actuator electrical connector before the HVAC module connector(s). This allows the actuator to be calibrated when the HVAC module is reconnected. TEST the system for normal operation. |
| **No** | REPAIR the circuits. |

**C5 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR A SHORT TO VOLTAGE**

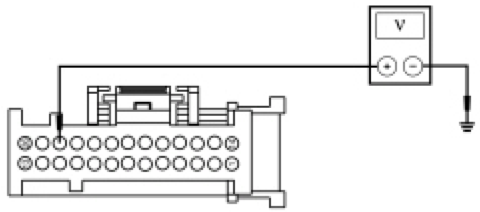
- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: HVAC Module — Base EMTC C294B .
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357B .
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- **For vehicles with Base EMTC**




N0147995

Measure the voltage between ground and HVAC module — EMTC C294A Pin 24, circuit LH111 (BN/WH), harness side.

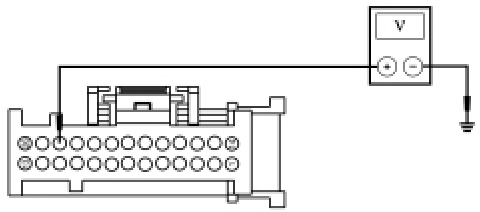
- **For vehicles with Remote Mount EMTC**




N0114403

Measure the voltage between ground and HVAC module — Remote Mount EMTC C2357A Pin 24, circuit LH111 (BN/WH), harness side.

- **For vehicles with EATC**




N0114403

Measure the voltage between ground and HVAC module — EATC C228A Pin 24, circuit LH111 (BN/WH), harness side.

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit. |
| **No** | GO to C6. |

## C6 CHECK FOR CORRECT HVAC MODULE OPERATION

- Ignition OFF.
- Disconnect and inspect all HVAC module connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC module. REFER to Section 412-01. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test D: DTC B1A61:11 or B1A61:15**

Refer to Wiring Diagrams Cell 54 , Manual Climate Control System for schematic and connector information.

Refer to Wiring Diagrams Cell 55 , Automatic Climate Control System for schematic and connector information.

**Normal Operation**

The in-vehicle temperature and humidity sensor receives a ground from the HVAC module. The sensor varies its resistance with the temperature. As the temperature rises, the resistance falls. As the temperature falls the resistance rises. The HVAC module measures this resistance to determine the temperature at the sensor.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B1A61:11 | Cabin Temperature Sensor: Circuit Short to Ground | This CMDTC and on demand DTC sets when the HVAC module senses lower than expected voltage on the sensor feedback circuit, indicating a short to ground. |
| B1A61:15 | Cabin Temperature Sensor: Circuit Short to Battery or Open | This CMDTC and on demand DTC sets when the HVAC module senses greater than expected voltage on the sensor feedback circuit, indicating a short to voltage or an open circuit or sensor. |

**This pinpoint test is intended to diagnose the following:**

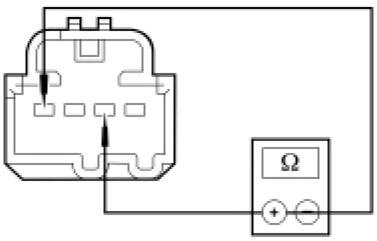
- Wiring, terminals or connectors
- In-vehicle temperature and humidity sensor
- HVAC module

**PINPOINT TEST D : DTC B1A61:11 OR B1A61:15**

**D1 CHECK THE IN-VEHICLE TEMPERATURE SENSOR RESISTANCE**

- Ignition OFF.
- Disconnect: In-Vehicle Temperature and Humidity Sensor C2247 .
- Measure the resistance between the in-vehicle temperature and humidity sensor C2247 Pin 1, component side and C2247 Pin 3, component side and compare to the table below.

| Ambient Temperature | Resistance (Ohms) |
|---|---|
| 0-10°C (32-50°F) | 57,950-97,640 |
| 10-20°C (50-68°F) | 36,880-59,920 |
| 20-25°C (68-77°F) | 29,700-37,790 |
| 25-30°C (77-86°F) | 23,960-30,300 |
| 30-35°C (86-95°F) | 19,440-24,550 |
| 35-40°C (95-104°F) | 15,860-20,000 |
| 40-50°C (104-122°F) | 10,740-16,390 |



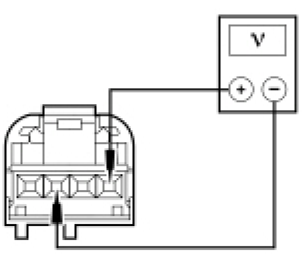

- N0119920

**Is the resistance within the specified values for the temperatures?**

| | |
|---|---|
| **Yes** | GO to D2. |
| **No** | INSTALL a new in-vehicle temperature and humidity sensor. REFER to Section 412-01. |

**D2 CHECK THE HVAC MODULE OUTPUT VOLTAGE**

- Ignition ON.
- Measure the voltage between in-vehicle temperature and humidity sensor C2247 Pin 1, circuit VH414 (GN/BU), harness side and C2247 Pin 3, circuit RH111 (GY/BU), harness side.



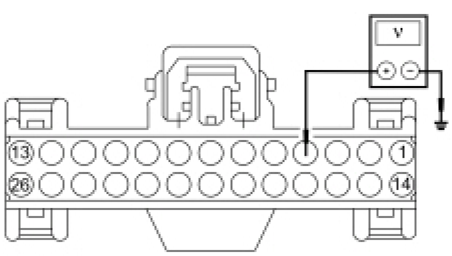

- N0119921

**Is the voltage between 4.7 and 5.1 volts?**

| | |
|---|---|
| **Yes** | INSTALL a new in-vehicle temperature and humidity sensor. REFER to Section 412-01. CLEAR the DTCs. REPEAT the self-test. If the DTC returns, GO to D8. |
| **No** | For DTC B1A61:15, GO to D3.<br>For DTC B1A61:11, GO to D6. |

| D3 CHECK THE IN-VEHICLE TEMPERATURE SENSOR FEEDBACK CIRCUIT FOR A SHORT TO VOLTAGE |
|---|
| |

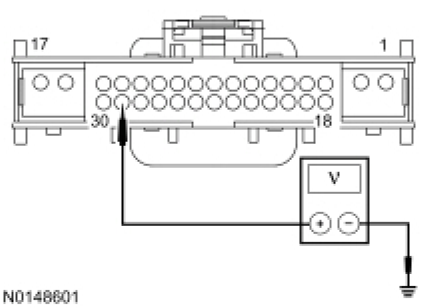
- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: HVAC Module — Base EMTC C294B .
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357B .
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Ignition ON.
- **For Base EMTC vehicles**



N0148600

Measure the voltage between ground and HVAC module — Base EMTC C294A Pin 4, circuit VH414 (GN/BU), harness side.

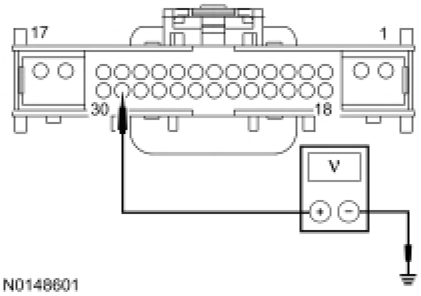
- **For Remote Mount EMTC vehicles**



N0148601

Measure the voltage between ground and HVAC module — Remote Mount EMTC C2357A Pin 29, circuit VH414 (GN/BU), harness side.

- **For EATC vehicles**




N0148601

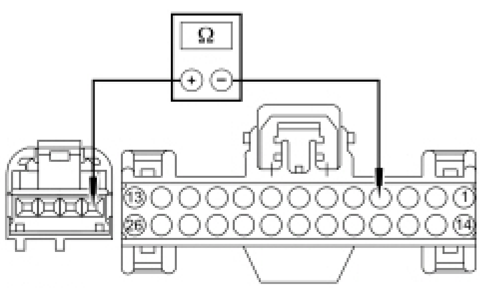
Measure the voltage between ground and HVAC module — EATC C228B Pin 29, circuit VH414 (GN/BU), harness side.

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit. |
| **No** | GO to D4. |

## D4 CHECK THE IN-VEHICLE TEMPERATURE SENSOR FEEDBACK CIRCUIT FOR AN OPEN

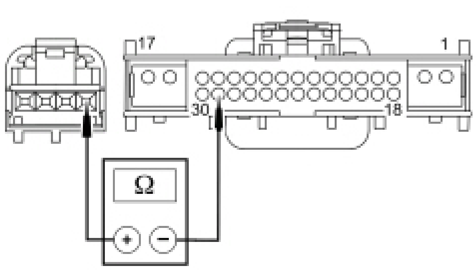
- Ignition OFF.
- **For Base EMTC vehicles**




N0148602

Measure the resistance between HVAC module — Base EMTC C294A Pin 4, circuit VH414 (GN/BU), harness side and in-vehicle temperature and humidity sensor C2247 Pin 1, circuit VH414 (GN/BU), harness side.

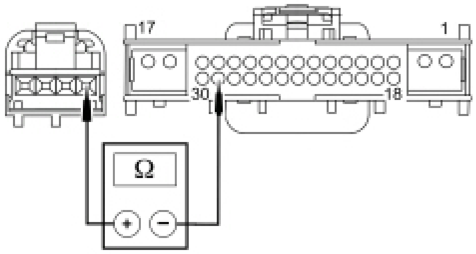
- **For Remote Mount EMTC vehicles**




N0148603

Measure the resistance between HVAC module — Remote Mount EMTC C2357A Pin 29, circuit VH414 (GN/BU), harness side and in-vehicle temperature and humidity sensor C2247 Pin 1, circuit VH414 (GN/BU), harness side.

- **For EATC vehicles**




N0148603

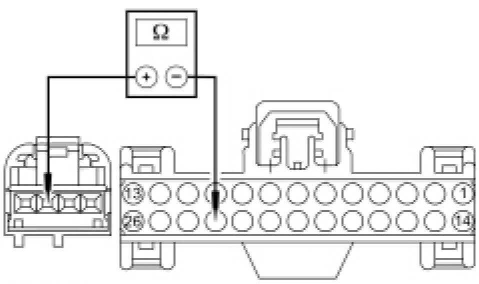
Measure the resistance between HVAC module — EATC C228B Pin 29, circuit VH414 (GN/BU), harness side and in-vehicle temperature and humidity sensor C2247 Pin 1, circuit VH414 (GN/BU), harness side.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to D5. |
| **No** | REPAIR the circuit. |

**D5 CHECK THE SIGNAL RETURN CIRCUIT FOR AN OPEN**

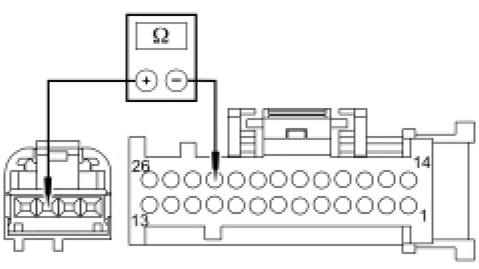
- **For Base EMTC vehicles**




N0148604

Measure the resistance between HVAC module — Base EMTC C294A Pin 23, circuit RH111 (GY/BU), harness side and in-vehicle temperature and humidity sensor C2247 Pin 3, circuit RH111 (GY/BU), harness side.

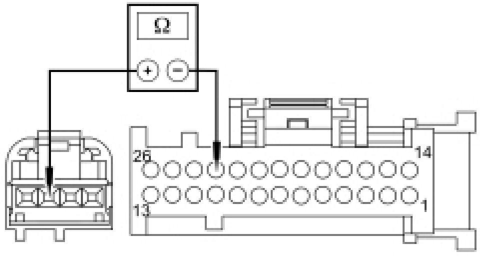
- **For Remote Mount EMTC vehicles**




N0148608

Measure the resistance between HVAC module — Remote Mount EMTC C2357A Pin 23, circuit RH111 (GY/BU), harness side and in-vehicle temperature and humidity sensor C2247 Pin 3, circuit RH111 (GY/BU), harness side.

- **For EATC vehicles**



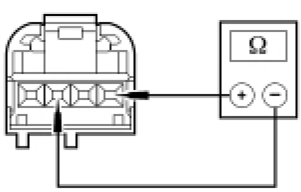

Measure the resistance between HVAC module — EATC C228A Pin 23, circuit RH111 (GY/BU), harness side and in-vehicle temperature and humidity sensor C2247 Pin 3, circuit RH111 (GY/BU), harness side.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to D8. |
| **No** | REPAIR the circuit. |

**D6 CHECK THE IN-VEHICLE TEMPERATURE SENSOR FEEDBACK CIRCUIT FOR A SHORT TO THE SIGNAL RETURN CIRCUIT**

- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: HVAC Module — Base EMTC C294B .
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357B .
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Measure the resistance between in-vehicle temperature and humidity sensor C2247 Pin 1, circuit VH414 (GN/BU), harness side and C2247 Pin 3, circuit RH111 (GY/BU), harness side.



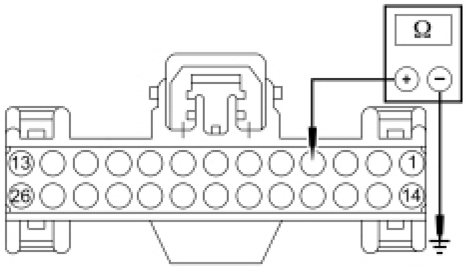
- N0098162

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to D7. |
| **No** | REPAIR the circuits. |

**D7 CHECK THE IN-VEHICLE TEMPERATURE SENSOR FEEDBACK CIRCUIT FOR A SHORT TO GROUND**

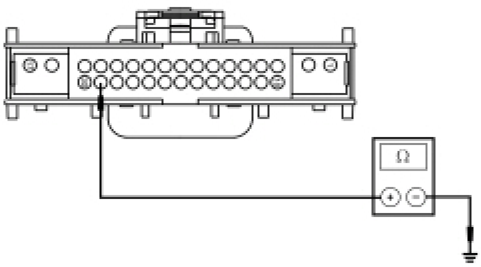
- **For Base EMTC vehicles**




N0148612

Measure the resistance between HVAC module — Base EMTC [C294A](#) Pin 4, circuit VH414 (GN/BU), harness side and ground.

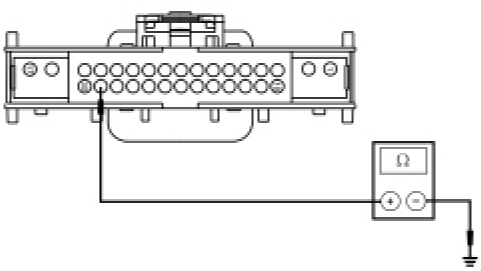
- **For Remote Mount EMTC vehicles**




N0114407

Measure the resistance between HVAC module — Remote Mount EMTC [C2357A](#) Pin 29, circuit VH414 (GN/BU), harness side and ground.

- **For EATC vehicles**




N0114407

Measure the resistance between HVAC module — EATC [C228B](#) Pin 29, circuit VH414 (GN/BU), harness side and ground.

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to [D8](#). |
| **No** | REPAIR the circuit. |

**D8 CHECK FOR CORRECT HVAC MODULE OPERATION**

- Ignition OFF.

- Disconnect and inspect all HVAC module connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC module. REFER to Section 412-01. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test E: DTC B1A69:11 or B1A69:15**

Refer to Wiring Diagrams Cell 54 , Manual Climate Control System for schematic and connector information.

Refer to Wiring Diagrams Cell 55 , Automatic Climate Control System for schematic and connector information.

**Normal Operation**

The in-vehicle temperature and humidity sensor receives a ground from the HVAC module. The sensor varies its resistance with the humidity inside the vehicle. As the humidity rises, the resistance falls. As the humidity falls the resistance rises. The HVAC module measures this resistance to determine the humidity at the sensor.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B1A69:11 | Humidity Sensor: Circuit Short to Ground | This DTC sets when the HVAC module senses lower than expected voltage on the sensor feedback circuit, indicating a short to ground. |
| B1A69:15 | Humidity Sensor: Circuit Short to Battery or Open | This DTC sets when the HVAC module senses greater than expected voltage on the sensor feedback circuit, indicating a short to voltage or an open circuit or sensor. |

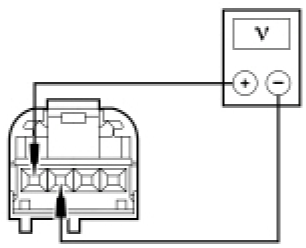
**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- In-vehicle temperature and humidity sensor
- HVAC module

# PINPOINT TEST E : DTC B1A69:11 OR B1A69:15

## E1 CHECK THE HVAC MODULE OUTPUT VOLTAGE

- Ignition OFF.
- Disconnect: In-Vehicle Temperature and Humidity Sensor C2247 .
- Ignition ON.
- Measure the voltage between in-vehicle temperature and humidity sensor C2247 Pin 4, circuit VH413 (WH/BU), harness side and C2247 Pin 3, circuit RH111 (GY/BU), harness side.



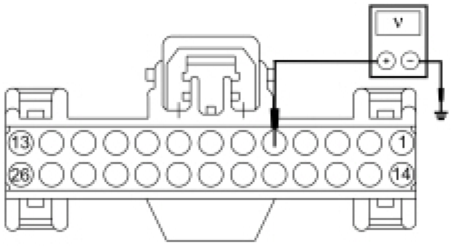

- N0119929

**Is the voltage between 4.7 and 5.1 volts?**

| | |
|---|---|
| **Yes** | INSTALL a new in-vehicle temperature and humidity sensor. REFER to Section 412-01. CLEAR the DTCs. REPEAT the self-test. If the DTC returns, GO to E7. |
| **No** | For DTC B1A69:15, GO to E2.<br>For DTC B1A69:11, GO to E5. |

## E2 CHECK THE HUMIDITY SENSOR FEEDBACK CIRCUIT FOR A SHORT TO VOLTAGE

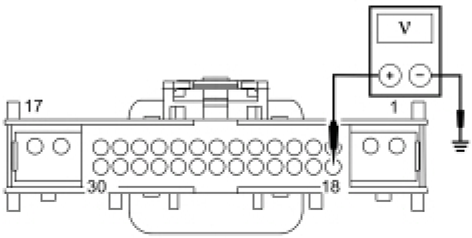
- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: HVAC Module — Base EMTC C294B .
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357B .
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Ignition ON.
- **For Base EMTC vehicles**




N0148613

Measure the voltage between ground and HVAC module — Base EMTC C294A Pin 5, circuit VH413 (WH/BU), harness side.

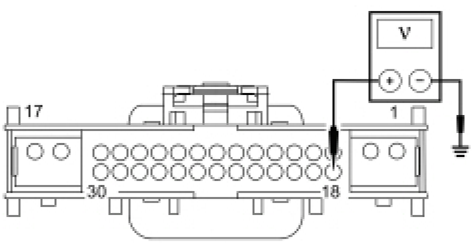
- **For Remote Mount EMTC vehicles**




N0148614

Measure the voltage between ground and HVAC module — Remote Mount EMTC C2357A Pin 18, circuit VH413 (WH/BU), harness side.

- **For EATC vehicles**




N0148614

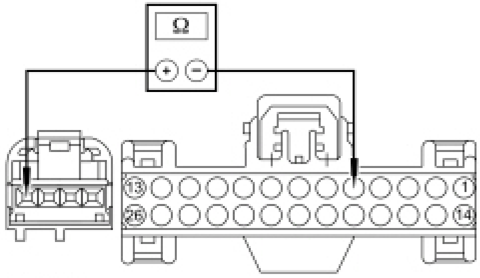
Measure the voltage between ground and HVAC module — EATC C228B Pin 18, circuit VH413 (WH/BU), harness side.

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit. |
| **No** | GO to E3. |

**E3 CHECK THE HUMIDITY SENSOR FEEDBACK CIRCUIT FOR AN OPEN**

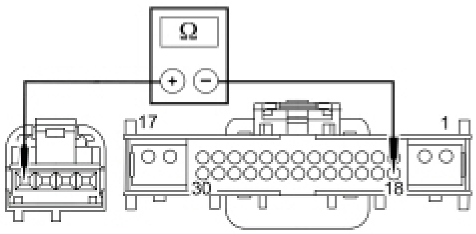
- Ignition OFF.
- **For Base EMTC vehicles**




N0148615

Measure the resistance between HVAC module — Base EMTC C294A Pin 5, circuit VH413 (WH/BU), harness side and in-vehicle temperature and humidity sensor C2247 Pin 4, circuit VH413 (WH/BU), harness side.

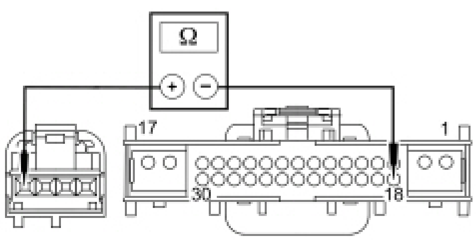
- **For Remote Mount EMTC vehicles**




N0148616

Measure the resistance between HVAC module — Remote Mount EMTC C2357A Pin 18, circuit VH413 (WH/BU), harness side and in-vehicle temperature and humidity sensor C2247 Pin 4, circuit VH413 (WH/BU), harness side.

- **For EATC vehicles**




N0148616

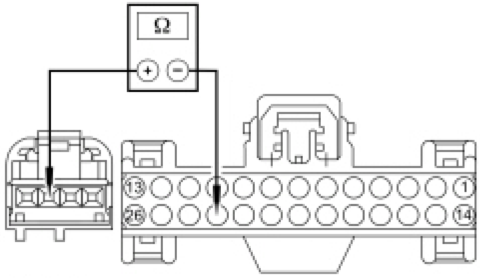
Measure the resistance between HVAC module — EATC C228B Pin 18, circuit VH413 (WH/BU), harness side and in-vehicle temperature and humidity sensor C2247 Pin 4, circuit VH413 (WH/BU), harness side.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to E4. |
| **No** | REPAIR the circuit. |

**E4 CHECK THE SIGNAL RETURN CIRCUIT FOR AN OPEN**

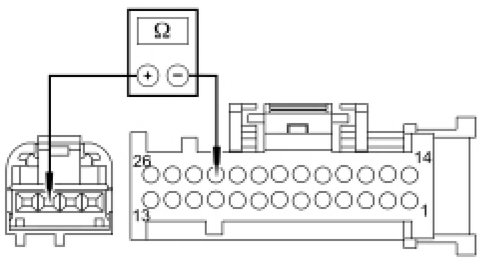
- **For Base EMTC vehicles**




N0148604

Measure the resistance between HVAC module — Base EMTC C294A Pin 23, circuit RH111 (GY/BU), harness side and in-vehicle temperature and humidity sensor C2247 Pin 3, circuit RH111 (GY/BU), harness side.

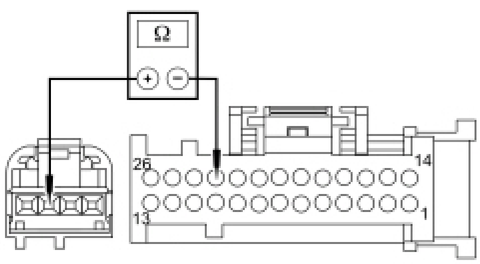
- **For Remote Mount EMTC vehicles**




N0148608

Measure the resistance between HVAC module — For Remote Mount EMTC C2357A Pin 23, circuit RH111 (GY/BU), harness side and in-vehicle temperature and humidity sensor C2247 Pin 3, circuit RH111 (GY/BU), harness side.

- **For EATC vehicles**




N0148608

Measure the resistance between HVAC module — EATC C228A Pin 23, circuit RH111 (GY/BU), harness side and in-vehicle temperature and humidity sensor C2247 Pin 3, circuit RH111 (GY/BU), harness side.

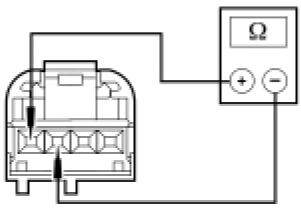
**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to E7. |
| **No** | REPAIR the circuit. |

**E5 CHECK THE HUMIDITY SENSOR FEEDBACK CIRCUIT FOR A SHORT TO THE SIGNAL RETURN CIRCUIT**

- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .

- Disconnect: HVAC Module — Base EMTC C294B .
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357B .
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Measure the resistance between in-vehicle temperature and humidity sensor C2247 Pin 4, circuit VH413 (WH/BU), harness side and C2247 Pin 3, circuit RH111 (GY/BU), harness side.



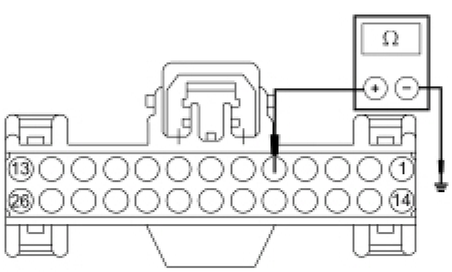

- N0119936

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to E6. |
| **No** | REPAIR the circuits. |

**E6 CHECK THE HUMIDITY SENSOR FEEDBACK CIRCUIT FOR A SHORT TO GROUND**

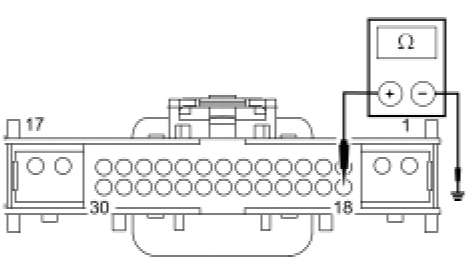
- **For Base EMTC vehicles**




N0148635

Measure the resistance between HVAC module — Base EMTC C294A Pin 5, circuit VH413 (WH/BU), harness side and ground.

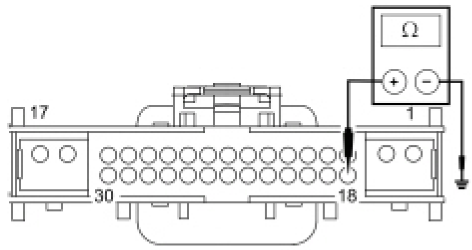
- **For Remote Mount EMTC vehicles**




N0148636

Measure the resistance between HVAC module — Remote Mount EMTC C2357B Pin 18, circuit VH413 (WH/BU), harness side and ground.

- **For EATC vehicles**




N0148636

Measure the resistance between HVAC module — EATC C228B Pin 18, circuit VH413 (WH/BU), harness side and ground.

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to E7. |
| **No** | REPAIR the circuit. |

**E7 CHECK FOR CORRECT HVAC MODULE OPERATION**

- Ignition OFF.
- Disconnect and inspect all HVAC module connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC module. REFER to Section 412-01. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test F: DTC B1A63:11, B1A63:15, B1A64:11 or DTC B1A64:15**

Refer to Wiring Diagrams Cell 55 , Automatic Climate Control System for schematic and connector information.

**Normal Operation**

The autolamp/autolamp/sunload sensor is grounded. As the light applied to the sensor changes, the HVAC module detects the change through the passenger side and driver side feedback circuits.

**DTC Fault Trigger Conditions**

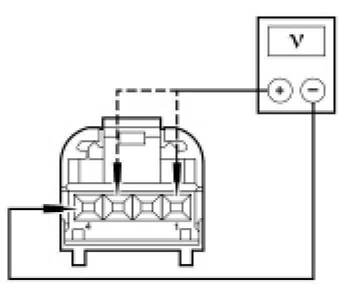
| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B1A63:11 | Right Solar Sensor: Circuit Short to Ground | This DTC sets when the HVAC module senses lower than expected voltage on the passenger side sensor feedback circuit, indicating a short to ground. |
| B1A63:15 | Right Solar Sensor: Circuit Short to Battery or Open | This DTC sets when the HVAC module senses greater than expected voltage on the passenger side sensor feedback circuit, indicating a short to voltage or an open circuit or sensor. |
| B1A64:11 | Left Solar Sensor: Circuit Short to Ground | This DTC sets when the HVAC module senses lower than expected voltage on the driver side sensor feedback circuit, indicating a short to ground. |
| B1A64:15 | Left Solar Sensor: Circuit Short to Battery or Open | This DTC sets when the HVAC module senses greater than expected voltage on the driver side sensor feedback circuit, indicating a short to voltage or an open circuit or sensor. |

**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- Autolamp/autolamp/sunload sensor
- HVAC module

**PINPOINT TEST F : DTC B1A63:11, B1A63:15, B1A64:11 OR DTC B1A64:15**

**F1 CHECK THE AUTOLAMP/SUNLOAD SENSOR REFERENCE VOLTAGE**

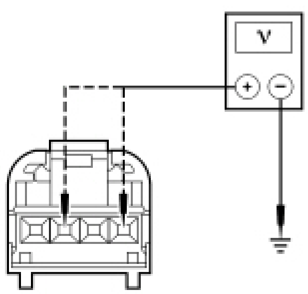
- Ignition OFF.
- Disconnect: Autolamp/Sunload Sensor C286 .
- Ignition ON.
- Press the AUTO button.
- Measure the voltage between autolamp/sunload sensor C286 Pin 4, circuit RH111 (GY/BU), harness side and autolamp/sunload sensor:
  - For DTC B1A63:11 or B1A63:15, C286 Pin 1, circuit VH417 (YE/OG), harness side.
  - For DTC B1A64:11 or B1A64:15, C286 Pin 3, circuit VH416 (VT), harness side.
- 

N0100611

**Is the voltage between 4.7 and 5.1 volts?**

| | |
|---|---|
| **Yes** | INSTALL a new autolamp/sunload sensor. REFER to Section 417-01. CLEAR the DTCs. REPEAT the self-test. If the code returns, GO to F7. |
| **No** | If diagnosing DTC B1A63:15 or DTC B1A64:15, GO to F2.<br>If diagnosing DTC B1A63:11 or DTC B1A64:11, GO to F5. |

**F2 CHECK THE AUTOLAMP/SUNLOAD SENSOR FEEDBACK CIRCUITS FOR A SHORT TO VOLTAGE**

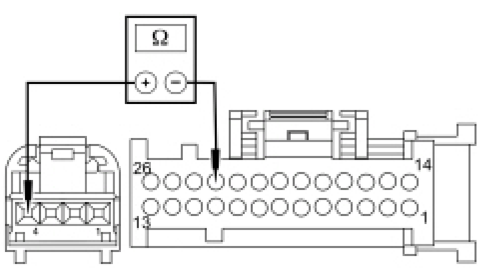
- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Ignition ON.
- Measure the voltage between ground and autolamp/sunload sensor:
  - For DTC B1A64:15, C286 Pin 3, circuit VH416 (VT), harness side.
  - For DTC B1A63:15, C286 Pin 1, circuit VH417 (YE/OG), harness side.
- 

N0086033

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |
| **No** | GO to F3. |

**F3 CHECK THE AUTOLAMP/SUNLOAD SENSOR GROUND CIRCUIT FOR AN OPEN**

- Ignition OFF.
- 

N0148637

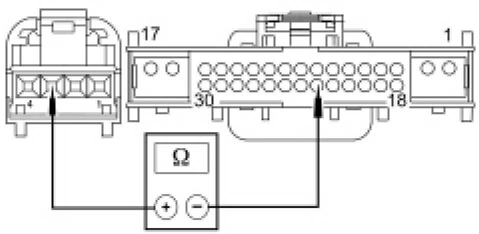
  Measure the resistance between autolamp/sunload sensor C286 Pin 4, circuit RH111 (GY/BU), harness side and HVAC module — EATC C228A Pin 23, circuit RH111 (GY/BU), harness side.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to F4. |
| **No** | REPAIR the circuit. |

## F4 CHECK THE AUTOLAMP/SUNLOAD SENSOR FEEDBACK CIRCUITS FOR AN OPEN

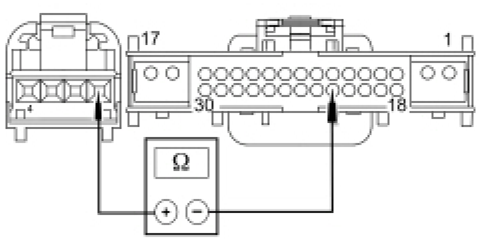
- **For DTC B1A64:15**




N0148638

Measure the resistance between HVAC module — EATC C228B Pin 23, circuit VH416 (VT), harness side and autolamp/sunload sensor C286 Pin 3, circuit VH416 (VT), harness side.

- **For DTC B1A63:15**




N0148639

Measure the resistance between HVAC module — EATC C228B Pin 22, circuit VH417 (YE/OG), harness side and autolamp/sunload sensor C286 Pin 1, circuit VH417 (YE/OG), harness side.

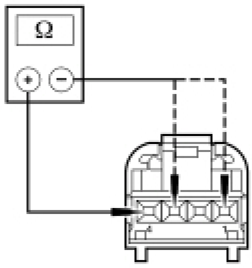
**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to F7. |
| **No** | REPAIR the circuit. |

## F5 CHECK THE AUTOLAMP/SUNLOAD SENSOR FEEDBACK CIRCUITS FOR A SHORT TO THE AUTOLAMP/SUNLOAD SENSOR GROUND CIRCUIT

- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Measure the resistance between autolamp/sunload sensor C286 Pin 4, circuit RH111 (GY/BU), harness side and autolamp/sunload sensor:

- For DTC B1A64:11, C286 Pin 3, circuit VH416 (VT), harness side.
- For DTC B1A63:11, C286 Pin 1, circuit VH417 (YE/OG), harness side.

- 

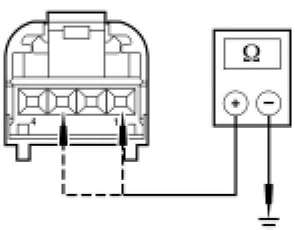
N0084815

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to F6. |
| **No** | REPAIR the circuit in question. |

| **F6 CHECK THE AUTOLAMP/SUNLOAD SENSOR FEEDBACK CIRCUITS FOR A SHORT TO GROUND** |
|---|

- Measure the resistance between ground and autolamp/sunload sensor:
  - For DTC B1A64:11, C286 Pin 3, circuit VH416 (VT), harness side.
  - For DTC B1A63:11, C286 Pin 1, circuit VH417 (YE/OG), harness side.

- 

N0083695

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to F7. |
| **No** | REPAIR the circuit in question. |

| **F7 CHECK FOR CORRECT HVAC MODULE OPERATION** |
|---|

- Ignition OFF.
- Disconnect and inspect all HVAC module connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC module. REFER to Section 412-01. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

## Pinpoint Test G: DTC B1B71:11 or B1B71:15

Refer to Wiring Diagrams Cell 54 , Manual Climate Control System for schematic and connector information.

Refer to Wiring Diagrams Cell 55 , Automatic Climate Control System for schematic and connector information.

### Normal Operation

The evaporator temperature sensor receives a ground from the HVAC module. The sensor varies its resistance with the temperature. As the temperature rises, the resistance falls. As the temperature falls the resistance rises. The HVAC module measures this resistance to determine the temperature at the sensor.

An accurate evaporator temperature is critical to prevent evaporator icing. The HVAC module uses the temperature measurement to turn off the A/C compressor before the evaporator temperatures are cold enough to freeze the condensation. This prevents ice blockage of airflow over the evaporator core.

### DTC Fault Trigger Conditions

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B1B71:11 | Evaporator Temperature Sensor: Circuit Short to Ground | This CMDTC and on demand DTC sets when the HVAC module senses lower than expected voltage on the sensor feedback circuit, indicating a short to ground. |
| B1B71:15 | Evaporator Temperature Sensor: Circuit Short to Battery or Open | This CMDTC and on demand DTC sets when the HVAC module senses greater than expected voltage on the sensor feedback circuit, indicating a short to voltage or an open circuit or sensor. |

**This pinpoint test is intended to diagnose the following:**

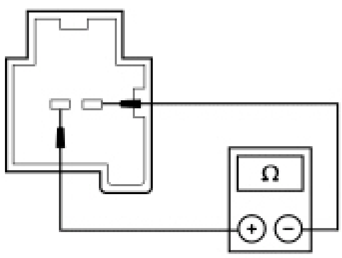
- Wiring, terminals or connectors
- Evaporator temperature sensor
- HVAC module

**PINPOINT TEST G : DTC B1B71:11 OR B1B71:15**

| G1 CHECK THE EVAPORATOR TEMPERATURE SENSOR RESISTANCE |
|---|
| • Ignition OFF. |

- Disconnect: Evaporator Temperature Sensor C296 .
- Measure the resistance between the evaporator temperature sensor C296 Pin 1, component side and evaporator temperature sensor C296 Pin 2, component side and compare to the table below.

| Ambient Temperature | Resistance (Ohms) |
|---|---|
| -40°C (-40°F) | 1,005,00-1,104,000 |
| -20°C (-4°F) | 287,811-307,800 |
| 0°C (32°F) | 96,890-101,300 |
| 20°C (68°F) | 37,080-38,000 |
| 25°C (77°F) | 29,700-30,300 |
| 30°C (86°F) | 23,840-24,430 |
| 35°C (95°F) | 19,260-19,820 |
| 40°C (104°F) | 15,660-16,180 |
| 60°C (140°F) | 7,239-7,594 |
| 100°C (212°F) | 1,943-2,091 |

- 

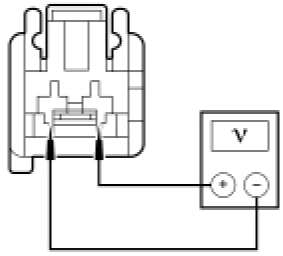
A0004761

**Is the resistance within the specified values for the temperatures?**

| | |
|---|---|
| **Yes** | GO to G2. |
| **No** | INSTALL a new heater core and evaporator core housing. REFER to Section 412-01. |

**G2 CHECK THE HVAC MODULE OUTPUT VOLTAGE**

- Ignition ON.
- Select PANEL on the HVAC controls.
- Measure the voltage between evaporator temperature sensor C296 Pin 1, circuit VH406 (VT/BN), harness side and C296 Pin 2, circuit RH111 (GY/BU), harness side.

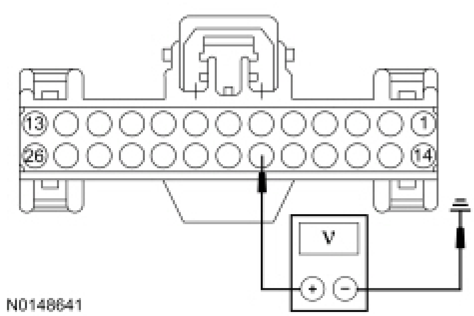



**Is the voltage between 4.7 and 5.1 volts?**

| | |
|---|---|
| **Yes** | GO to G8. |
| **No** | For DTC B1B71:11, GO to G6.<br>For DTC B1B71:15, GO to G3. |

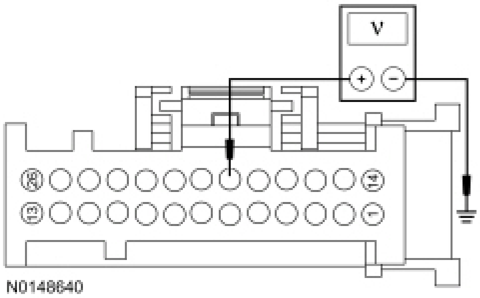
**G3 CHECK THE EVAPORATOR TEMPERATURE SENSOR FEEDBACK CIRCUIT FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: HVAC Module — Base EMTC C294B .
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357B .
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Ignition ON.
- **For Base EMTC vehicles**



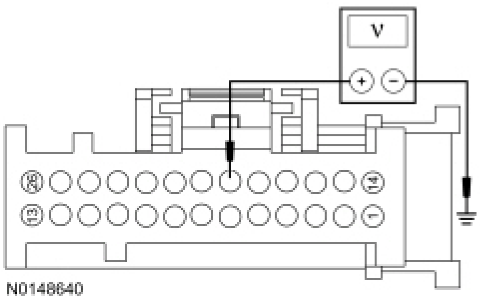

Measure the voltage between ground and HVAC module — Base EMTC C294A Pin 19, circuit VH406 (VT), harness side and ground.

- **For Remote Mount EMTC vehicles**




Measure the voltage between ground and HVAC module — Remote Mount EMTC C2357A Pin 19, circuit VH406 (VT/BN), harness side and ground.

- **For EATC vehicles**



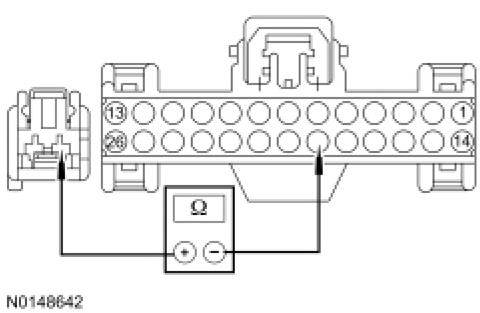

Measure the voltage between ground and HVAC module — EATC C228A Pin 19, circuit VH406 (VT/BN), harness side and ground.

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit. |
| **No** | GO to G4. |

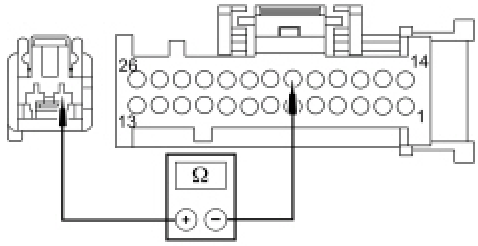
**G4 CHECK THE EVAPORATOR TEMPERATURE SENSOR FEEDBACK CIRCUIT FOR AN OPEN**

- Ignition OFF.
- **For Base EMTC vehicles**




Measure the resistance between HVAC module — Base EMTC C294A Pin 19, circuit VH406 (VT), harness side and evaporator temperature sensor C296 Pin 1, circuit VH406 (VT/BN), harness side.

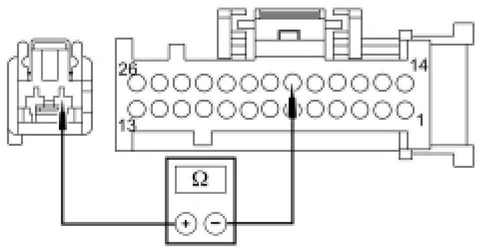
- **For Remote Mount EMTC vehicles**




N0148649

Measure the resistance between HVAC module — Remote Mount EMTC C2357A Pin 19, circuit VH406 (VT/BN), harness side and evaporator temperature sensor C296 Pin 1, circuit VH406 (VT/BN), harness side.

- **For EATC vehicles**




N0148649

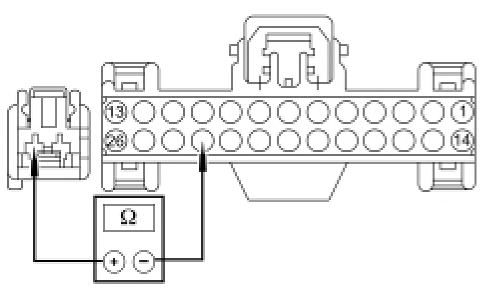
Measure the resistance between HVAC module — EATC C228A Pin 19, circuit VH406 (VT/BN), harness side and evaporator temperature sensor C296 Pin 1, circuit VH406 (VT/BN), harness side.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to G5. |
| **No** | REPAIR the circuit. |

**G5 CHECK THE SIGNAL RETURN CIRCUIT FOR AN OPEN**

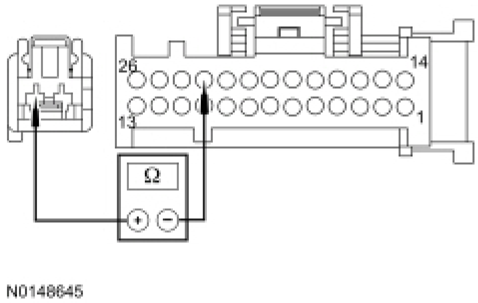
- **For Base EMTC vehicles**




N0148644

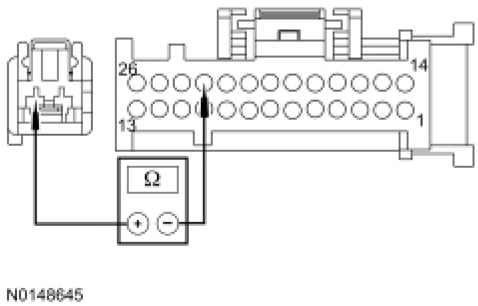
Measure the resistance between HVAC module — Base EMTC C294A Pin 23, circuit RH111 (GY/BU), harness side and evaporator temperature sensor C296 Pin 2, circuit RH111 (GY/BU), harness side.

- **For Remote Mount EMTC vehicles**




Measure the resistance between HVAC module — Remote Mount EMTC C2357A Pin 23, circuit RH111 (GY/BU), harness side and evaporator temperature sensor C296 Pin 2, circuit RH111 (GY/BU), harness side.

- **For EATC vehicles**



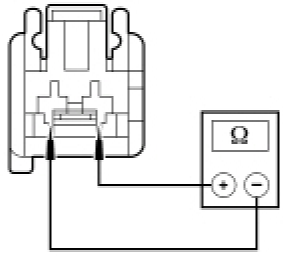

Measure the resistance between HVAC module — EATC C228A Pin 23, circuit RH111 (GY/BU), harness side and evaporator temperature sensor C296 Pin 2, circuit RH111 (GY/BU), harness side.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to G8. |
| **No** | REPAIR the circuit. |

**G6 CHECK THE EVAPORATOR TEMPERATURE SENSOR FEEDBACK CIRCUIT FOR A SHORT TO THE SIGNAL RETURN CIRCUIT**

- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: HVAC Module — Base EMTC C294B .
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357B .
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Measure the resistance between evaporator temperature sensor C296 Pin 1, circuit VH406 (VT/BN), harness side and C296 Pin 2, circuit RH111 (GY/BU), harness side.

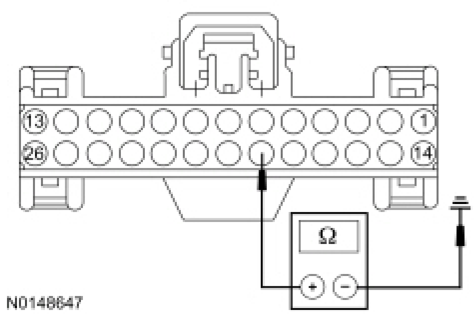



**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to G7. |
| **No** | REPAIR the circuits. |

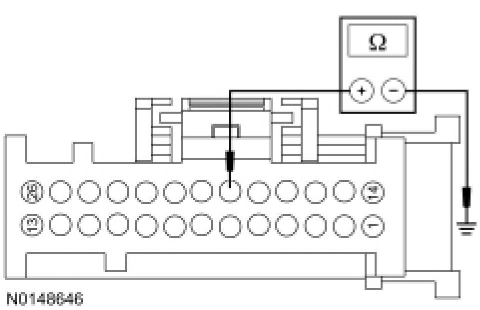
**G7 CHECK THE EVAPORATOR TEMPERATURE SENSOR FEEDBACK CIRCUIT FOR A SHORT TO GROUND**

- **For Base EMTC vehicles**



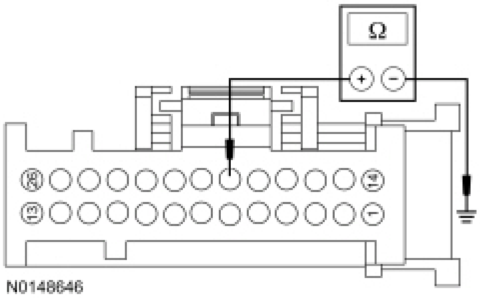

Measure the resistance between HVAC module — EMTC C294A Pin 19, circuit VH406 (VT), harness side and ground.

- **For Remote Mount EMTC vehicles**




Measure the resistance between HVAC module — Remote Mount EMTC C2357A Pin 19, circuit VH406 (VT/BN), harness side and ground.

- **For EATC vehicles**




Measure the resistance between HVAC module — EATC C228A Pin 19, circuit VH406 (VT/BN), harness side and ground.

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to G8. |
| **No** | REPAIR the circuit. |

**G8 CHECK FOR CORRECT HVAC MODULE OPERATION**

- Ignition OFF.
- Disconnect and inspect all HVAC module connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC module. REFER to Section 412-01. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test H: DTC U3003:16 or U3003:17**

Refer to Wiring Diagrams Cell 54 , Manual Climate Control System for schematic and connector information.

Refer to Wiring Diagrams Cell 55 , Automatic Climate Control System for schematic and connector information.

**NOTE:** *DTC U3003 can be set if the vehicle has been recently jump started, the battery has been recently charged or the battery has been discharged. The battery may become discharged due to excessive load(s) on the charging system from* *aftermarket accessories or if the battery has been left unattended with the accessories on.*

**NOTE:** *Carry out a thorough inspection and verification before proceeding with the pinpoint test. Refer to Inspection and Verification in this section.*

**Normal Operation**

The HVAC module is supplied constant battery voltage and ground.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| U3003:16 | Battery Voltage: Circuit Voltage Below Threshold | This CMDTC sets when the HVAC module senses battery voltage is less than 8 volts for more than 10 seconds. |
| U3003:17 | Battery Voltage: Circuit Voltage Above Threshold | This CMDTC sets when the HVAC module senses battery voltage is greater than 16 volts for more than 10 seconds. |

**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- HVAC module

**PINPOINT TEST H : U3003:16 OR U3003:17**

**H1 RETRIEVE PCM DTCS**

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test — PCM .

**Are any charging system DTCs present in the PCM?**

| | |
|---|---|
| **Yes** | REFER to Section 414-00 for diagnosis of the battery and charging system. |
| **No** | GO to H2. |

**H2 CHECK BATTERY CONDITION**

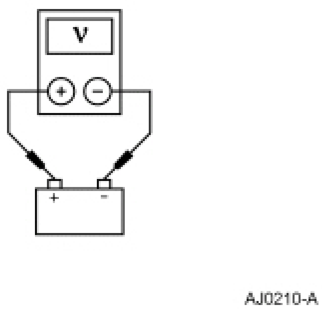
- Ignition OFF.
- Refer to Section 414-01 and carry out the Battery — Condition Test

**Does the battery pass the condition test?**

| | |
|---|---|
| **Yes** | If the battery passed the condition test but required a recharge, REFER to Section 414-00 to diagnose the charging system. CLEAR all Continuous Memory Diagnostic Trouble Codes (CMDTCs) . TEST the system for normal operation.<br>If the battery passed the condition test and did not require a recharge, GO to H3. |
| **No** | INSTALL a new battery. REFER to Section 414-01. |

## H3 CHECK THE CHARGING SYSTEM VOLTAGE

**NOTE:** *Do not allow the engine rpm to exceed 2,000 rpm while performing this step or the generator may self-excite, resulting in default charging system output voltage. If engine rpm has exceeded 2,000 rpm, shut the vehicle off and restart the engine before performing this step.*

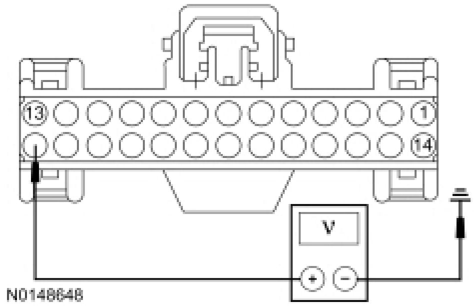
- Start the engine.
- Measure the voltage of the battery:
  - For DTC U3003:17, turn off all accessories and run the engine at 1,500 rpm for a minimum of 2 minutes.
  - For DTC U3003:16, turn on headlights and HVAC fan on high and run engine at 1,500 rpm for a minimum of 2 minutes.
- 


**Is the battery voltage between 13-15.2 volts?**

| | |
|---|---|
| **Yes** | For DTC U3003:16, GO to H4.<br>For DTC U3003:17, GO to H6. |
| **No** | REFER to Section 414-00 to diagnose the charging system. |

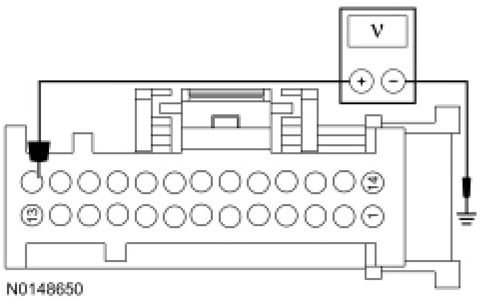
## H4 CHECK THE HVAC MODULE VOLTAGE SUPPLY CIRCUIT FOR HIGH RESISTANCE

- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: HVAC Module — Base EMTC C294B .
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357B .
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Ignition ON.
- **For Base EMTC vehicles**



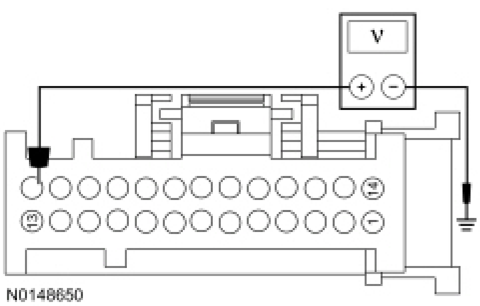

Measure the voltage between ground and HVAC module — EMTC C294A Pin 19, circuit VH406 (VT), harness side and ground.

- **For Remote Mount EMTC vehicles**




Measure the voltage between ground and HVAC module — Remote Mount EMTC C2357A Pin 19, circuit VH406 (VT/BN), harness side and ground.

- **For EATC vehicles**



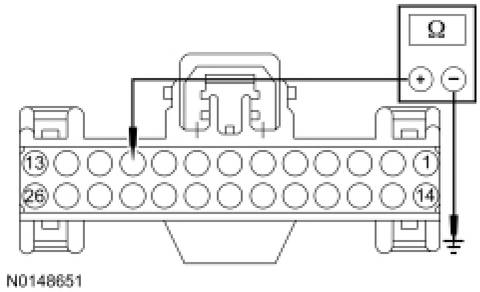

Measure the voltage between ground and HVAC module — EATC C228A Pin 19, circuit VH406 (VT/BN), harness side and ground.

**Is the voltage greater than 11 volts?**

| | |
|---|---|
| **Yes** | GO to H5. |
| **No** | REPAIR the circuit. |

**H5 CHECK THE HVAC MODULE GROUND CIRCUIT FOR HIGH RESISTANCE**

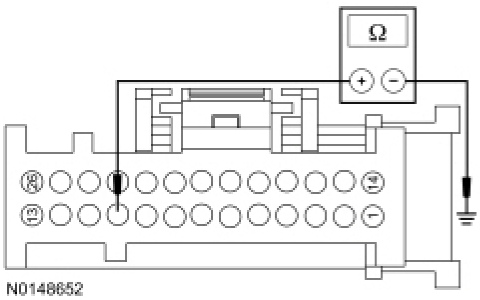
- Ignition OFF.
- **For Base EMTC vehicles**




Measure the resistance between HVAC module — Base EMTC C294A Pin 10, circuit GD133 (BK), harness side and ground.

- Repeat this measurement while wiggling the harness.

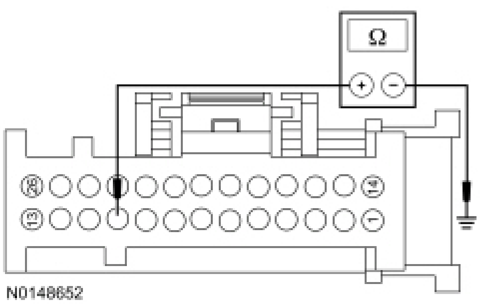
- **For For Remote Mount EMTC vehicles**




Measure the resistance between HVAC module — For Remote Mount EMTC C2357A Pin 10, circuit GD133 (BK), harness side and ground.

- Repeat this measurement while wiggling the harness.

- **For EATC vehicles**




Measure the resistance between HVAC module — EATC C228A Pin 10, circuit GD133 (BK), harness side and ground.

- Repeat this measurement while wiggling the harness.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to H6. |
| **No** | REPAIR the circuit. |

**H6 CHECK FOR CORRECT HVAC MODULE OPERATION**

- Ignition OFF.
- Disconnect and inspect all HVAC module connectors.

- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC module. REFER to Section 412-01. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test I: Unable to Duplicate the Customer Concern and No DTCs Present**

**Normal Operation**

This pinpoint test tests the functions of the HVAC system and identifies the correct HVAC symptom pinpoint test.

**This pinpoint test is intended to diagnose the following:**

- HVAC system and/or related components

**PINPOINT TEST I : UNABLE TO DUPLICATE THE CUSTOMER CONCERN AND NO DTCS PRESENT**

**NOTE:** *Before carrying out the following test, diagnose any HVAC or PCM DTCs.*

**I1 CHECK THE BLOWER MOTOR OPERATION**

- Ignition ON.
- Select PANEL mode.
- Observe blower motor operation and select each blower motor speed.

**Does the blower motor operate in all selections and change speed in each?**

| | |
|---|---|
| **Yes** | GO to I2. |
| **No** | For EMTC base with 4 blower motor speed positions, if the blower motor does not operate in any setting, GO to Pinpoint Test W.<br>For EATC dual zone or EMTC with 4.2 inch display with 7 blower motor speed positions, if the blower motor does not operate in any setting, GO to Pinpoint Test Y.<br>For EMTC base with 4 blower motor speed positions, if the blower motor does not properly change speeds or shut OFF, GO to Pinpoint Test X.<br>For EATC dual zone or EMTC with 4.2 inch display with 7 blower motor speed positions, if the blower motor does not properly change speeds or shut OFF, GO to Pinpoint Test Z. |

**I2 CHECK AIRFLOW OPERATION**

- Select the highest blower motor setting.
- **NOTE:** *Refer to Description and Operation in this section for proper airflow descriptions*.

  While observing the airflow, select each of the airflow positions (PANEL, PANEL/FLOOR, FLOOR, FLOOR/DEFROST, DEFROST).

**Is the airflow directed to the proper outlets?**

| | |
|---|---|
| **Yes** | GO to I3. |
| **No** | EMTC base, GO to Pinpoint Test L.<br>For other systems, GO to Pinpoint Test M. |

**I3 VERIFY TEMPERATURE CONTROL OPERATION**

- Start the vehicle and allow it to reach normal operating temperature.
- With the A/C OFF, select PANEL mode.
- Change the temperature setting from the coldest to the warmest and back to the coldest.

**Does the temperature change between very warm to cool?**

| | |
|---|---|
| **Yes** | GO to I4. |
| **No** | If the temperature does not get very warm, GO to Pinpoint Test N.<br>For EMTC base with 4 blower motor speed positions, if the temperature does not change at all, GO to Pinpoint Test R.<br>For EMTC with 4.2 inch display with 7 blower motor speed positions, if the temperature does not change at all, GO to Pinpoint Test S.<br>For EATC dual zone, if the driver side temperature does not change at all, GO to Pinpoint Test U.<br>For EATC dual zone, if the passenger side temperature does not change at all, GO to Pinpoint Test V. |

**I4 VERIFY THE A/C CLUTCH DOES NOT ENGAGE WITH A/C OFF**

- With the engine running and the A/C off, select PANEL mode.
- Select the coldest temperature setting.

**Is the outlet temperature close to ambient temperature?**

| | |
|---|---|
| **Yes** | GO to I5. |
| **No** | For EMTC base with 4 blower motor speed positions, if the temperature is warmer than ambient temperature, GO to Pinpoint Test R and diagnose for inoperative blend door.<br>For EMTC with 4.2 inch display with 7 blower motor speed positions, if the temperature is warmer than ambient temperature, GO to Pinpoint Test S and diagnose for inoperative blend door.<br>For EATC dual zone, if the driver side temperature is warmer than ambient temperature, GO to Pinpoint Test U and diagnose for an inoperative blend door.<br>For EATC dual zone, if the passenger side temperature is warmer than ambient temperature, GO to Pinpoint Test V and diagnose for an inoperative blend door.<br>If the outlet temperature is significantly colder than ambient temperature and the A/C compressor |

| | |
|---|---|
| | clutch cycles normally, GO to Pinpoint Test Q.<br>If the outlet temperature is significantly colder than ambient temperature and the A/C compressor clutch does not cycle, GO to Pinpoint Test P. |

**I5 VERIFY A/C CLUTCH ENGAGEMENT IN THE A/C MODE**

- Make sure the ambient air temperature is above 2°C (35°F).
- With the engine running, select PANEL mode.
- Press the A/C button (indicator ON).

**Does the A/C clutch engage when the PANEL and A/C button (indicator ON) is pressed?**

| | |
|---|---|
| **Yes** | GO to I6. |
| **No** | GO to Pinpoint Test O. |

**I6 CHECK THE RECIRC OPERATION**

- Press the RECIRC button (indicator OFF).
- Select PANEL mode.
- Select the highest blower motor setting.
- Observe airflow noise.
- Press the RECIRC button (indicator ON).

**Does the airflow noise increase when the RECIRC mode is selected (indicator ON)?**

| | |
|---|---|
| **Yes** | The system is operating normally. |
| **No** | EMTC base with 4 blower motor speed positions, GO to Pinpoint Test J.<br>For other systems, GO to Pinpoint Test K. |

### Pinpoint Test J: The Air Inlet Door Is Inoperative — EMTC Base With 4 Blower Motor Speed Positions

Refer to Wiring Diagrams Cell 54 , Manual Climate Control System for schematic and connector information.

**Normal Operation**

To rotate the air inlet door actuator, the HVAC module supplies voltage and ground to the air inlet door actuator through the door actuator motor circuits. To reverse the air inlet door actuator rotation, the HVAC module reverses the voltage and ground circuits.

The air inlet door actuator feedback resistors are supplied a ground from the HVAC module by the air inlet door actuator return circuits and a 5-volt reference voltage on the air inlet door actuator reference circuits. The HVAC module reads the voltage on the air inlet door actuator feedback circuits to determine the air inlet door actuator position by the position of the actuator feedback resistor wiper arm.

During an actuator calibration cycle, the HVAC module drives the air inlet door until the door reaches both internal stops in the HVAC case. If the air inlet door is temporarily obstructed or binding during a calibration cycle, the HVAC module may interpret this as the actual end of travel for the door. When this condition occurs and the HVAC module commands the actuator to its end of travel, the air intake may not be from the expected source.

In FLOOR and FLOOR/DEFROST modes, the RECIRC request will time-out after 5 minutes and return to fresh air mode. RECIRC mode can be selected again after the time-out.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B1083:00 | Recirculation Damper Motor: No Sub Type Information | The HVAC module senses no changes in the feedback circuit when motor movement is commanded and no motor electrical DTCs are present. This indicates either an open in an actuator motor circuit, an actuator internal failure or a binding or stuck linkage or door. |
| B1083:11 | Recirculation Damper Motor: Circuit Short to Ground | The HVAC module senses lower than expected voltage on an actuator motor circuit when voltage is applied to drive the motor, indicating a short to ground. |
| B1083:12 | Recirculation Damper Motor: Circuit Short to Battery | The HVAC module senses higher than expected voltage on the actuator motor circuit when ground is applied to drive the motor, indicating a short to voltage. |
| B1083:13 | Recirculation Damper Motor: Circuit Open | The HVAC module senses no voltage on the actuator motor circuit when ground is applied to drive the motor, indicating an open circuit. |
| B11F0:11 | Air Intake Damper Position Sensor: Circuit Short to Ground | The HVAC module senses less than 1 volt on the actuator feedback circuit, indicating a short to ground. |
| B11F0:15 | Air Intake Damper Position Sensor: Circuit Short to Battery or Open | The HVAC module senses greater than 4 volts on the actuator feedback circuit, indicating an open circuit or a short to voltage. |

**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- Air inlet mode door actuator
- HVAC module
- Stuck or bound linkage or door

**PINPOINT TEST J : THE AIR INLET DOOR IS INOPERATIVE — EMTC BASE WITH 4 BLOWER MOTOR SPEED POSITIONS**

**J1 CHECK THE HVAC MODULE FOR DTCS**

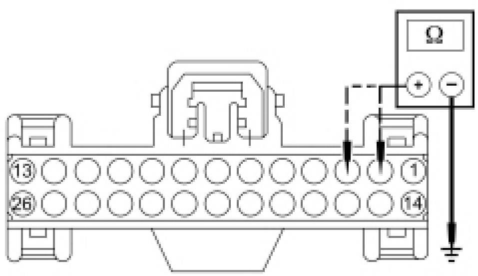
- Ignition ON.
- Check the HVAC module for DTCs.

**Are any DTCs present?**

| | |
|---|---|
| **Yes** | For DTC B1083:00, INSPECT for binding/damaged linkage or door. REPAIR as necessary. If no condition is found, GO to J4. |

| | |
|---|---|
| | For DTC B1083:11, GO to J2.<br>For DTC B1083:12, GO to J3.<br>For DTC B1083:13, GO to J4.<br>For DTC B11F0:11, GO to J6.<br>For DTC B11F0:15, GO to J9. |
| **No** | INSPECT for broken linkage or door. REPAIR as necessary. If no condition is found, GO to J13. |

**J2 CHECK THE AIR INLET MODE DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TO GROUND**

- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: HVAC Module — Base EMTC C294B .
- Measure the resistance between ground and:
  - HVAC module — EMTC C294A Pin 2, circuit CH207 (BU), harness side.
  - HVAC module — EMTC C294A Pin 3, circuit CH208 (GN), harness side.
- 

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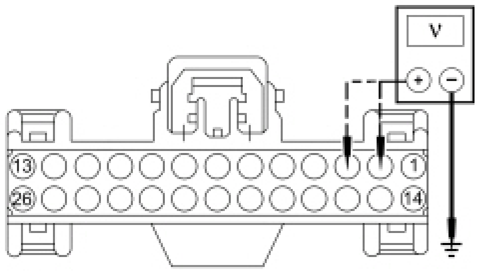
**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to J5. |
| **No** | REPAIR the circuit in question. |

**J3 CHECK THE AIR INLET MODE DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: HVAC Module — Base EMTC C294B .
- Ignition ON.
- Measure the voltage between ground and:
  - HVAC module — EMTC C294A Pin 2, circuit CH207 (BU), harness side.
  - HVAC module — EMTC C294A Pin 3, circuit CH208 (GN), harness side.

- 

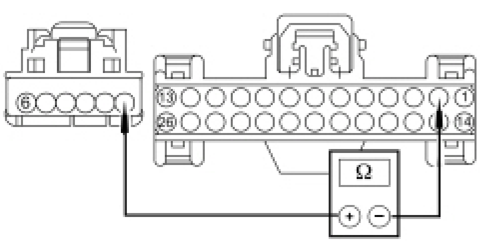
N0148654

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |
| **No** | GO to J5. |

## J4 CHECK THE AIR INLET MODE DOOR ACTUATOR MOTOR CIRCUITS FOR AN OPEN

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: HVAC Module — Base EMTC C294B .
- Disconnect: Air Inlet Mode Door Actuator C289 .
- Measure the resistance between HVAC module — EMTC C294A Pin 2, circuit CH207 (BU), harness side and air inlet mode door actuator C289 Pin 1, circuit CH207 (BU/GY), harness side.
- 

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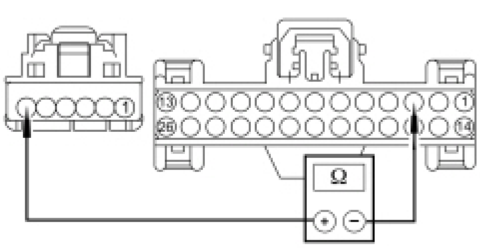
- 

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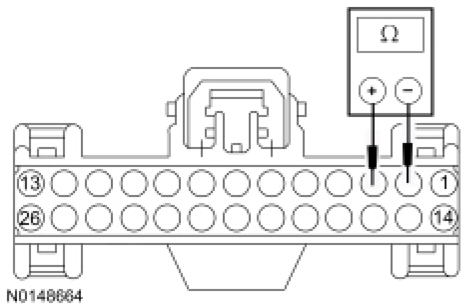
  Measure the resistance between HVAC module — EMTC C294A Pin 3, circuit CH208 (GN), harness side and air inlet mode door actuator C289 Pin 6, circuit CH208 (GN/OG), harness side.

**Are the resistances less than 3 ohms?**

| Yes | GO to J13. |
|---|---|
| No | REPAIR the circuit in question. |

**J5 CHECK THE AIR INLET MODE DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TOGETHER**

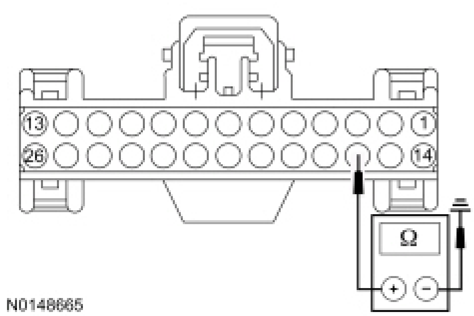
**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Ignition OFF.
- Disconnect: Air Inlet Mode Door Actuator C289 .
- Measure the resistance between HVAC module — EMTC C294A Pin 2, circuit CH207 (BU), harness side and HVAC module — EMTC C294A Pin 3 , circuit CH208 (GN), harness side.
- 


**Is the resistance greater than 10,000 ohms?**

| Yes | GO to J13. |
|---|---|
| No | REPAIR the circuits. |

**J6 CHECK THE AIR INLET MODE DOOR ACTUATOR FEEDBACK CIRCUIT FOR A SHORT TO GROUND**

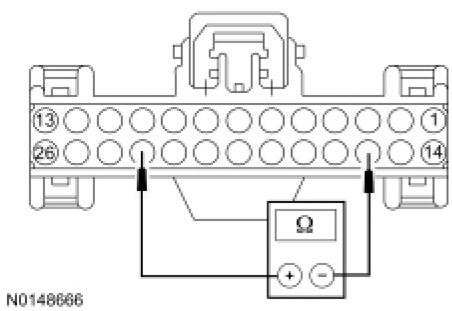
- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: HVAC Module — Base EMTC C294B .
- Measure the resistance between ground and HVAC Module — EMTC C294A Pin 16, circuit VH438 (GN), harness side.
- 

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to J7. |
| **No** | REPAIR the circuit. |

**J7 CHECK THE SIGNAL RETURN CIRCUIT FOR A SHORT TO THE AIR INLET MODE DOOR ACTUATOR FEEDBACK CIRCUIT**

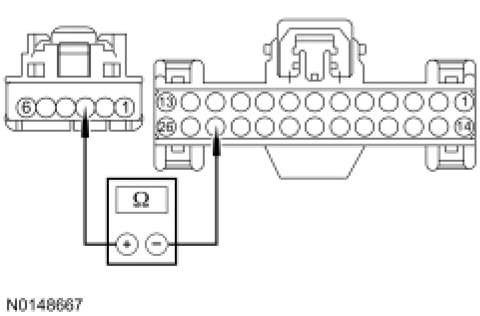
**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Disconnect: Air Inlet Mode Door Actuator C289 .
- Measure the resistance between HVAC Module — EMTC C294A Pin 16, circuit VH438 (GN), harness side and HVAC module — EMTC C294A Pin 23, circuit RH111 (GY/BU), harness side.
- 


**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to J8. |
| **No** | REPAIR the circuits. |

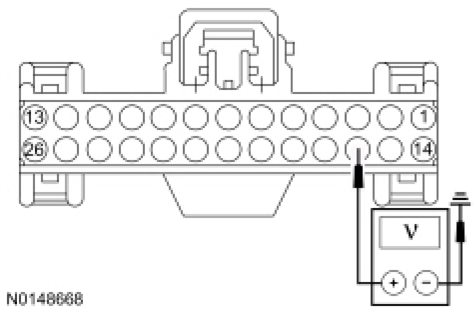
**J8 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR AN OPEN**

- Measure the resistance between HVAC Module — EMTC C294A Pin 24, circuit LH111 (BN/WH), harness side and air inlet door actuator C289 Pin 3, circuit LH111 (BN/WH), harness side.
- 


**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to J13. |
| **No** | REPAIR the circuit. |

## J9 CHECK THE AIR INLET MODE DOOR ACTUATOR FEEDBACK CIRCUIT FOR A SHORT TO VOLTAGE

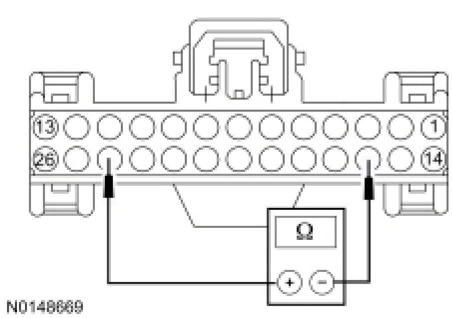
- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: HVAC Module — Base EMTC C294B .
- Ignition ON.
- Measure the voltage between ground and HVAC Module — EMTC C294A Pin 16, circuit VH438 (GN), harness side.
- 


**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit. |
| **No** | GO to J10. |

## J10 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR A SHORT TO THE AIR INLET MODE DOOR ACTUATOR FEEDBACK CIRCUIT

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

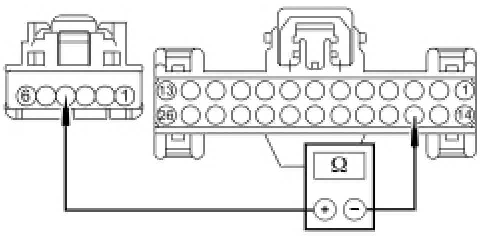
- Ignition OFF.
- Disconnect: Air Inlet Mode Door Actuator C289 .
- Measure the resistance between HVAC Module — EMTC C294A Pin 16, circuit VH438 (GN), harness side and HVAC module — EMTC C294A Pin 24, circuit LH111 (BN/WH), harness side.
- 


**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to J11. |
| **No** | REPAIR the circuits. |

**J11 CHECK THE AIR INLET MODE DOOR ACTUATOR FEEDBACK CIRCUIT FOR AN OPEN**

- Measure the resistance between HVAC Module — EMTC C294A Pin 16, circuit VH438 (GN), harness side and air inlet door actuator C289 Pin 4, circuit VH438 (GN/VT), harness side.



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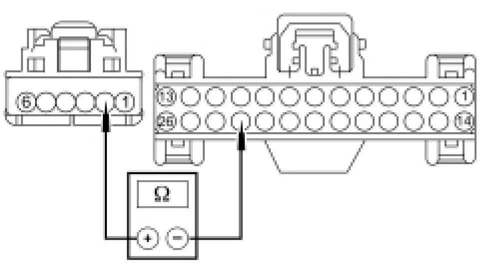
**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to J12. |
| **No** | REPAIR the circuit. |

**J12 CHECK THE AIR INLET MODE DOOR ACTUATOR RETURN CIRCUIT FOR AN OPEN**

- Measure the resistance between HVAC Module — EMTC C294A Pin 23, circuit RH111 (GY/BU), harness side and air inlet door actuator C289 Pin 2, circuit RH111 (GY/BU), harness side.



- N0148671

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to J13. |
| **No** | REPAIR the circuit. |

**J13 CHECK THE AIR INLET MODE DOOR ACTUATOR CONNECTION**

- Ignition OFF.
- Disconnect and inspect the air inlet mode door actuator connector.
- Repair:

- corrosion (install new connector or terminals - clean module pins)
- damaged or bent pins - install new terminals/pins
- pushed-out pins - install new pins as necessary

- Reconnect the air inlet mode door actuator connector. Make sure it seats and latches correctly.
- Clear the DTCs.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CARRY OUT the Air Inlet Mode Door Actuator component test in this section. If the actuator tests OK, GO to J14. When installing an actuator, CONNECT the actuator electrical connector before the HVAC module connector(s). This allows the actuator to be calibrated when the HVAC module is reconnected. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

## J14 MODULE ACTUATOR POSITION CALIBRATION

**NOTE:** *The purpose of the module actuator position calibration is to allow the HVAC module to reinitialize and calibrate the actuator stop points. To carry out the calibration, follow the steps below.*

- Ignition OFF.
- Disconnect and inspect all HVAC module connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Ignition ON.
- Clear the DTCs.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Select any position except OFF.
- **NOTE:** *The HVAC module will now initialize and calibrate the actuators. Calibration of the actuators takes approximately 30 seconds.*

  Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC module. REFER to Section 412-01. |
| | |

| | |
|---|---|
| **No** | The system is now operating correctly at this time. The concern may have been caused by a foreign object in the HVAC case or temporary binding that restricted actuator door travel. CHECK any actuator external linkage. If condition recurs, INSPECT actuator linkage and door for binding and CHECK HVAC case for foreign objects. |

**Pinpoint Test K: The Air Inlet Door Is Inoperative — EATC Dual Zone And EMTC With 4.2 Inch Display System With 7 Blower Motor Speed Positions**

Refer to Wiring Diagrams Cell 54 , Automatic Climate Control System for schematic and connector information.

Refer to Wiring Diagrams Cell 55 , Automatic Climate Control System for schematic and connector information.

**Normal Operation**

To rotate the air inlet door actuator, the HVAC module supplies voltage and ground to the air inlet door actuator through the door actuator motor circuits. To reverse the air inlet door actuator rotation, the HVAC module reverses the voltage and ground circuits.

The air inlet door actuator feedback resistors are supplied a ground from the HVAC module by the air inlet door actuator return circuits and a 5-volt reference voltage on the air inlet door actuator reference circuits. The HVAC module reads the voltage on the air inlet door actuator feedback circuits to determine the air inlet door actuator position by the position of the actuator feedback resistor wiper arm.

During an actuator calibration cycle, the HVAC module drives the air inlet door until the door reaches both internal stops in the HVAC case. If the air inlet door is temporarily obstructed or binding during a calibration cycle, the HVAC module may interpret this as the actual end of travel for the door. When this condition occurs and the HVAC module commands the actuator to its end of travel, the air intake may not be from the expected source.

In FLOOR and FLOOR/DEFROST modes, the RECIRC request will time-out after 5 minutes and return to fresh air mode. RECIRC mode can be selected again after the time-out.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B1083:00 | Recirculation Damper Motor: No Sub Type Information | The HVAC module senses no changes in the feedback circuit when motor movement is commanded and no motor electrical DTCs are present. This indicates either an open in an actuator motor circuit, an actuator internal failure or a binding or stuck linkage or door. |
| B1083:11 | Recirculation Damper Motor: Circuit Short to Ground | The HVAC module senses lower than expected voltage on an actuator motor circuit when voltage is applied to drive the motor, indicating a short to ground. |
| B1083:12 | Recirculation Damper Motor: Circuit Short to Battery | The HVAC module senses higher than expected voltage on the actuator motor circuit when ground is applied to drive the motor, indicating a short to voltage. |
| B1083:13 | Recirculation Damper Motor: Circuit Open | The HVAC module senses no voltage on the actuator motor circuit when ground is applied to drive the motor, indicating an open circuit. |

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B11F0:11 | Air Intake Damper Position Sensor: Circuit Short to Ground | The HVAC module senses less than 1 volt on the actuator feedback circuit, indicating a short to ground. |
| B11F0:15 | Air Intake Damper Position Sensor: Circuit Short to Battery or Open | The HVAC module senses greater than 4 volts on the actuator feedback circuit, indicating an open circuit or a short to voltage. |

**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- Air inlet mode door actuator
- HVAC module
- FCIM
- Stuck or bound linkage or door

**PINPOINT TEST K : THE AIR INLET DOOR IS INOPERATIVE — EATC DUAL ZONE AND EMTC WITH 4.2 INCH DISPLAY SYSTEM WITH 7 BLOWER MOTOR SPEED POSITIONS**

**K1 CHECK THE HVAC MODULE FOR DTCS**

- Ignition ON.
- Check the HVAC module for DTCs.

**Are any DTCs present?**

| | |
|---|---|
| **Yes** | For DTC B1083:00, INSPECT for binding/damaged linkage or door. REPAIR as necessary. If no condition is found, GO to K5.<br>For DTC B1083:11, GO to K3.<br>For DTC B1083:12, GO to K4.<br>For DTC B1083:13, GO to K5.<br>For DTC B11F0:11, GO to K7.<br>For DTC B11F0:15, GO to K10. |
| **No** | INSPECT for broken/binding linkage or door. REPAIR as necessary. If no condition is found, GO to K2. |

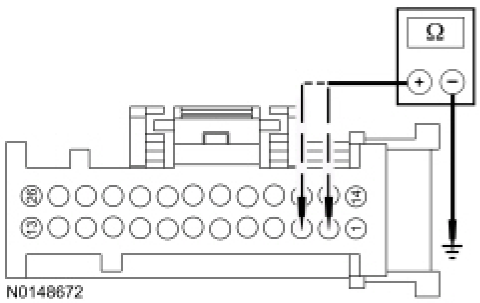
**K2 CHECK THE AIR INLET MODE DOOR ACTUATOR OPERATION USING THE HVAC A/C RECIRCULATING SERVO POSITION (RECIRC_DPOS) ACTIVE COMMAND**

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: HVAC Module DataLogger .
- Select PANEL mode on the HVAC controls.
- Select the highest blower motor setting.
- Observe airflow noise while using the scan tool to operate the HVAC module active command RECIRC_DPOS between Fresh Air and RECIRC.

**Does the airflow noise change when the active command is changed?**

| | |
|---|---|
| **Yes** | INSTALL a new FCIM . Refer to the appropriate section in Group 415 for the procedure.. |
| **No** | INSPECT for broken/binding linkage or door. REPAIR as necessary. If no condition is found, GO to K14. |

**K3 CHECK THE AIR INLET MODE DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TO GROUND**

- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Disconnect: HVAC Module Remote Mount EMTC C2357A .
- Disconnect: HVAC Module Remote Mount EMTC C2357B .
- 

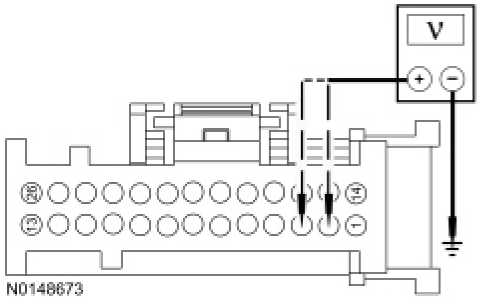

  Measure the resistance between ground and:
  - HVAC module EATC C228A Pin 3, circuit CH208 (GN), harness side.
  - HVAC module EATC C228A Pin 2, circuit CH207 (BU), harness side.
  - HVAC module EMTC C2357A Pin 3, circuit CH208 (GN), harness side.
  - HVAC module EMTC C2357A Pin 2, circuit CH207 (BU), harness side.

**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to K6. |
| **No** | REPAIR the circuit in question. |

**K4 CHECK THE AIR INLET MODE DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TO VOLTAGE**

- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Disconnect: HVAC Module Remote Mount EMTC C2357A .
- Disconnect: HVAC Module Remote Mount EMTC C2357B .
- Ignition ON.

- 
N0148673

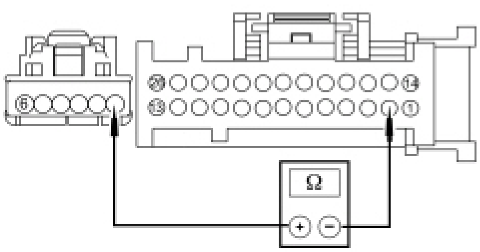
  Measure the voltage between ground and:
  - HVAC module EATC C228A Pin 3, circuit CH208 (GN), harness side.
  - HVAC module EATC C228A Pin 2, circuit CH207 (BU), harness side.
  - HVAC module EMTC C2357A Pin 3, circuit CH208 (GN), harness side.
  - HVAC module EMTC C2357A Pin 2, circuit CH207 (BU), harness side.

**Is any voltage present?**

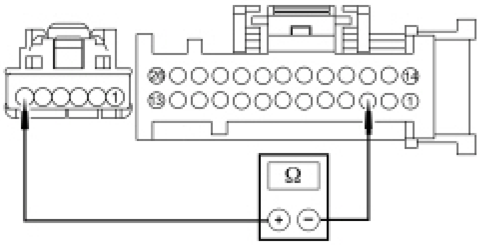
| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |
| **No** | GO to K6. |

**K5 CHECK THE AIR INLET MODE DOOR ACTUATOR MOTOR CIRCUITS FOR AN OPEN**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Disconnect: HVAC Module Remote Mount EMTC C2357A .
- Disconnect: HVAC Module Remote Mount EMTC C2357B .
- Disconnect: Air Inlet Mode Door Actuator C289 .
- 
N0148674

  Measure the resistance between:
  - HVAC module EATC C228A Pin 2, circuit CH207 (BU), harness side and air inlet mode door actuator C289 Pin 1, circuit CH207 (BU/GY), harness side.
  - HVAC module EMTC C2357A Pin 2, circuit CH207 (BU), harness side and air inlet mode door actuator C289 Pin 1, circuit CH207 (BU/GY), harness side.



- N0148675

  Measure the resistance between:
  - HVAC module EATC C228A Pin 3, circuit CH208 (GN), harness side and air inlet mode door actuator C289 Pin 6, circuit CH208 (GN/OG), harness side.
  - HVAC module EMTC C2357A Pin 3, circuit CH208 (GN), harness side and air inlet mode door actuator C289 Pin 6, circuit CH208 (GN/OG), harness side.

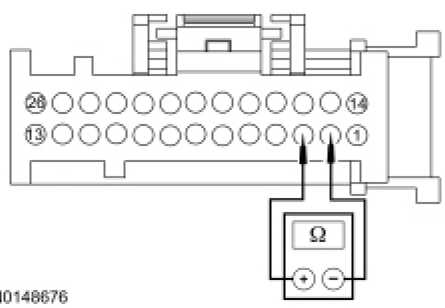
**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to K14. |
| **No** | REPAIR the circuit in question. |

**K6 CHECK THE AIR INLET MODE DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TOGETHER**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Ignition OFF.
- Disconnect: Air Inlet Mode Door Actuator C289 .



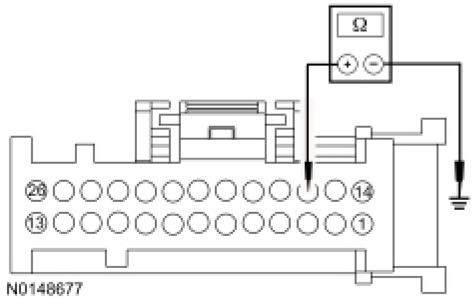
- N0148676

  Measure the resistance between:
  - HVAC module EATC C228A Pin 2, circuit CH207 (BU), harness side and HVAC module C228A Pin 3, circuit CH208 (GN), harness side.
  - HVAC module EMTC C2357A Pin 2, circuit CH207 (BU), harness side and HVAC module C228A Pin 3, circuit CH208 (GN), harness side.

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to K14. |
| **No** | REPAIR the circuits. |

**K7 CHECK THE AIR INLET MODE DOOR ACTUATOR FEEDBACK CIRCUIT FOR A SHORT TO GROUND**

- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Disconnect: HVAC Module Remote Mount EMTC C2357A .
- Disconnect: HVAC Module Remote Mount EMTC C2357B .
- 

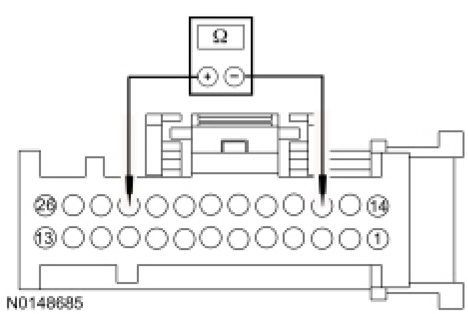

  Measure the resistance between ground and:
  - HVAC module EATC C228A Pin 16, circuit VH438 (GN), harness side.
  - HVAC module EMTC C2357A Pin 16, circuit VH438 (GN), harness side.

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to K8. |
| **No** | REPAIR the circuit. |

**K8 CHECK THE SIGNAL RETURN CIRCUIT FOR A SHORT TO THE AIR INLET MODE DOOR ACTUATOR FEEDBACK CIRCUIT**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

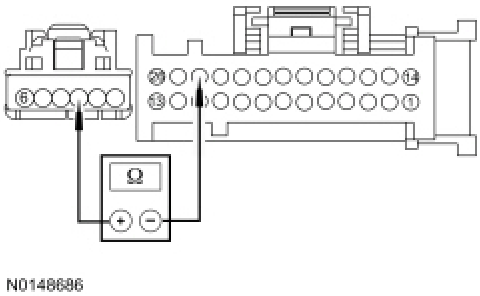
- Disconnect: Air Inlet Mode Door Actuator C289 .
- 


  Measure the resistance between:
  - HVAC module EATC C228A Pin 16, circuit VH438 (GN), harness side and HVAC module C228A Pin 23, circuit RH111 (GY/BU), harness side.
  - HVAC module C2357A Pin 16, circuit VH438 (GN), harness side and HVAC module C228A Pin 23, circuit RH111 (GY/BU), harness side.

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to K9. |
| **No** | REPAIR the circuits. |

## K9 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR AN OPEN

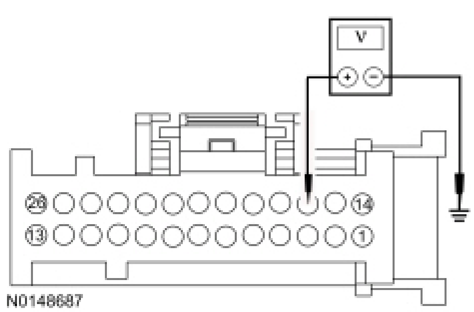
- 


  Measure the resistance between:
  - HVAC module EATC C228A Pin 24, circuit LH111 (BN/WH), harness side and air inlet door actuator C289 Pin 3, circuit LH111 (BN/WH), harness side.
  - HVAC module EMTC C2357A Pin 24, circuit LH111 (BN/WH), harness side and air inlet door actuator C289 Pin 3, circuit LH111 (BN/WH), harness side.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to K14. |
| **No** | REPAIR the circuit. |

## K10 CHECK THE AIR INLET MODE DOOR ACTUATOR FEEDBACK CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Disconnect: HVAC Module Remote Mount EMTC C2357A .
- Disconnect: HVAC Module Remote Mount EMTC C2357B .
- Ignition ON.
- 

Measure the voltage between ground and:

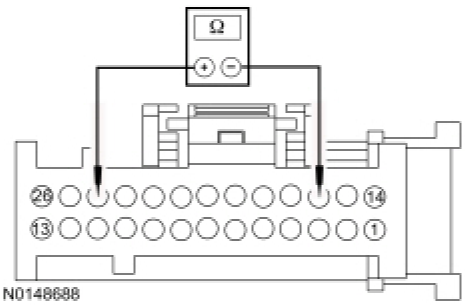
- HVAC module EATC C228A Pin 16, circuit VH438 (GN), harness side.
- HVAC module EMTC C2357A Pin 16, circuit VH438 (GN), harness side.

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit. |
| **No** | GO to K11. |

**K11 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR A SHORT TO THE AIR INLET MODE DOOR ACTUATOR FEEDBACK CIRCUIT**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Ignition OFF.
- Disconnect: Air Inlet Mode Door Actuator C289 .
- 

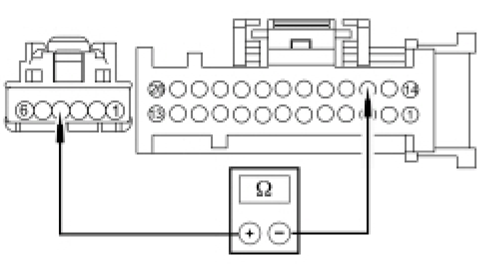
N0148688

  Measure the resistance between:
  - HVAC module EATC C228A Pin 24, circuit LH111 (BN/WH), harness side and HVAC module C228A Pin 16, circuit VH438 (GN), harness side.
  - HVAC module EMTC C2357A Pin 24, circuit LH111 (BN/WH), harness side and HVAC module C2357A Pin 16, circuit VH438 (GN), harness side.

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to K12. |
| **No** | REPAIR the circuits. |

**K12 CHECK THE AIR INLET MODE DOOR ACTUATOR FEEDBACK CIRCUIT FOR AN OPEN**

- 

N0148689

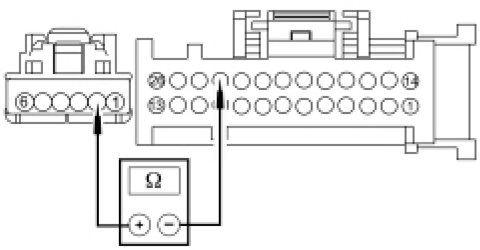
  Measure the resistance between:

- HVAC module EATC C228A Pin 16, circuit VH438 (GN), harness side and air inlet mode door actuator C289 Pin 4, circuit VH438 (GN/VT), harness side.
- HVAC module EMTC C2357A Pin 16, circuit VH438 (GN), harness side and air inlet mode door actuator C289 Pin 4, circuit VH438 (GN/VT), harness side.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to K13. |
| **No** | REPAIR the circuit. |

**K13 CHECK THE AIR INLET MODE DOOR ACTUATOR RETURN CIRCUIT FOR AN OPEN**

- 

N0148692

  Measure the resistance between:
  - HVAC module EATC C228A Pin 23, circuit RH111 (GY/BU), harness side and inlet door mode door actuator C289 Pin 2, circuit RH111 (GY/BU), harness side.
  - HVAC module EMTC C2357A Pin 23, circuit RH111 (GY/BU), harness side and inlet door mode door actuator C289 Pin 2, circuit RH111 (GY/BU), harness side.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to K14. |
| **No** | REPAIR the circuit. |

**K14 CHECK THE AIR INLET MODE DOOR ACTUATOR CONNECTION**

- Disconnect and inspect the air inlet mode door actuator connector.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect the air inlet mode door actuator connector. Make sure they seat and latch correctly.
- Clear the DTCs.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CARRY OUT the Air Inlet Mode Door Actuator component test in this section. If the actuator tests OK, GO to K15. When installing an actuator, CONNECT the actuator electrical connector before the HVAC module connector(s). This allows the actuator to be calibrated when the HVAC module is reconnected. |
| **No** | The system is now operating correctly at this time. The concern may have been caused by a foreign object in the HVAC case or temporary binding that restricted actuator door travel. CHECK any actuator external linkage. If condition recurs, INSPECT actuator linkage and door for binding and CHECK HVAC case for foreign objects. ADDRESS the root cause of any connector or pin issues. |

**K15 MODULE ACTUATOR POSITION CALIBRATION**

**NOTE:** *The purpose of the module actuator position calibration is to allow the HVAC module to reinitialize and calibrate the actuator stop points. To carry out the calibration, follow the steps below.*

- Ignition OFF.
- Disconnect and inspect all HVAC module connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Ignition ON.
- Clear the DTCs.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Select any position except OFF.
- **NOTE:** *The HVAC module will now initialize and calibrate the actuators. Calibration of the actuators takes approximately 30 seconds.*

  Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC module. REFER to Section 412-01. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test L: Incorrect/Erratic Direction of Airflow From Outlets — EMTC Base**

Refer to Wiring Diagrams Cell 54 , Manual Climate Control System for schematic and connector information.

**Normal Operation**

To rotate the defrost/panel/floor mode door actuator, the HVAC module supplies voltage and ground to the defrost/panel/floor mode door actuator motor through the actuator motor circuits. To reverse the defrost/panel/floor mode door actuator rotation, the HVAC module reverses the voltage and ground circuits.

The defrost/panel/floor mode door actuator feedback resistor is supplied a 5-volt reference voltage and ground from the HVAC module. The HVAC module measures the resistance on the defrost/panel/floor mode door actuator feedback circuit to determine the defrost/panel/floor mode door actuator position by the position of the actuator feedback resistor wiper arm.

During an actuator calibration cycle, the HVAC module drives the defrost/panel/floor mode door until the door reaches both internal stops in the HVAC case. If the defrost/panel/floor mode door is temporally obstructed or binding during a calibration cycle, the HVAC module may interpret this as the actual end of travel for the door. When this condition occurs and the HVAC module commands the actuator to it's end of travel, the airflow may not be from the expected outlets.

### DTC Fault Trigger Conditions

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B1086:00 | Air Distribution Damper Motor: No Sub Type Information | The HVAC module senses no changes in the feedback circuit when motor movement is commanded and no motor electrical DTCs are present. This indicates either an open in an actuator motor circuit, an actuator internal failure or a binding or stuck linkage or door. |
| B1086:11 | Air Distribution Damper Motor: Circuit Short to Ground | The HVAC module senses lower than expected voltage on an actuator motor circuit when voltage is applied to drive the motor, indicating a short to ground. |
| B1086:12 | Air Distribution Damper Motor: Circuit Short to Battery | The HVAC module senses greater than expected voltage on the actuator motor circuit when ground is applied to drive the motor, indicating a short to voltage. |
| B1086:13 | Air Distribution Damper Motor: Circuit Open | The HVAC module senses no voltage on the actuator motor circuit when ground is applied to drive the motor, indicating an open circuit. |
| B11E7:11 | Air Distribution Damper Position Sensor: Circuit Short to Ground | The HVAC module senses less than 1 volt on the actuator feedback circuit, indicating a short to ground. |
| B11E7:15 | Air Distribution Damper Position Sensor: Circuit Short to Battery or Open | The HVAC module senses greater than 4.88 volts on the actuator feedback circuit, indicating an open circuit or a short to voltage. |

**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- Defrost/panel/floor mode door actuator
- HVAC module
- Stuck or bound linkage or door

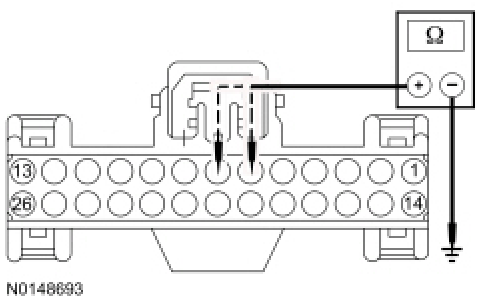
### PINPOINT TEST L : INCORRECT/ERRATIC DIRECTION OF AIRFLOW FROM OUTLETS — EMTC BASE

| L1 CHECK THE HVAC MODULE FOR DTCS |
|---|
| • Ignition ON.<br>• Check the HVAC module for DTCs. |

**Are any DTCs present?**

| | |
|---|---|
| **Yes** | For DTC B1086:00, INSPECT for binding/damaged linkage or door. REPAIR as necessary. If no condition is found, GO to L4.<br>For DTC B1086:11, GO to L2.<br>For DTC B1086:12, GO to L3.<br>For DTC B1086:13, GO to L4.<br>For DTC B11E7:11, GO to L6.<br>For DTC B11E7:15, GO to L9. |
| **No** | INSPECT for broken linkage or door. REPAIR as necessary. If no condition is found, GO to L13. |

**L2 CHECK THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TO GROUND**

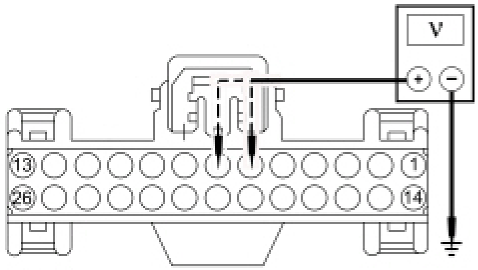
- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: HVAC Module — Base EMTC C294B .
- Measure the resistance between ground and:
  - HVAC module — EMTC C294A Pin 6, circuit CH229 (WH), harness side.
  - HVAC module — EMTC C294A Pin 7, circuit CH228 (YE), harness side.
- 


**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to L5. |
| **No** | REPAIR the circuit in question. |

**L3 CHECK THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: HVAC Module — Base EMTC C294B .
- Ignition ON.
- Measure the voltage between ground and:
  - HVAC module — EMTC C294A Pin 6, circuit CH229 (WH), harness side.
  - HVAC module — EMTC C294A Pin 7, circuit CH228 (YE), harness side.

- 

N0148694

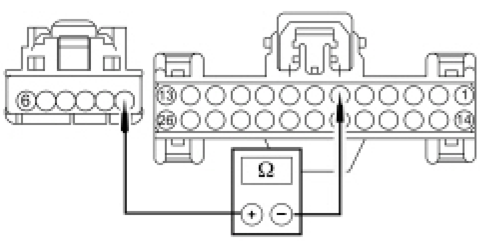
**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |
| **No** | GO to L5. |

**L4 CHECK THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR MOTOR CIRCUITS FOR AN OPEN**

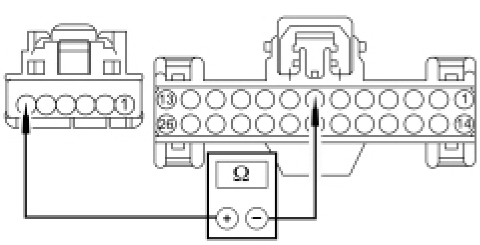
**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: HVAC Module — Base EMTC C294B .
- Disconnect: Defrost/Panel/Floor Mode Door Actuator C236 .
- Measure the resistance between HVAC module — EMTC C294A Pin 6, circuit CH229 (WH), harness side and defrost/panel/floor mode door actuator C236 Pin 1, circuit CH229 (WH/BU), harness side.

- 

N0148695

- Measure the resistance between HVAC module — EMTC C294A Pin 7, circuit CH228 (YE), harness side and defrost/panel/floor mode door actuator C236 Pin 6, circuit CH228 (YE/GY), harness side.

- 

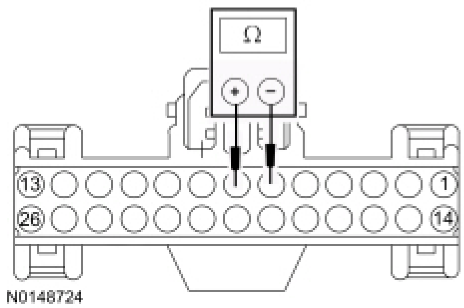
N0148696

**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to L13. |
| **No** | REPAIR the circuit in question. |

**L5 CHECK THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TOGETHER**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

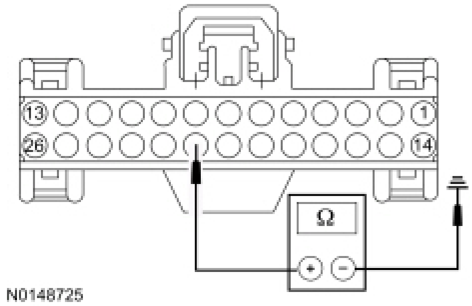
- Ignition OFF.
- Disconnect: Defrost/Panel/Floor Mode Door Actuator C236 .
- Measure the resistance between HVAC module — EMTC C294A Pin 7, circuit CH228 (YE), harness side and HVAC module — EMTC C294A Pin 6, circuit CH229 (WH), harness side.
- 


**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to L13. |
| **No** | REPAIR the circuits. |

**L6 CHECK THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR FEEDBACK CIRCUIT FOR A SHORT TO GROUND**

- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: HVAC Module — Base EMTC C294B .
- Measure the resistance between ground and HVAC module — EMTC C294A Pin 21, circuit VH436 (YE), harness side.

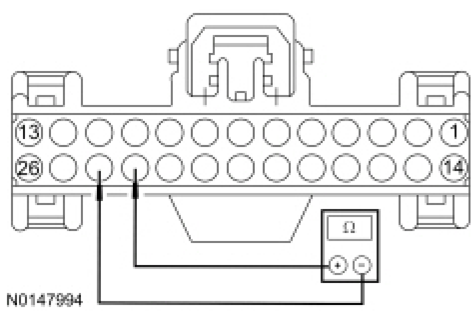
- 


**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to L7. |
| **No** | REPAIR the circuit. |

**L7 CHECK THE SIGNAL RETURN CIRCUIT FOR A SHORT TO THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR FEEDBACK CIRCUIT**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

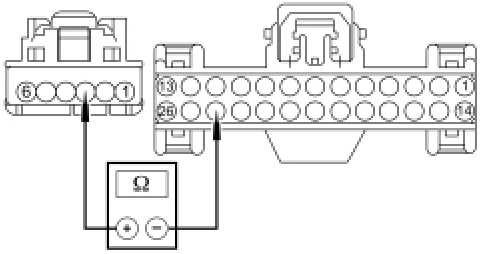
- Disconnect: Defrost/Panel/Floor Mode Door Actuator C236 .
- Measure the resistance between HVAC module — EMTC C294A Pin 23, circuit RH111 (GY/BU), harness side and HVAC module — EMTC C294A Pin 21, circuit VH436 (YE), harness side.
- 


**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to L8. |
| **No** | REPAIR the circuits. |

**L8 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR AN OPEN**

- Measure the resistance between HVAC module — EMTC C294A Pin 24, circuit LH111 (BN/WH), harness side and defrost/panel/floor mode door actuator C236 Pin 3, circuit LH111 (BN/WH), harness side.



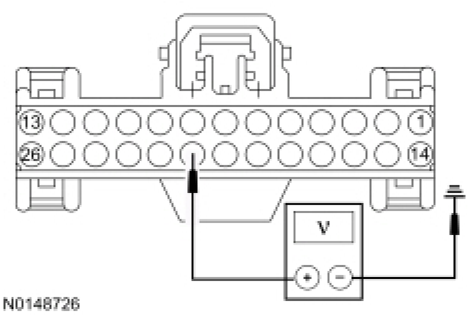

- N0148667

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to L13. |
| **No** | REPAIR the circuit. |

**L9 CHECK THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR FEEDBACK CIRCUIT FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: HVAC Module — Base EMTC C294B .
- Ignition ON.
- Measure the voltage between ground and HVAC module — EMTC C294A Pin 21, circuit VH436 (YE), harness side.




- N0148726

**Is any voltage present?**

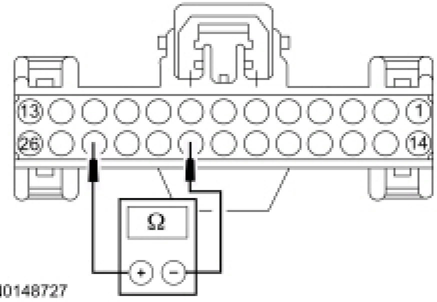
| | |
|---|---|
| **Yes** | REPAIR the circuit. |
| **No** | GO to L10. |

**L10 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR A SHORT TO THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR FEEDBACK CIRCUIT**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Ignition OFF.
- Disconnect: Defrost/Panel/Floor Mode Door Actuator C236 .

- Measure the resistance between HVAC module — EMTC C294A Pin 24, circuit LH111 (BN/WH), harness side and HVAC module — EMTC C294A Pin 21, circuit VH436 (YE), harness side.

- 

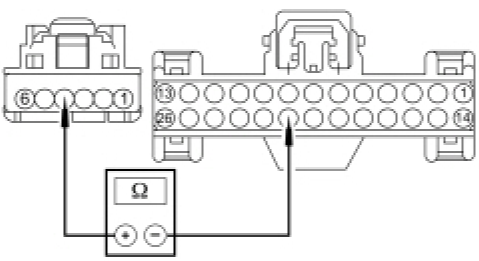
N0148727

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to L11. |
| **No** | REPAIR the circuits. |

## L11 CHECK THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR FEEDBACK CIRCUIT FOR AN OPEN

- Measure the resistance between HVAC module — EMTC C294A Pin 21, circuit VH436 (YE), harness side and defrost/panel/floor mode door actuator C236 Pin 4, circuit VH436 (YE/VT), harness side.

- 

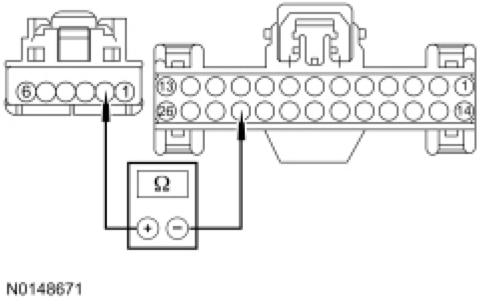
N0148728

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to L12. |
| **No** | REPAIR the circuit. |

## L12 CHECK THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR RETURN CIRCUIT FOR AN OPEN

- Measure the resistance between HVAC module — EMTC C294A Pin 23, circuit RH111 (GY/BU), harness side and defrost/panel/floor mode door actuator C236 Pin 2, circuit RH111 (GY/BU), harness side.




**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to L13. |
| **No** | REPAIR the circuit. |

**L13 CHECK FOR CORRECT DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR OPERATION**

- Disconnect and inspect the defrost/panel/floor mode door actuator connector.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Reconnect the defrost/panel/floor mode door actuator connector. Make sure it seats and latches correctly.
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Clear the DTCs.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CARRY OUT the Defrost/Panel/Floor Mode Door Actuator component test in this section. If the actuator tests OK, GO to L14. When installing an actuator, CONNECT the actuator electrical connector before the HVAC module connector(s). This allows the actuator to be calibrated when the HVAC module is reconnected. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

**L14 MODULE ACTUATOR POSITION CALIBRATION**

**NOTE:** *The purpose of the module actuator position calibration is to allow the HVAC module to reinitialize and calibrate the actuator stop points. To carry out the calibration, follow the steps below.*

- Ignition OFF.
- Disconnect and inspect all HVAC module connectors.
- Repair:

- corrosion (install new connector or terminals - clean module pins)
- damaged or bent pins - install new terminals/pins
- pushed-out pins - install new pins as necessary

- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Ignition ON.
- Clear the DTCs.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Select any position except OFF.
- **NOTE:** *The HVAC module will now initialize and calibrate the actuators. Calibration of the actuators takes approximately 30 seconds.*

  Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC module. REFER to Section 412-01. |
| **No** | The system is now operating correctly at this time. The concern may have been caused by a foreign object in the HVAC case or temporary binding that restricted actuator door travel. CHECK any actuator external linkage. If condition recurs, INSPECT actuator linkage and door for binding and CHECK HVAC case for foreign objects. |

**Pinpoint Test M: Incorrect/Erratic Direction of Airflow From Outlets — EATC Dual Zone and EMTC With 4.2 Inch Display System With 7 Blower Motor Speed Positions**

Refer to Wiring Diagrams Cell 54 , Automatic Climate Control System for schematic and connector information.

Refer to Wiring Diagrams Cell 55 , Automatic Climate Control System for schematic and connector information.

**Normal Operation**

To rotate the defrost/panel/floor mode door actuator, the HVAC module supplies voltage and ground to the defrost/panel/floor mode door actuator motor through the actuator motor circuits. To reverse the defrost/panel/floor mode door actuator rotation, the HVAC module reverses the voltage and ground circuits.

The defrost/panel/floor mode door actuator feedback resistor is supplied a 5-volt reference voltage and ground from the HVAC module. The HVAC module measures the resistance on the defrost/panel/floor mode door actuator feedback circuit to determine the defrost/panel/floor mode door actuator position by the position of the actuator feedback resistor wiper arm.

During an actuator calibration cycle, the HVAC module drives the defrost/panel/floor mode door until the door reaches both internal stops in the HVAC case. If the defrost/panel/floor mode door is temporally obstructed or binding during a calibration cycle, the HVAC module may interpret this as the actual end of travel for the door. When this condition occurs and the HVAC module commands the actuator to it's end of travel, the airflow may not be from the expected outlets.

## DTC Fault Trigger Conditions

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B1086:00 | Air Distribution Damper Motor: No Sub Type Information | The HVAC module senses no changes in the feedback circuit when motor movement is commanded and no motor electrical DTCs are present. This indicates either an open in an actuator motor circuit, an actuator internal failure or a binding or stuck linkage or door. |
| B1086:11 | Air Distribution Damper Motor: Circuit Short to Ground | The HVAC module senses lower than expected voltage on an actuator motor circuit when voltage is applied to drive the motor, indicating a short to ground. |
| B1086:12 | Air Distribution Damper Motor: Circuit Short to Battery | The HVAC module senses greater than expected voltage on the actuator motor circuit when ground is applied to drive the motor, indicating a short to voltage. |
| B1086:13 | Air Distribution Damper Motor: Circuit Open | The HVAC module senses no voltage on the actuator motor circuit when ground is applied to drive the motor, indicating an open circuit. |
| B11E7:11 | Air Distribution Damper Position Sensor: Circuit Short to Ground | The HVAC module senses less than 1 volt on the actuator feedback circuit, indicating a short to ground. |
| B11E7:15 | Air Distribution Damper Position Sensor: Circuit Short to Battery or Open | The HVAC module senses greater than 4.88 volts on the actuator feedback circuit, indicating an open circuit or a short to voltage. |

**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- Defrost/panel/floor mode door actuator
- HVAC module
- Front Controls Interface Module (FCIM)
- Stuck or bound linkage or door

### PINPOINT TEST M : INCORRECT/ERRATIC DIRECTION OF AIRFLOW FROM OUTLETS — EATC DUAL ZONE AND EMTC WITH 4.2 INCH DISPLAY SYSTEM WITH 7 BLOWER MOTOR SPEED POSITIONS

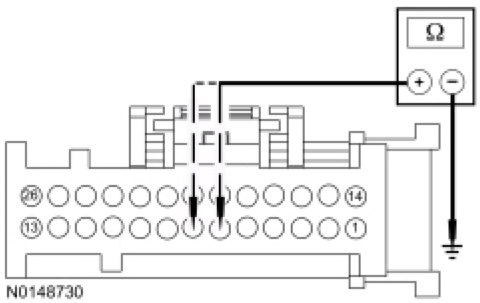
**M1 CHECK THE HVAC MODULE FOR DTCS**

- Ignition ON.
- Check the HVAC module for DTCs.

**Are any DTCs present?**

| | |
|---|---|
| **Yes** | For DTC B1086:00, INSPECT for binding/damaged linkage or door. REPAIR as necessary. If no condition is found, GO to M4.<br>For DTC B1086:11, GO to M2.<br>For DTC B1086:12, GO to M3.<br>For DTC B1086:13, GO to M4.<br>For DTC B11E7:11, GO to M6.<br>For DTC B11E7:15, GO to M9. |
| **No** | INSPECT for binding/damaged linkage or door. REPAIR as necessary. If no condition is found, GO to M13. |

## M2 CHECK THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Disconnect: HVAC Module — EMTC C2357A .
- Disconnect: HVAC Module — EMTC C2357B .
- 

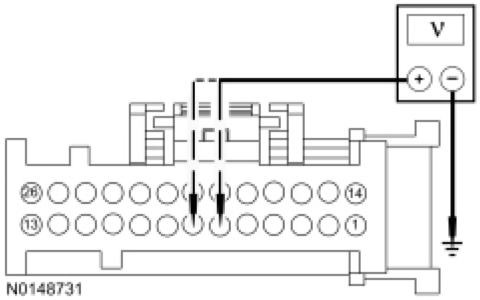

  Measure the resistance between ground and:
  - HVAC module EATC C228A Pin 6, circuit CH229 (WH/BU), harness side.
  - HVAC module EATC C228A Pin 7, circuit CH228 (YE/GY), harness side.
  - HVAC module EMTC C2357A Pin 6, circuit CH229 (WH/BU), harness side.
  - HVAC module EMTC C2357A Pin 7, circuit CH228 (YE/GY), harness side.

**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to M5. |
| **No** | REPAIR the circuit in question. |

## M3 CHECK THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357B .
- Ignition ON.

- 
  
  N0148731

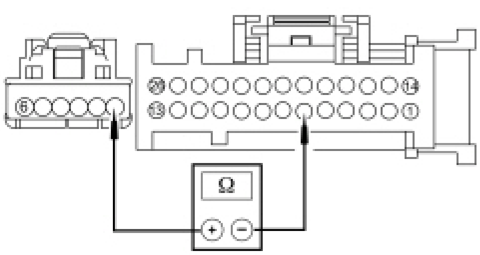
  Measure the voltage between ground and:
  - HVAC module EATC C228A Pin 6, circuit CH229 (WH/BU), harness side.
  - HVAC module EATC C228A Pin 7, circuit CH228 (YE/GY), harness side.
  - HVAC module EMTC C2357A Pin 6, circuit CH229 (WH/BU), harness side.
  - HVAC module EMTC C2357A Pin 7, circuit CH228 (YE/GY), harness side.

**Is any voltage present?**

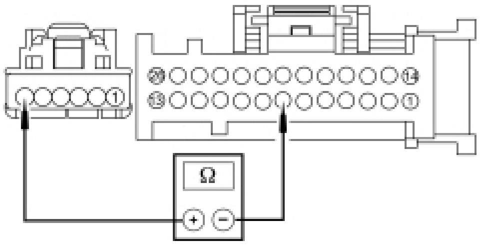
| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |
| **No** | GO to M5. |

**M4 CHECK THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR MOTOR CIRCUITS FOR AN OPEN**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357B .
- Disconnect: Defrost/Panel/Floor Mode Door Actuator C236 .
- 
  
  N0148732

  Measure the resistance between:
  - HVAC module EATC C228A Pin 6, circuit CH229 (WH), harness side and defrost/panel/floor mode door actuator C236 Pin 1, circuit CH229 (WH/BU), harness side.
  - HVAC module EMTC C2357A Pin 6, circuit CH229 (WH), harness side and defrost/panel/floor mode door actuator C236 Pin 1, circuit CH229 (WH/BU), harness side.




- N0148733

  Measure the resistance between:
  - HVAC module EATC C228A Pin 7, circuit CH228 (YE), harness side and defrost/panel/floor mode door actuator C236 Pin 6, circuit CH228 (YE/GY), harness side.
  - HVAC module EMTC C2357A Pin 7, circuit CH228 (YE), harness side and defrost/panel/floor mode door actuator C236 Pin 6, circuit CH228 (YE/GY), harness side.

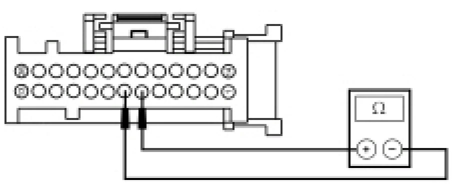
**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to M13. |
| **No** | REPAIR the circuit in question. |

**M5 CHECK THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TOGETHER**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Disconnect: Defrost/Panel/Floor Mode Door Actuator C236 .
- Ignition OFF.



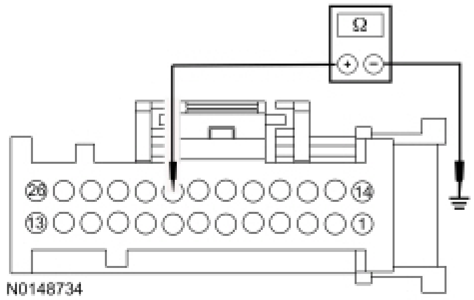

- N0114426

  Measure the resistance between:
  - HVAC module EATC C228A Pin 7, circuit CH228 (YE), harness side and HVAC module C228A Pin 6, circuit CH229 (WH), harness side.
  - HVAC module EMTC C2357A Pin 7, circuit CH228 (YE), harness side and HVAC module C228A Pin 6, circuit CH229 (WH), harness side.

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to M13. |
| **No** | REPAIR the circuits. |

**M6 CHECK THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR FEEDBACK CIRCUIT FOR A SHORT TO GROUND**

- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357B .
- 

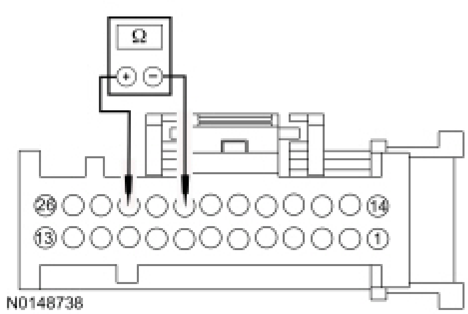

  Measure the resistance between ground and:
  - HVAC module EATC C228A Pin 21, circuit VH436 (YE/VT), harness side.
  - HVAC module EMTC C2357A Pin 21, circuit VH436 (YE/VT), harness side.

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to M7. |
| **No** | REPAIR the circuit. |

**M7 CHECK THE SIGNAL RETURN CIRCUIT FOR A SHORT TO THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR FEEDBACK CIRCUIT**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

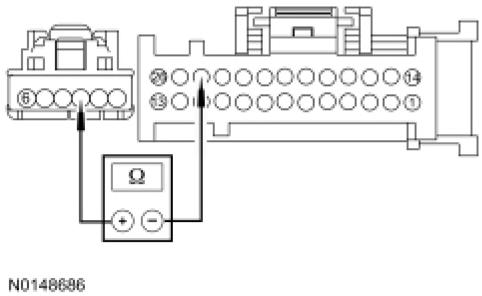
- Disconnect: Defrost/Panel/Floor Mode Door Actuator C236 .
- 


  Measure the resistance between:
  - HVAC module EATC C228A Pin 23, circuit RH111 (GY/BU), harness side and HVAC module C228A Pin 21, circuit VH436 (YE/VT), harness side.
  - HVAC module EMTC C2357A Pin 23, circuit RH111 (GY/BU), harness side and HVAC module C2357A Pin 21, circuit VH436 (YE/VT), harness side.

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to M8. |
| **No** | REPAIR the circuits. |

**M8 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR AN OPEN**

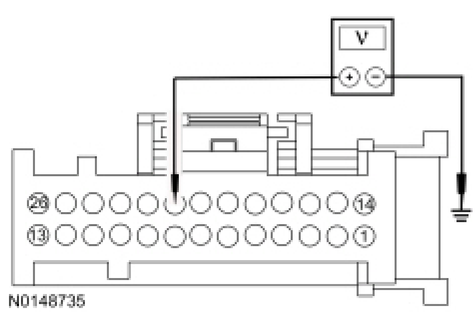
- 


  Measure the resistance between:
  - HVAC module EATC C228A Pin 24, circuit LH111 (BN/WH), harness side and defrost/panel/floor mode door actuator C236 Pin 3, circuit LH111 (BN/WH), harness side.
  - HVAC module EMTC C2357A Pin 24, circuit LH111 (BN/WH), harness side and defrost/panel/floor mode door actuator C236 Pin 3, circuit LH111 (BN/WH), harness side.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to M13. |
| **No** | REPAIR the circuit. |

**M9 CHECK THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR FEEDBACK CIRCUIT FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357B .
- Ignition ON.
- 

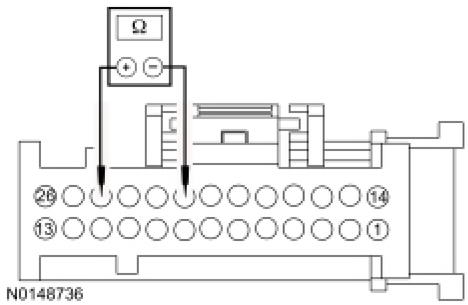
Measure the voltage between ground and:
  - HVAC module EATC C228A Pin 21, circuit VH436 (YE/VT), harness side.
  - HVAC module EMTC C2357A Pin 21, circuit VH436 (YE/VT), harness side.

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit. |
| **No** | GO to M10. |

**M10 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR A SHORT TO THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR FEEDBACK CIRCUIT**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

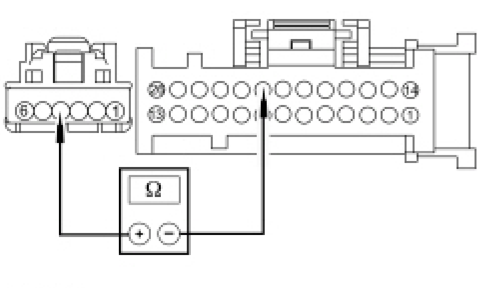
- Ignition OFF.
- Disconnect: Defrost/Panel/Floor Mode Door Actuator C236 .
- 


  Measure the resistance between:
  - HVAC module EATC C228A Pin 24, circuit LH111 (BN/WH), harness side and HVAC module C228A Pin 21, circuit VH436 (YE/VT), harness side.
  - HVAC module EMTC C2357A Pin 24, circuit LH111 (BN/WH), harness side and HVAC module C2357A Pin 21, circuit VH436 (YE/VT), harness side.

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to M11. |
| **No** | REPAIR the circuits. |

**M11 CHECK THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR FEEDBACK CIRCUIT FOR AN OPEN**

- 

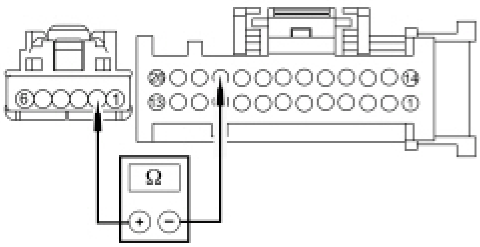
Measure the resistance between:

- HVAC module EATC C228A Pin 21, circuit VH436 (YE/VT), harness side and defrost/panel/floor mode door actuator C236 Pin 4, circuit VH436 (YE/VT), harness side.
- HVAC module EMTC C2357A Pin 21, circuit VH436 (YE/VT), harness side and defrost/panel/floor mode door actuator C236 Pin 4, circuit VH436 (YE/VT), harness side.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to M12. |
| **No** | REPAIR the circuit. |

## M12 CHECK THE DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR RETURN CIRCUIT FOR AN OPEN

- 

N0148692

Measure the resistance between:

- HVAC module EATC C228A Pin 23, circuit RH111 (GY/BU), harness side and defrost/panel/floor mode door actuator C236 Pin 2, circuit RH111 (GY/BU), harness side.
- HVAC module EMTC C2357A Pin 23, circuit RH111 (GY/BU), harness side and defrost/panel/floor mode door actuator C236 Pin 2, circuit RH111 (GY/BU), harness side.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to M13. |
| **No** | REPAIR the circuit. |

## M13 CHECK FOR CORRECT DEFROST/PANEL/FLOOR MODE DOOR ACTUATOR OPERATION

- Disconnect and inspect the defrost/panel/floor mode door actuator connector.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Reconnect the defrost/panel/floor mode door actuator connector. Make sure it seats and latches correctly.
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Clear the DTCs.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CARRY OUT the Defrost/Panel/Floor Mode Door Actuator component test in this section. If the actuator tests OK, GO to M14. When installing an actuator, CONNECT the actuator electrical connector before the HVAC module connector(s). This allows the actuator to be calibrated when the HVAC module is reconnected. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

**M14 MODULE ACTUATOR POSITION CALIBRATION**

**NOTE:** *The purpose of the module actuator position calibration is to allow the HVAC module to reinitialize and calibrate the actuator stop points. To carry out the calibration, follow the steps below.*

- Ignition OFF.
- Disconnect and inspect all HVAC module connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Ignition ON.
- Clear the DTCs.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Select any position except OFF.
- **NOTE:** *The HVAC module will now initialize and calibrate the actuators. Calibration of the actuators takes approximately 30 seconds.*

  Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC module. REFER to Section 412-01. |
| **No** | The system is now operating correctly at this time. The concern may have been caused by a foreign object in the HVAC case or temporary binding that restricted actuator door travel. CHECK any actuator external linkage. If condition recurs, INSPECT actuator linkage and door for binding and CHECK HVAC case for foreign objects. |

**Pinpoint Test N: Insufficient, Erratic or No Heat**

**Normal Operation**

Warm coolant flows from the engine through the heater core and back to the engine. Proper coolant temperatures are critical for good heater performance.

**This pinpoint test is intended to diagnose the following:**

- Plugged or partially plugged heater core
- Low engine coolant level
- Temperature blend door actuator(s)
- Temperature blend door(s) binding or stuck

**PINPOINT TEST N : INSUFFICIENT, ERRATIC OR NO HEAT**

**N1 CHECK FOR CORRECT ENGINE COOLANT LEVEL**

- Ignition OFF.
- Check the engine coolant level when hot and cold.

**Is the engine coolant at the correct level (hot/cold) as indicated on the engine coolant recovery reservoir?**

| | |
|---|---|
| **Yes** | GO to N3. |
| **No** | GO to N2. |

**N2 CHECK THE ENGINE COOLING SYSTEM FOR LEAKS**

- Pressure test the cooling system for leaks. Refer to Section 303-03.

**Does the engine cooling system leak?**

| | |
|---|---|
| **Yes** | REPAIR the engine coolant leak. TEST the system for normal operation. |
| **No** | FILL and BLEED the cooling system. REFER to Section 303-03. After filling and bleeding the cooling system, GO to N3. |

**N3 CHECK FOR COOLANT FLOW TO THE HEATER CORE**

- Run the engine until it reaches normal operation temperature. Select the FLOOR position on the control assembly. Set the temperature control to full warm and the blower to the lowest setting.
- Using a suitable temperature measuring device, check the heater core inlet hose to see if it is hot.

**Is the heater core inlet hose temperature above 65.5°C (150°F)?**

| | |
|---|---|
| **Yes** | GO to N4. |
| **No** | REFER to Section 303-03 to check cooling system function. |

**N4 CHECK FOR A PLUGGED OR RESTRICTED HEATER CORE**

- Using a suitable temperature measuring device, measure the heater core outlet hose temperature.

**Is the heater core outlet hose temperature similar to the inlet hose temperature (within approximately 6-17°C**

[10-30°F])?

| | |
|---|---|
| **Yes** | For EMTC base with 4 blower motor speed positions, GO to Pinpoint Test R.<br>For EMTC with 4.2 inch display with 7 blower motor speed positions, GO to Pinpoint Test S.<br>For EATC dual zone, if the driver side temperature is not hot, GO to Pinpoint Test U.<br>For EATC dual zone, if the passenger side temperature is not hot, GO to Pinpoint Test V. |
| **No** | INSTALL a new heater core. REFER to Section 412-01. TEST the system for normal operation. |

**Pinpoint Test O: The Air Conditioning (A/C) is Inoperative**

Refer to Wiring Diagrams Cell 54 , Manual Climate Control System for schematic and connector information.

Refer to Wiring Diagrams Cell 55 , Automatic Climate Control System for schematic and connector information.

**Normal Operation**

For vehicles equipped with EMTC base system with 4 blower motor speed positions, when A/C button is pressed, a message is sent from the HVAC module over the MS-CAN bus to the BCM , then from the BCM through the HS-CAN bus to the PCM.

For vehicles equipped with EMTC with non touch 4.2" display (FCDIM ) system with 7 blower motor speed positions, when A/C is requested, the A/C request message is sent over I-CAN to the FCIM . The FCIM sends a message over the MS-CAN bus to the HVAC module that A/C has been selected. The HVAC module then sends an A/C request message over the MS-CAN bus to the BCM , then from the BCM through the HS-CAN bus to the PCM.

For vehicles equipped with 8" touchscreen, when A/C is requested by touching the FDIM , a message is sent from the APIM over the MS-CAN bus to the HVAC module. The HVAC module then sends an A/C request message over the MS-CAN bus to the BCM , then from the BCM through the HS-CAN bus to the PCM.

Voltage is provided to the A/C clutch relay coil from BJB fuse 77 (10A) and switch contacts from BJB fuse 30 (10A). When A/C is requested and A/C line pressures allow, a ground is provided to the A/C clutch relay coil from the PCM, energizing the A/C clutch relay. When the PCM energizes the relay, voltage is supplied to the A/C compressor clutch field coil from the relay. Ground is supplied for the A/C compressor clutch field coil.

An accurate evaporator temperature is critical for compressor engagement. The PCM uses the temperature measurement to turn the A/C compressor off before the evaporator temperatures are cold enough to freeze the condensation on the evaporator core.

When an A/C request is received by the PCM, the A/C compressor clutch will only be engaged through the A/C clutch relay if all of the following conditions are met:

- The PCM does not detect excessively high or low refrigerant pressure from the A/C pressure transducer.
- The PCM does not detect excessively high engine coolant temperature.
- The PCM does not detect an ambient air temperature below -1°C (30°F).
- The PCM has not detected a WOT condition.
- The HVAC module does not detect an evaporator discharge air temperature below 2°C (36°F).

**This pinpoint test is intended to diagnose the following:**

- Fuse(s)
- Wiring, terminals or connectors
- A/C system discharged/low charge
- HVAC module

- FCIM
- Evaporator temperature sensor
- A/C pressure transducer
- A/C compressor clutch field coil
- A/C clutch relay
- A/C compressor clutch air gap

**PINPOINT TEST O : THE AIR CONDITIONING (A/C) IS INOPERATIVE**

**NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may cause damage to the connector.**

**O1 CHECK THE A/C SYSTEM PRESSURE**

- Ignition OFF.
- With the manifold gauge set connected, check the A/C system pressure.

**Is the A/C system pressure above 290 kPa (42 psi) ?**

| | |
|---|---|
| **Yes** | GO to O2. |
| **No** | CHECK the A/C system for leaks. REFER to Electronic Leak Detection or Fluorescent Dye Leak Detection in this section. RECHARGE the A/C system. REFER to Air Conditioning (A/C) System Recovery, Evacuation and Charging. TEST the system for normal operation. |

**O2 CHECK THE COMMUNICATION NETWORK**

- Ignition ON.
- Using a diagnostic scan tool, carry out the network test.

**Do the FCIM , BCM and the PCM pass the network test?**

| | |
|---|---|
| **Yes** | GO to O3. |
| **No** | DIAGNOSE the FCIM , BCM or PCM does not communicate with the diagnostic scan tool. REFER to: Section 418-00. |

**O3 CHECK THE PCM (A/C) PRESSURE SENSOR (ACP_PRESS) PID**

- Using a diagnostic scan tool, monitor the ACP_PRESS PCM PID .
- With the manifold gauge set connected, compare the pressure readings of the manifold gauge set and the ACP_PRESS PID .

**Are the pressure values of the manifold gauge set and the ACP_PRESS PID within ± 15 psi (103 kPa)?**

| | |
|---|---|
| **Yes** | GO to O4. |
| **No** | GO to Pinpoint Test A. |

**O4 COMPARE THE PCM INTAKE AIR TEMPERATURE (IAT) PID AND THE OTHER TEMPERATURE SENSOR READINGS TO THE PCM AMBIENT AIR TEMPERATURE (AAT) PID**

**NOTE:** *Compare multiple engine sensor readings to the ambient temperature to determine sensors are reading correctly. A faulty sensor can cause the PCM to disable the A/C with or without a DTC.*

- Allow the vehicle exterior and interior to stabilize to ambient temperature. This can take a soak period of at least 6 hours.
- Using a diagnostic scan tool, view PCM PIDs .
- Monitor the AAT, CACT, CHT, ECT, IAT, IAT2, MAF, MAPT, TCB and TCIPT PIDs (as applicable).

**Are the temperature values similar (typically within 18ºC (32.4ºF)?**

| | |
|---|---|
| **Yes** | GO to O5. |
| **No** | DIAGNOSE the suspect temperature sensor. REFER to Section 413-01 for Ambient Air Temperature (AAT) sensor or Outside Air Temperature Display concerns. For all other temperature sensor concerns, REFER to Powertrain Control/Emissions Diagnosis (PC/ED) manual. |

**O5 CHECK THE HVAC EVAPORATOR TEMPERATURE (EVAP_TEMP) PID**

- Using a diagnostic scan tool, view HVAC control module PIDs .
- Monitor the EVAP_TEMP PID .

**Does the evaporator temperature sensor PID read similar (typically within 18ºC (32.4ºF) to the ambient temperature?**

| | |
|---|---|
| **Yes** | GO to O6. |
| **No** | DIAGNOSE the evaporator temperature sensor. GO to Pinpoint Test G. |

**O6 CHECK THE FCIM (A/C) SWITCH (CC_SW_AC) PID WITH THE A/C ON**

- Start the engine.
- Using a diagnostic scan tool, monitor the FCIM CC_SW_AC PID .
- Select PANEL and press the A/C button (indicator on) on the HVAC controls.

**Does the PID display Active when the button is pressed and the indicator is on?**

| | |
|---|---|
| **Yes** | GO to O7. |
| **No** | GO to O16. |

**O7 CHECK THE PCM AIR CONDITIONING REQUEST SIGNAL (AC_REQ) PID WITH THE A/C ON**

- Using a diagnostic scan tool, monitor the PCM AC_REQ PID .
- Select PANEL and press the A/C button (indicator on) on the HVAC controls.

**Does the AC_REQ PID read YES when the button is pressed and the indicator is on?**

| | |
|---|---|
| **Yes** | GO to O9. |
| **No** | GO to O8. |

**O8 RESET THE AMBIENT AIR TEMPERATURE (AAT) SENSOR**

- On the HVAC controls, press the A/C and Recirc buttons simultaneously, then release both. Within 2 seconds press A/C button again.
- Start the engine.
- On the HVAC controls, set the temperature to full cold, select PANEL and select the A/C button (indicator on).

**Does the A/C compressor turn on?**

| | |
|---|---|
| **Yes** | RETEST the A/C system for normal operation. CARRY OUT the refrigerant system tests. Refer to the appropriate Refrigerant System Tests procedure in Group 412. |
| **No** | GO to O9. |

**O9 CHECK THE PCM AIR CONDITIONING COMPRESSOR COMMANDED STATE (ACC_CMD) PID WITH THE A/C COMMANDED ON**

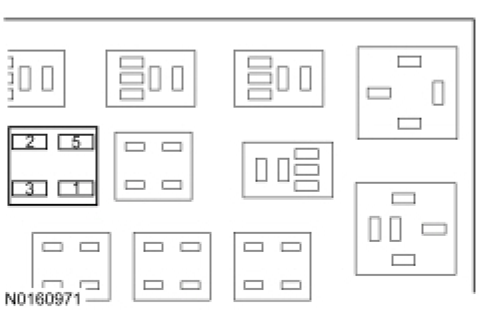
- Using a diagnostic scan tool, activate the PCM ACC_CMD PID .

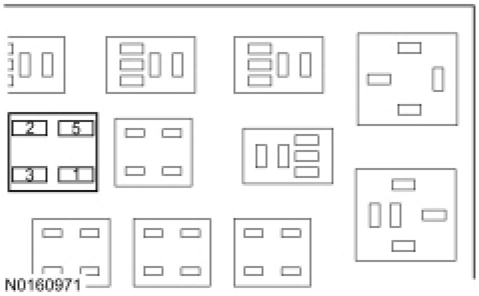
**Does the A/C compressor turn on?**

| | |
|---|---|
| **Yes** | GO to O17. |
| **No** | GO to O10. |

**O10 CHECK THE VOLTAGE TO THE A/C CLUTCH RELAY**

- Ignition OFF.
- Disconnect: A/C Clutch Relay .
- Ignition ON.
- Measure:

| Positive Lead | Measurement / Action | Negative Lead |
|---|---|---|
|  | V | Ground |

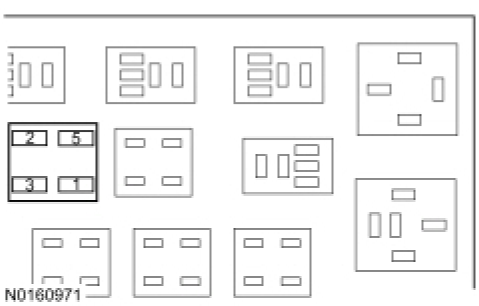
| Positive Lead | Measurement / Action | Negative Lead |
|---|---|---|
| A/C clutch relay socket 1 | | |
|  A/C clutch relay socket 3 | | Ground |

**Is the voltage greater than 11 volts?**

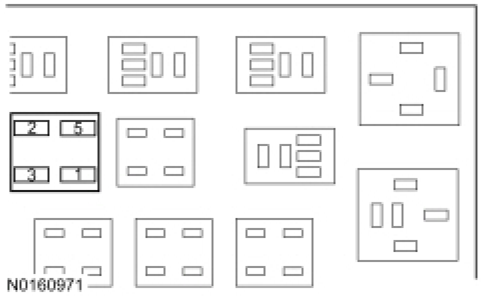
| | |
|---|---|
| **Yes** | GO to O11. |
| **No** | VERIFY Battery Junction Box (BJB) fuses 77 (10A) and 30 (10A) are OK. If OK, REPAIR the circuit in question. TEST the system for normal operation. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. |

## O11 CHECK THE A/C CLUTCH RELAY CONTROL GROUND CIRCUIT FOR AN OPEN

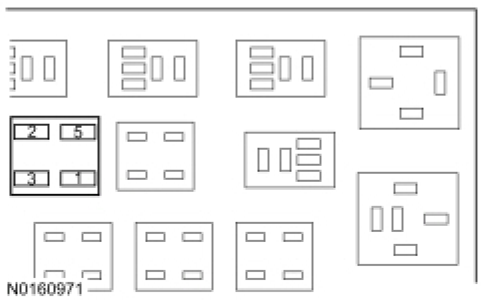
- Ignition OFF.
- Disconnect: PCM C1551B (3.5L) .
- Disconnect: PCM C1381B (5.0L, 6.2L) .
- Disconnect: PCM C175B (3.7L) .
- For 3.5L, measure:

| Positive Lead | Measurement / Action | Negative Lead |
|---|---|---|
|  A/C clutch relay socket 2 | Ω | C1551B Pin 2 |

- For 5.0L or 6.2L, measure:

| Positive Lead | Measurement / Action | Negative Lead |
|---|---|---|
|  A/C clutch relay socket 2 | Ω | C1318B Pin 12 |

- For 3.7L, measure:

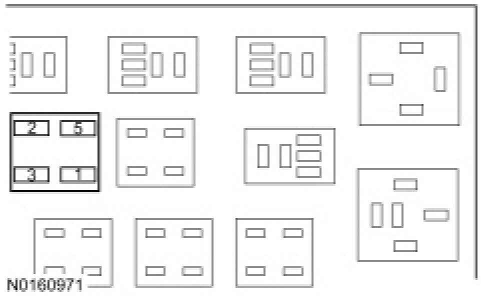
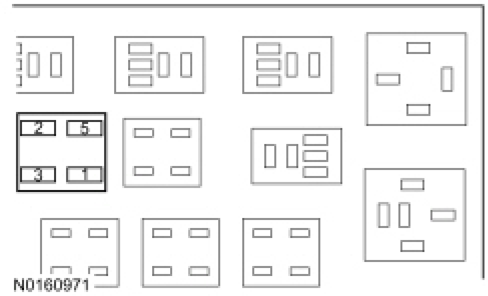
| Positive Lead | Measurement / Action | Negative Lead |
|---|---|---|
|  A/C clutch relay socket 2 | Ω | C175B Pin 12 |

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to O12. |
| **No** | REPAIR the circuit. |

**O12 BYPASS THE A/CA/C CLUTCH RELAY**

- Ignition OFF.
- Connect a fused jumper wire:

| Positive Lead | Measurement / Action | Negative Lead |
|---|---|---|
| <br>A/C clutch relay socket 3 |  | <br>A/C clutch relay socket 5 |

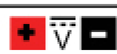
- Start the engine.

**Does the A/C compressor clutch engage?**

| | |
|---|---|
| **Yes** | REMOVE the fused jumper wire. INSTALL a new A/C clutch relay. |
| **No** | LEAVE the fused jumper wire installed. GO to O13. |

## O13 CHECK THE A/C COMPRESSOR CLUTCH FIELD COIL VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: A/C compressor clutch field coil C100 .
- Measure:

| Positive Lead | Measurement / Action | Negative Lead |
|---|---|---|
| C100 Pin 1 |  | Ground |

**Is the voltage greater than 11 volts?**

| | |
|---|---|
| **Yes** | REMOVE the fused jumper wire. GO to O14. |
| **No** | REMOVE the fused jumper wire. REPAIR the circuit. |

## O14 CHECK THE A/C COMPRESSOR CLUTCH FIELD COIL GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Measure:

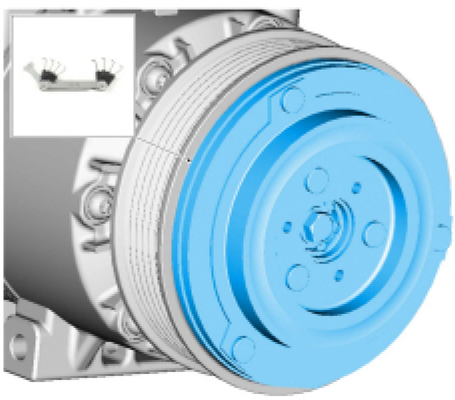
| Positive Lead | Measurement / Action | Negative Lead |
|---|---|---|
| C100 Pin 2 | Ω | Ground |

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to O15. |
| **No** | REPAIR the circuit. |

**O15 CHECK THE A/C COMPRESSOR CLUTCH AIR GAP**

- Measure the A/C compressor clutch air gap at 3 equally spaced locations between the clutch hub and the A/C compressor clutch pulley.



**Is the A/C compressor clutch air gap average greater than 0.65 mm (0.026 in)?**

| | |
|---|---|
| **Yes** | ADJUST the A/C compressor clutch gap. REFER to Air Conditioning (A/C) Clutch Air Gap Adjustment in this section. TEST the system for normal operation. |
| **No** | INSTALL a new A/C compressor clutch field coil. REFER to Section 412-01. TEST the system for normal operation. |

**O16 CHECK FOR CORRECT HVAC CONTROL MODULE OPERATION**

- Disconnect and inspect all HVAC control module connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Reconnect all HVAC control module connectors. Make sure they seat and latch correctly.
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC control module. REFER to Section 412-01. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

**O17 CHECK FOR AN INPUT DISABLING THE A/C CLUTCH RELAY**

**NOTE:** *The PCM strategy may disable the A/C compressor operation. If the A/C compressor clutch can be commanded on using IDS PCM Parameter Identifications (PIDs) Active Commands, PCM replacement will not resolve the condition.*

- Diagnose the PCM not energizing the A/C clutch relay using the suggestions in the table below. Refer to the reference value information in Section 6 of the Powertrain Control/Emissions Diagnosis (PC/ED) manual.

| |
|---|
| A/C Clutch Engagement is inhibited because strategy has not yet detected hardware that indicates vehicle is equipped with A/C |
| A/C Clutch Engagement is inhibited because the strategy is operating in Ignition System Failure Mode |
| A/C Clutch Engagement is inhibited because the (ECU) received a Request to Command the A/C Off |
| A/C Clutch Engagement is inhibited because the engine has not yet reached a stable running mode after starting |
| A/C Clutch Engagement is inhibited because the A/C Discharge (Head) Pressure is Too High |
| A/C Clutch Engagement is inhibited because the Engine Coolant Temperature is Too High |
| A/C Clutch Engagement is inhibited to Prevent Frost and Ice Build Up on the Evaporator |
| A/C Clutch Engagement is inhibited to Prevent an Engine Stall during a Low Engine Speed condition |
| A/C Clutch Engagement is inhibited to Protect the Compressor from a Compressor Over-speed condition |
| A/C Clutch Engagement is inhibited to temporarily make more power available when Accelerator Pedal is Fully Depressed |
| A/C Clutch Engagement is inhibited because Low A/C Refrigerant Charge has been detected |
| A/C Clutch Engagement is inhibited to Protect the Compressor from Operating at Too Low of an Ambient Temperature |
| A/C Clutch Engagement is inhibited due to Missing Climate Control Message |
| A/C Clutch Engagement is inhibited because the strategy is operating in Failsafe Cooling Mode |
| A/C Clutch Engagement is inhibited to Protect the Clutch from Damage because the Compressor Load and Speed are Too High |
| A/C Clutch Engagement is inhibited by the off portion of the A/C Cycling Strategy invoked to manage High Engine Temperature |
| A/C Clutch Engagement is inhibited due to Low Battery State of Charge |
| A/C Clutch Engagement is inhibited to Protect the Variable Displacement Compressor from Operating at Too Low of a Temperature |
| A/C Clutch Engagement is inhibited to improve Brake Booster Vacuum |
| A/C Clutch Engagement is inhibited because Evaporator Temperature is sufficiently low and compressor was at minimum displacement |
| A/C Clutch Engagement is inhibited (Disabled) because the Evaporator Temperature is sufficiently below the target temperature |
| A/C Clutch Engagement is inhibited to satisfy A/C Clutch minimum off time |

| |
|---|
| A/C Clutch Engagement is inhibited Due to Request from Torque Control Strategy (to Temporarily Make More Power Available) |
| A/C Clutch Engagement is inhibited to Prevent Engine Stalling |
| A/C Clutch Engagement is inhibited Due to Request to Disable A/C from Stop-Start Strategy |
| A/C Clutch Engagement is inhibited (Delayed) to make Power Available for Power Steering |
| A/C Clutch Engagement Is Inhibited Due To State Of Auxiliary A/C Disable (Typically A Pressure Or Temperature) Switch 1 input |
| A/C Clutch Engagement is inhibited due to state of auxiliary A/C Disable (Typically a Pressure or Temperature) Switch 2 input |

**Are any of the conditions described above not within normal parameters?**

| | |
|---|---|
| **Yes** | DIAGNOSE the condition found to be disabling the A/C clutch relay. REFER to the appropriate Workshop Manual (WSM) section or the Powertrain Control/Emissions Diagnosis (PC/ED) manual. |
| **No** | The A/C compressor clutch can be commanded on using the IDS PCM Active command PID . The A/C inoperative concern may be caused by an intermittent condition due to a component or module connection, wiring or pin issue. ADDRESS the root cause of any connector or pin issues. CHECK the vehicle service history for recent service actions that have replaced modules. This condition may be due to incomplete or incorrect PMI (programmable module installation) procedures. |

**Pinpoint Test P: The Air Conditioning (A/C) is Always On — A/C Compressor Does Not Cycle**

Refer to Wiring Diagrams Cell 54 , Manual Climate Control System for schematic and connector information.

Refer to Wiring Diagrams Cell 55 , Automatic Climate Control System for schematic and connector information.

**Normal Operation**

For vehicles equipped with EMTC base system with 4 blower motor speed positions, when A/C button is pressed, a message is sent from the HVAC module over the MS-CAN bus to the BCM , then from the BCM through the HS-CAN bus to the PCM.

For vehicles equipped with EMTC with non touch 4.2" display (FCDIM ) system with 7 blower motor speed positions, when A/C is requested, the A/C request message is sent over I-CAN to the FCIM . The FCIM sends a message over the MS-CAN bus to the HVAC module that A/C has been selected. The HVAC module then sends an A/C request message over the MS-CAN bus to the BCM , then from the BCM through the HS-CAN bus to the PCM.

For vehicles equipped with 8" touchscreen, when A/C is requested by touching the FDIM , a message is sent from the APIM over the MS-CAN bus to the HVAC module. The HVAC module then sends an A/C request message over the MS-CAN bus to the BCM , then from the BCM through the HS-CAN bus to the PCM.

Voltage is provided to the A/C clutch relay coil from Battery Junction Box (BJB) fuse 77 (10A) and switch contacts from Battery Junction Box (BJB) fuse 30 (10A). When A/C is requested and A/C line pressures allow, a ground is provided to the A/C clutch relay coil from the PCM, energizing the A/C clutch relay. When the PCM energizes the relay, voltage is supplied to the A/C compressor clutch field coil from the relay. Ground is supplied for the A/C compressor clutch field coil.

The evaporator temperature sensor is used for A/C compressor cycling. An accurate evaporator temperature is critical for compressor engagement. The PCM uses the temperature measurement to turn the A/C compressor off before the evaporator temperatures are cold enough to freeze the condensation on the evaporator core.

**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- PCM
- Evaporator temperature sensor
- A/C clutch relay
- A/C compressor clutch air gap

**PINPOINT TEST P : THE AIR CONDITIONING (A/C) IS ALWAYS ON — A/C COMPRESSOR DOES NOT CYCLE**

**NOTE:** *Before carrying out the following test, diagnose any HVAC or PCM or DTCs.*

**P1 CHECK THE A/C EVAPORATOR TEMPERATURE (EVAP_TEMP) HVAC PID**

- Ignition ON.
- Allow the vehicle exterior and interior to stabilize to the ambient temperature.
- Enter the following diagnostic mode on the scan tool: HVAC DataLogger .
- Monitor the EVAP_TEMP HVAC PID

**Does the EVAP_TEMP HVAC PID read similar to the ambient temperature?**

| | |
|---|---|
| **Yes** | GO to P2. |
| **No** | GO to Pinpoint Test G. |

**P2 CHECK THE AIR CONDITIONING COMPRESSOR COMMANDED STATE (ACC_CMD) PCM PID WITH THE A/C OFF**

- Enter the following diagnostic mode on the scan tool: PCM DataLogger .
- Monitor the ACC_CMD PCM PID
- Select PANEL press the A/C button (indicator off) on the HVAC controls.

**Does the ACC_CMD PCM PID read OFF?**

| | |
|---|---|
| **Yes** | GO to P3. |
| **No** | GO to P7. |

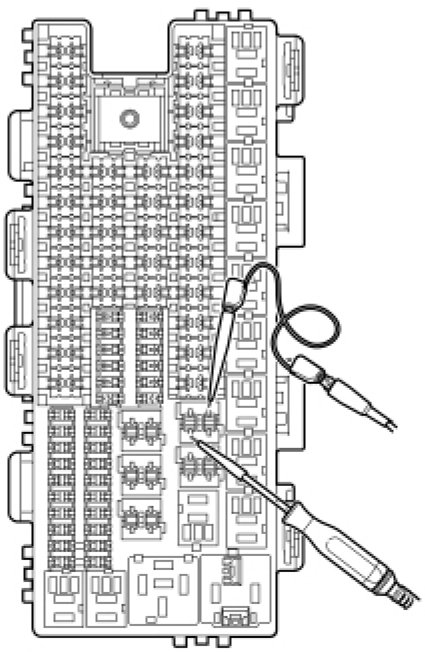
**P3 CHECK THE A/C COMPRESSOR CLUTCH ENGAGEMENT WITH RELAY DISCONNECTED**

- Ignition OFF.
- Disconnect: A/C Clutch Relay .
- Start the engine.
- With the engine running, observe the A/C compressor clutch.

**Is the compressor clutch disengaged?**

| | |
|---|---|
| **Yes** | GO to P4. |
| **No** | GO to P6. |

**P4 CHECK THE A/C CLUTCH RELAY**

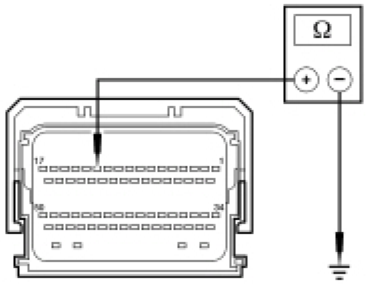
- Select PANEL and press the A/C button (indicator off) on the HVAC controls.
- With the engine running, connect a 12-volt test light between A/C clutch relay socket, circuit CBB77 (WH) and A/C clutch relay socket, circuit CH302 (WH/BN).
- 
  N0122684

**Does the test light illuminate?**

| | |
|---|---|
| **Yes** | GO to P5. |
| **No** | INSTALL a new A/C clutch relay. TEST the system for normal operation. |

**P5 CHECK THE A/C CLUTCH RELAY COIL SWITCHED GROUND CIRCUIT FOR A SHORT TO GROUND**

- Ignition OFF.
- Disconnect: PCM C1381B (5.0L, 6.2L) .
- Disconnect: PCM C175B (3.7L) .
- Disconnect: PCM C1551B (3.5L) .
- **For 3.7L, 5.0L and 6.2L**

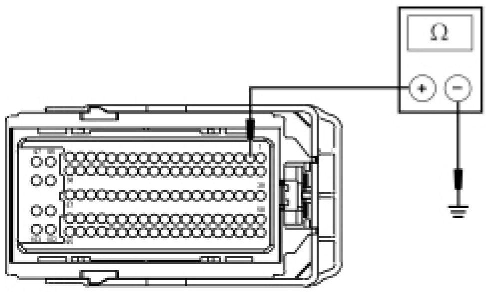



N0086590

Measure the resistance between ground and:

- For 3.7L, PCM C175B Pin 12, circuit CH302 (WH/BN), harness side.
- For 5.0L and 6.2L, PCM C1381B Pin 12, circuit CH302 (WH/BN), harness side.

- **For 3.5L**




N0122700

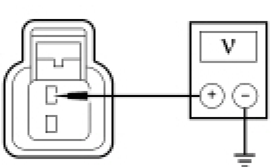
For 3.5L, measure the resistance between ground and PCM C1551B Pin 2, circuit CH302 (WH/BN), harness side.

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to P7. |
| **No** | REPAIR the circuit. |

**P6 CHECK FOR VOLTAGE TO THE A/C COMPRESSOR CLUTCH FIELD COIL**

- Ignition OFF.
- Disconnect: A/C Compressor Clutch Field Coil C100 .
- Ignition ON.
- Select the OFF position on the HVAC module.
- Measure the voltage between A/C compressor clutch field coil C100 Pin 1, circuit CH401 (VT/WH), harness side and ground.




- A0048576

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit. |
| **No** | ADJUST the A/C compressor clutch gap. REFER to Air Conditioning (A/C) Clutch Air Gap Adjustment in this section. TEST the system for normal operation. |

**P7 CHECK FOR CORRECT PCM OPERATION**

- Disconnect and inspect all PCM connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all PCM connectors. Make sure they seat and latch correctly.
- Clear the DTCs.
- Enter the following diagnostic mode on the scan tool: Self-Test — PCM .
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new PCM. REFER to Section 303-14. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

## Pinpoint Test Q: The Air Conditioning (A/C) is Always On — A/C Mode Always Commanded ON

Refer to Wiring Diagrams Cell 54 , Manual Climate Control System for schematic and connector information.

Refer to Wiring Diagrams Cell 55 , Automatic Climate Control System for schematic and connector information.

### Normal Operation

For vehicles equipped with EMTC base system with 4 blower motor speed positions, when A/C button is pressed, a message is sent from the HVAC module over the MS-CAN bus to the BCM , then from the BCM through the HS-CAN bus to the PCM.

For vehicles equipped with EMTC with non touch 4.2" display (FCDIM ) system with 7 blower motor speed positions, when A/C is requested, the A/C request message is sent over I-CAN to the FCIM . The FCIM sends a message over the MS-CAN bus to the HVAC module that A/C has been selected. The HVAC module then sends an A/C request message over the MS-CAN bus to the BCM , then from the BCM through the HS-CAN bus to the PCM.

For vehicles equipped with 8" touchscreen, when A/C is requested by touching the FDIM , a message is sent from the APIM over the MS-CAN bus to the HVAC module. The HVAC module then sends an A/C request message over the MS-CAN bus to the BCM , then from the BCM through the HS-CAN bus to the PCM.

**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- Front Controls Interface Module (FCIM)
- PCM
- HVAC module

**PINPOINT TEST Q : THE AIR CONDITIONING (A/C) IS ALWAYS ON — A/C MODE ALWAYS COMMANDED ON**

**NOTE:** *Before carrying out the following test, diagnose any HVAC or PCM DTCs.*

**Q1 CHECK THE AIR CONDITIONING REQUEST SIGNAL (AC_REQ) PCM PID WITH THE A/C OFF**

- Enter the following diagnostic mode on the scan tool: PCM DataLogger .
- Select PANEL press the A/C button (indicator off) on the HVAC controls and monitor the AC_REQ PCM PID.

**Does the AC_REQ PCM PID read NO?**

| | |
|---|---|
| **Yes** | GO to Q6. |
| **No** | GO to Q2. |

**Q2 CHECK THE AIR CONDITIONING SWITCH STATUS (CC_SW_AC) HVAC PID WITH THE A/C OFF**

- Enter the following diagnostic mode on the scan tool: HVAC DataLogger .
- Select PANEL on the HVAC controls and monitor the CC_SW_AC HVAC PID.

**Does the CC_SW_AC HVAC PID read Inactive?**

| | |
|---|---|
| **Yes** | GO to Q3. |
| **No** | GO to Q5. |

**Q3 CHECK THE AIR CONDITIONING SWITCH STATUS (CC_SW_AC) FCIM PID WITH THE A/C OFF**

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Front Controls Interface Module (FCIM) DataLogger .
- Monitor the CC_SW_AC FCIM PID
- Select PANEL on the HVAC controls and monitor the CC_SW_AC FCIM PID.

**Does the CC_SW_AC FCIM PID read Inactive?**

| | |
|---|---|
| **Yes** | GO to Q4. |
| **No** | INSTALL a new FCIM . REFER to Refer to the appropriate section in Group 415 for the procedure.. TEST the system for normal operation. |

## Q4 CHECK THE AIR CONDITIONING REQUEST SIGNAL (AC_REQ) PCM PID WITH HVAC MODULE DISCONNECTED

- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: HVAC Module — Base EMTC C294B .
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357B .
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Enter the following diagnostic mode on the scan tool: PCM DataLogger .
- Monitor the AC_REQ PCM PID

**Does the AC_REQ PCM PID read NO?**

| | |
|---|---|
| **Yes** | GO to Q5. |
| **No** | GO to Q6. |

## Q5 CHECK FOR CORRECT HVAC MODULE OPERATION

- Ignition OFF.
- Disconnect and inspect all HVAC module connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Ignition ON.
- Clear the DTCs.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC module. REFER to Section 412-01. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

| |
|---|
| |
| **Q6 CHECK FOR CORRECT PCM OPERATION** |
| • Disconnect and inspect all PCM connectors.<br>• Repair:<br>  ▪ corrosion (install new connector or terminals - clean module pins)<br>  ▪ damaged or bent pins - install new terminals/pins<br>  ▪ pushed-out pins - install new pins as necessary<br>• Connect all PCM connectors. Make sure they seat and latch correctly.<br>• Clear the DTCs.<br>• Enter the following diagnostic mode on the scan tool: Self-Test — PCM .<br>• Operate the system and verify the concern is still present.<br><br>**Is the concern still present?** |

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new PCM. REFER to Section 303-14. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

### Pinpoint Test R: Temperature Control Is Inoperative/Does Not Operate Correctly — EMTC Base With 4 Blower Motor Speed Positions

Refer to Wiring Diagrams Cell 54 , Manual Climate Control System for schematic and connector information.

**Normal Operation**

To rotate the temperature blend door actuator, the HVAC module supplies voltage and ground to the temperature blend door actuators through the door actuator motor circuits. To reverse the temperature blend door actuator rotation, the HVAC module reverses the voltage and ground circuits.

The temperature blend door actuator feedback resistor is supplied a ground from the HVAC module and a 5-volt reference voltage on the temperature blend door actuator reference circuits. The HVAC module reads the voltage on the temperature blend door actuator feedback circuits to determine the temperature blend door actuator position by the position of the actuator feedback resistor wiper arm.

During an actuator calibration cycle, the HVAC module drives the temperature blend door until the door reaches both internal stops in the HVAC case. If the temperature blend door is temporally obstructed or binding during a calibration cycle, the HVAC module may interpret this as the actual end of travel for the door. When this condition occurs and the HVAC module commands the actuator to it's end of travel, the airflow temperature may not be as warm or cold as expected.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B1081:00 | Left Temperature Damper Motor: No Sub Type Information | The HVAC module senses no changes in the feedback circuit when motor movement is commanded and no motor electrical DTCs are present. This indicates either an open in an actuator motor circuit, an actuator internal failure or a binding or stuck linkage or door. |
| B1081:11 | Left Temperature Damper Motor: Circuit Short to Ground | The HVAC module senses lower than expected voltage on an actuator motor circuit when voltage is applied to drive the motor, indicating a short to ground. |
| B1081:12 | Left Temperature Damper Motor: Circuit Short to Battery | The HVAC module senses greater than expected voltage on the actuator motor circuit when ground is applied to drive the motor, indicating a short to voltage. |
| B1081:13 | Left Temperature Damper Motor: Circuit Open | The HVAC module senses no voltage on the actuator motor circuit when ground is applied to drive the motor, indicating an open circuit. |
| B11E5:11 | Left Temperature Damper Position Sensor: Circuit Short to Ground | The HVAC module senses less than 1 volt on the actuator feedback circuit, indicating a short to ground. |
| B11E5:15 | Left Temperature Damper Position Sensor: Circuit Short to Battery or Open | The HVAC module senses greater than 4.88 volts on the actuator feedback circuit, indicating an open circuit or a short to voltage. |

**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- Temperature blend door actuator
- HVAC module
- Stuck or bound linkage or door

**PINPOINT TEST R : TEMPERATURE CONTROL IS INOPERATIVE/DOES NOT OPERATE CORRECTLY — EMTC BASE WITH 4 BLOWER MOTOR SPEED POSITIONS**

**R1 CHECK THE HVAC MODULE FOR DTCS**

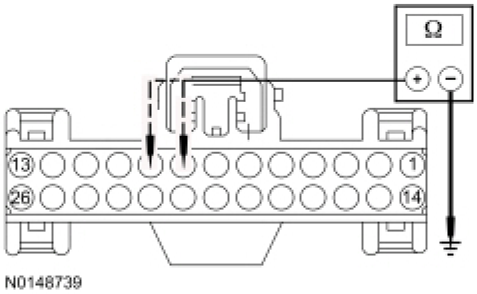
- Ignition ON.
- Check the HVAC module for DTCs.

**Are any DTCs present?**

| | |
|---|---|
| **Yes** | For DTC B1081:00, INSPECT for binding/damaged linkage or door. REPAIR as necessary. If no condition is found, GO to R4.<br>For DTC B1081:11, GO to R2.<br>For DTC B1081:12, GO to R3.<br>For DTC B1081:13, GO to R4.<br>For DTC B11E5:11, GO to R6.<br>For DTC B11E5:15, GO to R9. |
| **No** | INSPECT for broken linkage or door. REPAIR as necessary. If no condition is found, GO to R13. |

**R2 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TO GROUND**

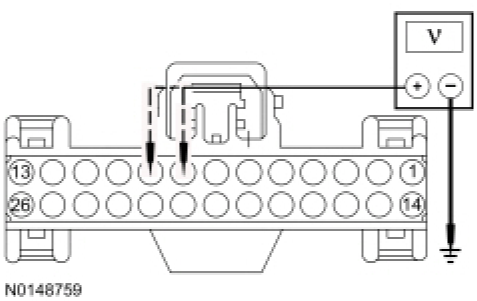
- Ignition OFF.

- Disconnect: HVAC Module — Base EMTC C294A .
- Measure the resistance between ground and:
  - HVAC module — EMTC C294A Pin 9, circuit CH238 (YE/OG), harness side.
  - HVAC module — EMTC C294A Pin 8, circuit CH239 (BU), harness side.
- 


**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to R5. |
| **No** | REPAIR the circuit in question. |

**R3 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Ignition ON.
- Measure the voltage between ground and:
  - HVAC module — EMTC C294A Pin 8, circuit CH239 (BU), harness side.
  - HVAC module — EMTC C294A Pin 9, circuit CH238 (YE/OG), harness side.
- 


**Are any voltages present?**

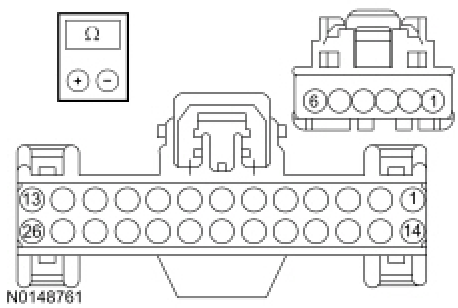
| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |
| **No** | GO to R5. |

**R4 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR MOTOR CIRCUITS FOR AN OPEN**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Disconnect: Temperature Blend Door Actuator C2381 .
- Measure the resistance between HVAC module — EMTC C294A, harness side and temperature blend door actuator C2381, harness side using the following chart:

| HVAC Module — EMTC | Circuit | Temperature Blend Door Actuator |
|---|---|---|
| C294A Pin 9 | CH238 (YE/OG) | C2381 Pin 6 |
| C294A Pin 8 | CH239 (BU) (BU/WH) | C2381 Pin 5 |

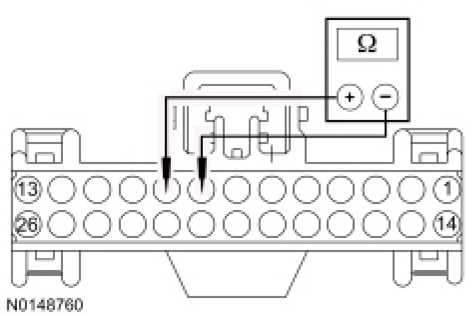
- 


**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to R13. |
| **No** | REPAIR the circuit in question. |

**R5 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TOGETHER**

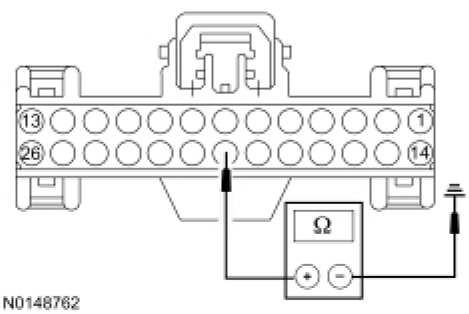
**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Ignition OFF.
- Disconnect: Temperature Blend Door Actuator C2381 .
- Measure the resistance between HVAC module — EMTC C294A Pin 8, circuit CH239 (BU), harness side and HVAC module — EMTC C294A Pin 9, circuit CH238 (YE/OG), harness side.
- 

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to R13. |
| **No** | REPAIR the circuits. |

**R6 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT FOR A SHORT TO GROUND**

- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Measure the resistance between ground and HVAC module — EMTC C294A Pin 20, circuit VH440 (BU), harness side.
- 

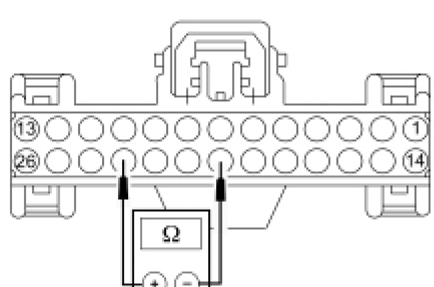
N0148762

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to R7. |
| **No** | REPAIR the circuit. |

**R7 CHECK THE SIGNAL RETURN CIRCUIT FOR A SHORT TO THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

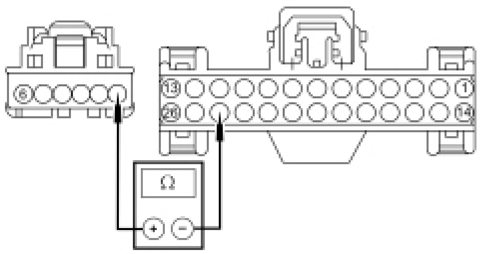
- Disconnect: Temperature Blend Door Actuator C2381 .
- Measure the resistance between HVAC module — EMTC C294A Pin 23, circuit RH111 (GY/BU) and HVAC module — EMTC C294A Pin 20, circuit VH440 (BU), harness side.
- 

N0148763

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to R8. |
| **No** | REPAIR the circuits. |

**R8 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR AN OPEN**

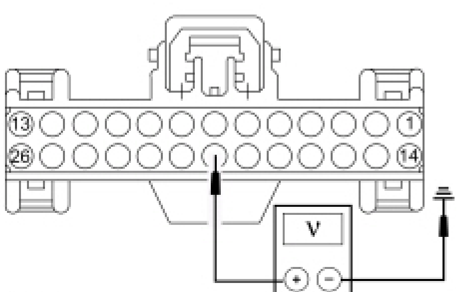
- Measure the resistance between HVAC module — EMTC C294A Pin 24, circuit LH111 (BN/WH) and temperature blend door actuator C2381 Pin 1, circuit LH111 (BN/WH), harness side.
- 

N0148766

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to R13. |
| **No** | REPAIR the circuit. |

**R9 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Ignition ON.
- Measure the voltage between ground and HVAC module — EMTC C294A Pin 20, circuit VH440 (BU), harness side.
- 

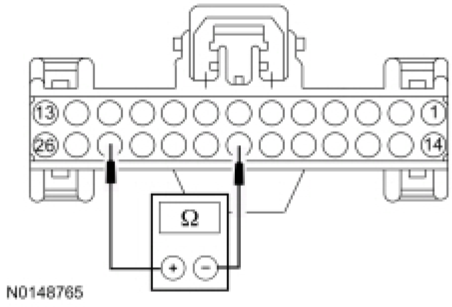
N0148764

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit. |
| **No** | GO to R10. |

**R10 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR A SHORT TO THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT**

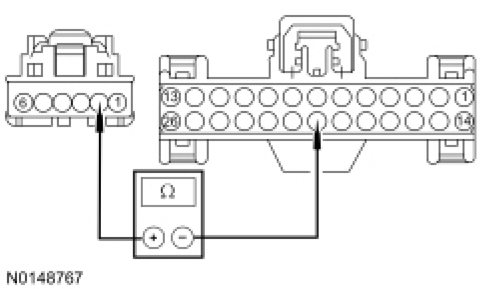
**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Ignition OFF.
- Disconnect: Temperature Blend Door Actuator C2381 .
- Measure the resistance between HVAC module — EMTC C294A Pin 24, circuit LH111 (BN/WH) and HVAC module — EMTC C294A Pin 20, circuit VH440 (BU), harness side.
- 


**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to R11. |
| **No** | REPAIR the circuits. |

**R11 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT FOR AN OPEN**

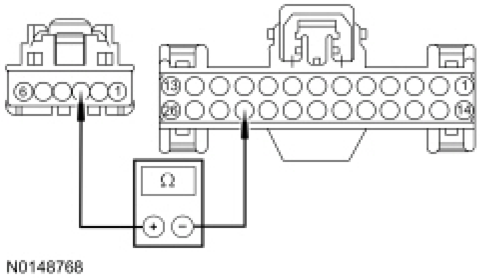
- Measure the resistance between HVAC module — EMTC C294A Pin 20, circuit VH440 (BU), harness side and driver side blend door actuator C2381 Pin 2, circuit VH440 (BU/BN), harness side.
- 


**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to R12. |
| **No** | REPAIR the circuit. |

**R12 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR RETURN CIRCUIT FOR AN OPEN**

- Measure the resistance between HVAC module — EMTC C294A Pin 23, circuit RH111 (GY/BU) and temperature blend door actuator C2381 Pin 3, circuit RH111 (GY/BU), harness side.

- 


**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to R13. |
| **No** | REPAIR the circuit. |

**R13 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR CONNECTION**

- Disconnect and inspect the temperature blend door actuator connector.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Reconnect the temperature blend door actuator connector. Make sure it seats and latches correctly.
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Clear the DTCs.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CARRY OUT the Temperature Blend Door Actuator component test in this section. If the actuator tests OK, GO to R14. When installing an actuator, CONNECT the actuator electrical connector before the HVAC module connector(s). This allows the actuator to be calibrated when the HVAC module is reconnected. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

**R14 MODULE ACTUATOR POSITION CALIBRATION**

**NOTE:** *The purpose of the module actuator position calibration is to allow the HVAC module to reinitialize and calibrate the actuator stop points. To carry out the calibration, follow the steps below.*

- Ignition OFF.
- Disconnect and inspect all HVAC module connectors.
- Repair:

- corrosion (install new connector or terminals - clean module pins)
- damaged or bent pins - install new terminals/pins
- pushed-out pins - install new pins as necessary

- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Ignition ON.
- Clear the DTCs.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Select any position except OFF.
- **NOTE:** *The HVAC module will now initialize and calibrate the actuators. Calibration of the actuators takes approximately 30 seconds.*

  Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC module. REFER to Section 412-01. |
| **No** | The system is now operating correctly at this time. The concern may have been caused by a foreign object in the HVAC case or temporary binding that restricted actuator door travel. CHECK any actuator external linkage. If condition recurs, INSPECT actuator linkage and door for binding and CHECK HVAC case for foreign objects. |

### Pinpoint Test S: Temperature Control Is Inoperative/Does Not Operate Correctly — EMTC With 4.2 Inch Display With 7 Blower Motor Speed Positions

Refer to Wiring Diagrams Cell 54 , Automatic Climate Control System for schematic and connector information.

Refer to Wiring Diagrams Cell 55 , Automatic Climate Control System for schematic and connector information.

**Normal Operation**

To rotate the temperature blend door actuator, the HVAC module supplies voltage and ground to the temperature blend door actuators through the door actuator motor circuits. To reverse the temperature blend door actuator rotation, the HVAC module reverses the voltage and ground circuits.

The temperature blend door actuator feedback resistor is supplied a ground from the HVAC module and a 5-volt reference voltage on the temperature blend door actuator reference circuits. The HVAC module reads the voltage on the temperature blend door actuator feedback circuits to determine the temperature blend door actuator position by the position of the actuator feedback resistor wiper arm.

During an actuator calibration cycle, the HVAC module drives the temperature blend door until the door reaches both internal stops in the HVAC case. If the temperature blend door is temporally obstructed or binding during a calibration cycle, the HVAC module may interpret this as the actual end of travel for the door. When this condition occurs and the HVAC module commands the actuator to it's end of travel, the airflow temperature may not be as warm or cold as expected.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B1081:00 | Left Temperature Damper Motor: No Sub Type Information | The HVAC module senses no changes in the feedback circuit when motor movement is commanded and no motor electrical DTCs are present. This indicates either an open in an actuator motor circuit, an actuator internal failure or a binding or stuck linkage or door. |
| B1081:11 | Left Temperature Damper Motor: Circuit Short to Ground | The HVAC module senses lower than expected voltage on an actuator motor circuit when voltage is applied to drive the motor, indicating a short to ground. |
| B1081:12 | Left Temperature Damper Motor: Circuit Short to Battery | The HVAC module senses greater than expected voltage on the actuator motor circuit when ground is applied to drive the motor, indicating a short to voltage. |
| B1081:13 | Left Temperature Damper Motor: Circuit Open | The HVAC module senses no voltage on the actuator motor circuit when ground is applied to drive the motor, indicating an open circuit. |
| B11E5:11 | Left Temperature Damper Position Sensor: Circuit Short to Ground | The HVAC module senses less than 1 volt on the actuator feedback circuit, indicating a short to ground. |
| B11E5:15 | Left Temperature Damper Position Sensor: Circuit Short to Battery or Open | The HVAC module senses greater than 4.88 volts on the actuator feedback circuit, indicating an open circuit or a short to voltage. |

**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- Temperature blend door actuator
- HVAC module
- Front Controls Interface Module (FCIM)
- Stuck or bound linkage or door

**PINPOINT TEST S : TEMPERATURE CONTROL IS INOPERATIVE/DOES NOT OPERATE CORRECTLY — EMTC WITH 4.2 INCH DISPLAY WITH 7 BLOWER MOTOR SPEED POSITIONS**

**S1 CHECK THE HVAC MODULE FOR DTCS**

- Ignition ON.
- Check the HVAC module for DTCs.

**Are any DTCs present?**

| | |
|---|---|
| **Yes** | For DTC B1081:00, INSPECT for binding/damaged linkage or door. REPAIR as necessary. If no condition is found, GO to S5.<br>For DTC B1081:11, GO to S3.<br>For DTC B1081:12, GO to S4.<br>For DTC B1081:13, GO to S5.<br>For DTC B11E5:11, GO to S7.<br>For DTC B11E5:15, GO to S10. |
| **No** | GO to S2. |

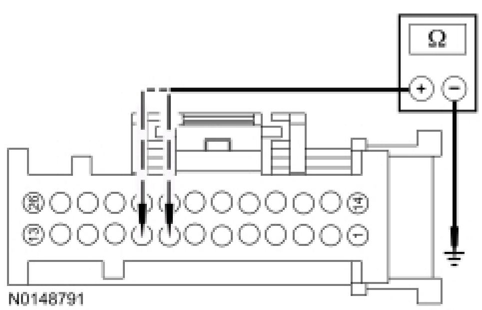
**S2 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR OPERATION USING THE HVAC LEFT BLEND DOOR POSITION (LEFT_BLEND) ACTIVE COMMAND**

- Start the engine.
- Run the engine until it reaches normal operating temperature.
- Enter the following diagnostic mode on the scan tool: HVAC Module DataLogger .
- Select PANEL mode on the HVAC controls.
- Select the highest blower motor setting.
- Observe airflow while using the scan tool to operate the HVAC module active command LEFT_BLEND, increasing and decreasing the percentage value.

**Does the airflow temperature change between warm and cold when the active command is changed?**

| | |
|---|---|
| **Yes** | INSTALL a new FCIM . REFER to Refer to the appropriate section in Group 415 for the procedure.. TEST the system for normal operation. |
| **No** | INSPECT for broken linkage or door. REPAIR as necessary. If no condition is found, GO to S14. |

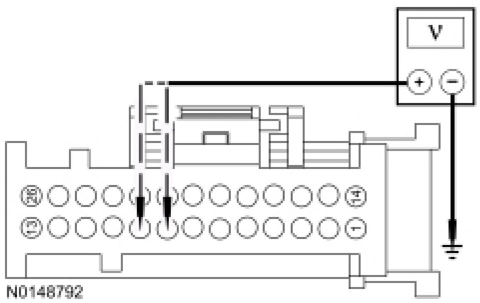
**S3 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TO GROUND**

- Ignition OFF.
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357B .
- Disconnect: Temperature Blend Door Actuator C2381 .
- Measure the resistance between ground and:
  - HVAC module C2357A Pin 8, circuit CH239 (BU), harness side.
  - HVAC module C2357A Pin 9, circuit CH238 (YE/OG), harness side.
- 


**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to S6. |
| **No** | REPAIR the circuit in question. |

## S4 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TO VOLTAGE

- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357B .
- Disconnect: Temperature Blend Door Actuator C2381 .
- Ignition ON.
- Measure the voltage between ground and:
  - HVAC module C2357A Pin 8, circuit CH239 (BU), harness side.
  - HVAC module C2357A Pin 9, circuit CH238 (YE/OG), harness side.
- 


**Are any voltages present?**

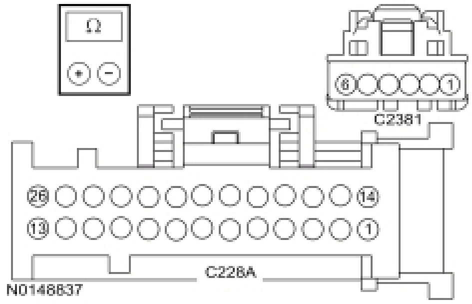
| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |
| **No** | GO to S6. |

## S5 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR MOTOR CIRCUITS FOR AN OPEN

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Ignition OFF.
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357B .
- Disconnect: Temperature Blend Door Actuator C2381 .
- Measure the resistance between HVAC module C228A, harness side and driver side temperature blend door actuator C2381, harness side using the following chart:

| HVAC Module | Circuit | Temperature Blend Door Actuator |
|---|---|---|
| C2357A Pin 9 | CH238 (YE) (YE/OG) | C2381 Pin 6 |
| C2357A Pin 8 | CH239 (BU) (BU/WH) | C2381 Pin 5 |

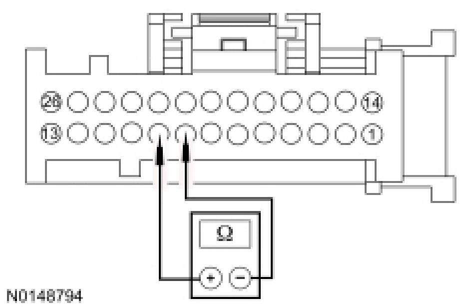
- 


**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to S14. |
| **No** | REPAIR the circuit in question. |

**S6 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TOGETHER**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Ignition OFF.
- Disconnect: Temperature Blend Door Actuator C2381 .
- Measure the resistance between HVAC module C2357A Pin 8, circuit CH239 (BU), harness side and HVAC module C2357A Pin 9, circuit CH238 (YE), harness side.
- 

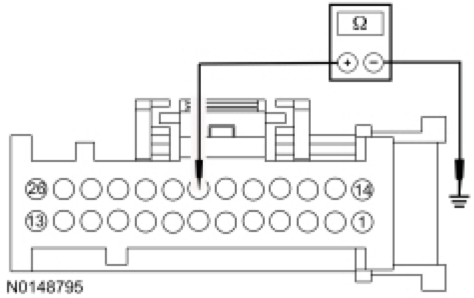

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to S14. |
| **No** | REPAIR the circuits. |

**S7 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT FOR A SHORT TO GROUND**

- Ignition OFF.
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357B .
- Disconnect: Temperature Blend Door Actuator C2381 .

- Measure the resistance between ground and HVAC module C2357A Pin 20, circuit VH440 (BU), harness side.

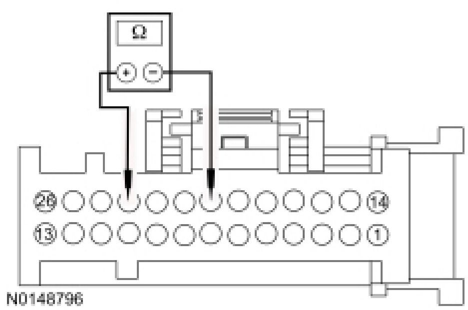
- 


**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to S8. |
| **No** | REPAIR the circuit. |

**S8 CHECK THE SIGNAL RETURN CIRCUIT FOR A SHORT TO THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

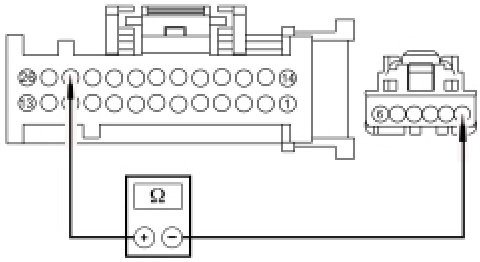
- Disconnect: Temperature Blend Door Actuator C2381 .
- Measure the resistance between HVAC module C2357A Pin 23, circuit RH111 (GY/BU) and HVAC module C2357A Pin 20, circuit VH440 (BU), harness side.
- 


**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to S9. |
| **No** | REPAIR the circuits. |

**S9 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR AN OPEN**

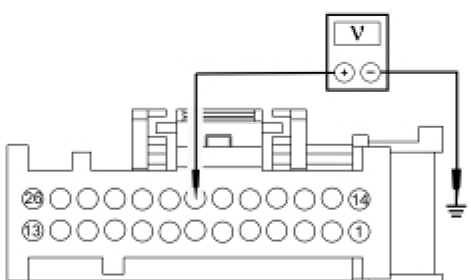
- Measure the resistance between HVAC module C2357A Pin 24, circuit LH111 (BN/WH) and temperature blend door actuator C2381 Pin 1, circuit LH111 (BN/WH), harness side.

- 
  
  N0148798

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to S14. |
| **No** | REPAIR the circuit. |

**S10 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357B .
- Ignition ON.
- Measure the voltage between ground and HVAC module C2357A Pin 20, circuit VH440 (BU), harness side.
- 
  
  N0148797

**Is any voltage present?**

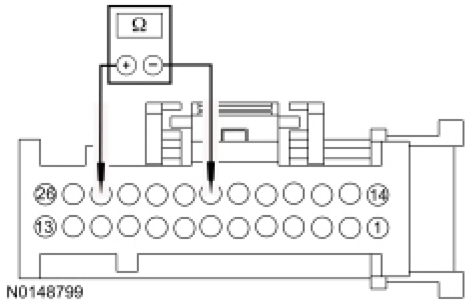
| | |
|---|---|
| **Yes** | REPAIR the circuit. |
| **No** | GO to S11. |

**S11 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR A SHORT TO THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Ignition OFF.
- Disconnect: Temperature Blend Door Actuator C2381 .

- Measure the resistance between HVAC module C2357A Pin 24, circuit LH111 (BN/WH) and HVAC module C2357A Pin 20, circuit VH440 (BU), harness side.

- 

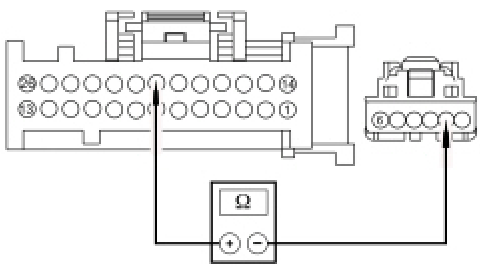
N0148799

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to S12. |
| **No** | REPAIR the circuits. |

**S12 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT FOR AN OPEN**

- Measure the resistance between HVAC module C2357A Pin 20, circuit VH440 (BU), harness side and temperature blend door actuator C2381 Pin 2, circuit VH440 (BU/BN), harness side.

- 

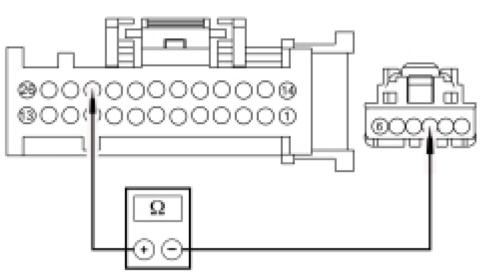
N0148802

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to S13. |
| **No** | REPAIR the circuit. |

**S13 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR RETURN CIRCUIT FOR AN OPEN**

- Measure the resistance between HVAC module C2357A Pin 23, circuit RH111 (GY/BU) and temperature blend door actuator C2381 Pin 3, circuit RH111 (GY/BU), harness side.

- 

N0148801

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to S14. |
| **No** | REPAIR the circuit. |

**S14 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR CONNECTION**

- Disconnect and inspect the temperature blend door actuator connector.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Reconnect the temperature blend door actuator connector. Make sure it seats and latches correctly.
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Clear the DTCs.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CARRY OUT the Temperature Blend Door Actuator component test in this section. If the actuator tests OK, GO to S15. When installing an actuator, CONNECT the actuator electrical connector before the HVAC module connector(s). This allows the actuator to be calibrated when the HVAC module is reconnected. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

**S15 MODULE ACTUATOR POSITION CALIBRATION**

**NOTE:** *The purpose of the module actuator position calibration is to allow the HVAC module to reinitialize and calibrate the actuator stop points. To carry out the calibration, follow the steps below.*

- Ignition OFF.
- Disconnect and inspect all HVAC module connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Ignition ON.
- Clear the DTCs.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .

- Select any position except OFF.
- **NOTE:** *The HVAC module will now initialize and calibrate the actuators. Calibration of the actuators takes approximately 30 seconds.*

  Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC module. REFER to Section 412-01. |
| **No** | The system is now operating correctly at this time. The concern may have been caused by a foreign object in the HVAC case or temporary binding that restricted actuator door travel. CHECK any actuator external linkage. If condition recurs, INSPECT actuator linkage and door for binding and CHECK HVAC case for foreign objects. |

### Pinpoint Test U: Temperature Control Is Inoperative/Does Not Operate Correctly — EATC , Dual Zone Driver Side

Refer to Wiring Diagrams Cell 55 , Automatic Climate Control System for schematic and connector information.

**Normal Operation**

To rotate the temperature blend door actuator, the HVAC module supplies voltage and ground to the temperature blend door actuators through the door actuator motor circuits. To reverse the temperature blend door actuator rotation, the HVAC module reverses the voltage and ground circuits.

The temperature blend door actuator feedback resistor is supplied a ground from the HVAC module and a 5-volt reference voltage on the temperature blend door actuator reference circuits. The HVAC module reads the voltage on the temperature blend door actuator feedback circuits to determine the temperature blend door actuator position by the position of the actuator feedback resistor wiper arm.

During an actuator calibration cycle, the HVAC module drives the temperature blend door until the door reaches both internal stops in the HVAC case. If the temperature blend door is temporally obstructed or binding during a calibration cycle, the HVAC module may interpret this as the actual end of travel for the door. When this condition occurs and the HVAC module commands the actuator to it's end of travel, the airflow temperature may not be as warm or cold as expected.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B1081:00 | Left Temperature Damper Motor: No Sub Type Information | The HVAC module senses no changes in the feedback circuit when motor movement is commanded and no motor electrical DTCs are present. This indicates either an open in an actuator motor circuit, an actuator internal failure or a binding or stuck linkage or door. |
| B1081:11 | Left Temperature Damper Motor: Circuit Short to Ground | The HVAC module senses lower than expected voltage on an actuator motor circuit when voltage is applied to drive the motor, indicating a short to ground. |

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B1081:12 | Left Temperature Damper Motor: Circuit Short to Battery | The HVAC module senses greater than expected voltage on the actuator motor circuit when ground is applied to drive the motor, indicating a short to voltage. |
| B1081:13 | Left Temperature Damper Motor: Circuit Open | The HVAC module senses no voltage on the actuator motor circuit when ground is applied to drive the motor, indicating an open circuit. |
| B11E5:11 | Left Temperature Damper Position Sensor: Circuit Short to Ground | The HVAC module senses less than 1 volt on the actuator feedback circuit, indicating a short to ground. |
| B11E5:15 | Left Temperature Damper Position Sensor: Circuit Short to Battery or Open | The HVAC module senses greater than 4.88 volts on the actuator feedback circuit, indicating an open circuit or a short to voltage. |

**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- Temperature blend door actuator
- HVAC module
- Front Controls Interface Module (FCIM)
- Stuck or bound linkage or door

**PINPOINT TEST U : TEMPERATURE CONTROL IS INOPERATIVE/DOES NOT OPERATE CORRECTLY — EATC , DUAL ZONE DRIVER SIDE**

**U1 CHECK THE HVAC MODULE FOR DTCS**

- Ignition ON.
- Check the HVAC module for DTCs.

**Are any DTCs present?**

| | |
|---|---|
| **Yes** | For DTC B1081:00, INSPECT for binding/damaged linkage or door. REPAIR as necessary. If no condition is found, GO to U5.<br>For DTC B1081:11, GO to U3.<br>For DTC B1081:12, GO to U4.<br>For DTC B1081:13, GO to U5.<br>For DTC B11E5:11, GO to U7.<br>For DTC B11E5:15, GO to U10. |
| **No** | GO to U2. |

**U2 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR OPERATION USING THE HVAC LEFT BLEND DOOR POSITION (LEFT_BLEND) ACTIVE COMMAND**

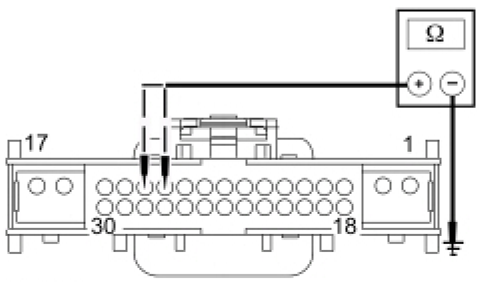
- Start the engine.
- Run the engine until it reaches normal operating temperature.
- Enter the following diagnostic mode on the scan tool: HVAC Module DataLogger .
- Select PANEL mode on the HVAC controls.
- Select the highest blower motor setting.

- Observe airflow while using the scan tool to operate the HVAC module active command LEFT_BLEND, increasing and decreasing the percentage value.

**Does the airflow temperature change between warm and cold when the active command is changed?**

| | |
|---|---|
| **Yes** | INSTALL a new FCIM . REFER to Refer to the appropriate section in Group 415 for the procedure.. TEST the system for normal operation. |
| **No** | INSPECT for broken linkage or door. REPAIR as necessary. If no condition is found, GO to U14. |

**U3 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TO GROUND**

- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Disconnect: Driver Side Temperature Blend Door Actuator C2091 .
- Measure the resistance between ground and:
  - HVAC module — EATC C228B Pin 12, circuit CH239 (BU), harness side.
  - HVAC module — EATC C228B Pin 13, circuit CH238 (YE), harness side.
- 

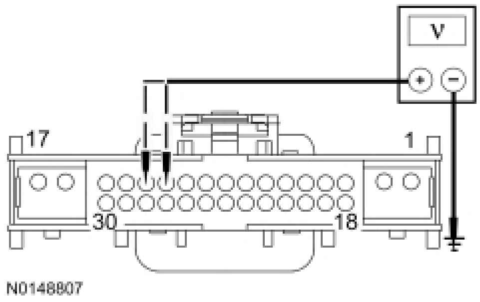
N0148806

**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to U6. |
| **No** | REPAIR the circuit in question. |

**U4 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TO VOLTAGE**

- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Disconnect: Driver Side Temperature Blend Door Actuator C2091 .
- Ignition ON.
- Measure the voltage between ground and:
  - HVAC module — EATC C228B Pin 12, circuit CH239 (BU), harness side.
  - HVAC module — EATC C228B Pin 13, circuit CH238 (YE), harness side.

- 


**Are any voltages present?**

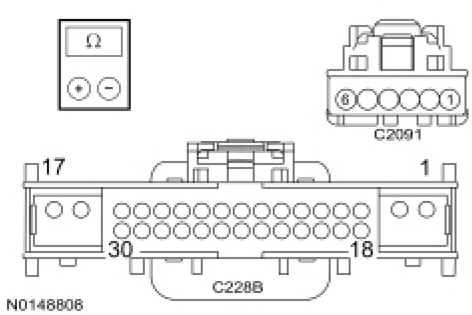
| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |
| **No** | GO to U6. |

| U5 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR MOTOR CIRCUITS FOR AN OPEN |
|---|

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Disconnect: Driver Side Temperature Blend Door Actuator C2091 .
- Measure the resistance between HVAC module — EATC C228B, harness side and driver side temperature blend door actuator C2091, harness side using the following chart:

| HVAC Module — EATC | Circuit | Driver side Temperature Blend Door Actuator |
|---|---|---|
| C228B Pin 13 | CH238 (YE) (YE/OG) | C2091 Pin 6 |
| C228B Pin 12 | CH239 (BU) (BU/WH) | C2091 Pin 5 |

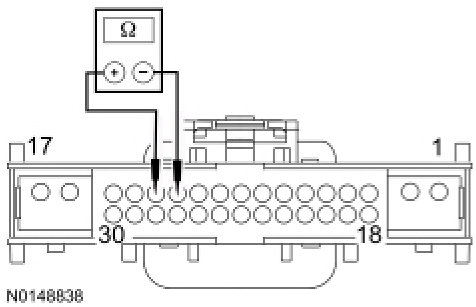
- 


**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to U14. |
| **No** | REPAIR the circuit in question. |

## U6 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TOGETHER

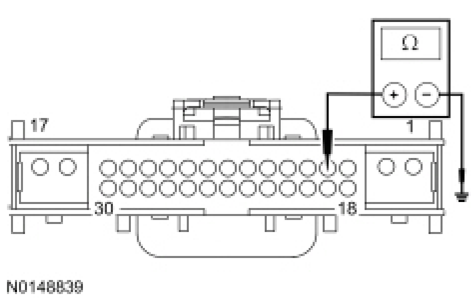
**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Ignition OFF.
- Disconnect: Driver side Temperature Blend Door Actuator C2091 .
- Measure the resistance between HVAC module — EATC C228B Pin 12, circuit CH239 (BU), harness side and HVAC module — EATC C228B Pin 13, circuit CH238 (YE), harness side.
- 


**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to U14. |
| **No** | REPAIR the circuits. |

## U7 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT FOR A SHORT TO GROUND

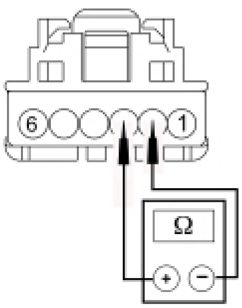
- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Disconnect: Driver Side Temperature Blend Door Actuator C2091 .
- Measure the resistance between ground and HVAC module — EATC C228B Pin 4, circuit VH440 (BU), harness side.
- 


**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to U8. |
| **No** | REPAIR the circuit. |

**U8 CHECK THE SIGNAL RETURN CIRCUIT FOR A SHORT TO THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

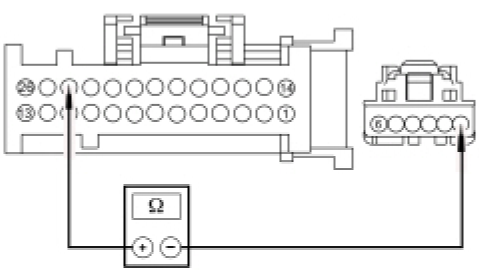
- Disconnect: Driver Side Temperature Blend Door Actuator C2091 .
- Measure the resistance between EATC C2091 Pin 3, circuit RH111 (GY/BU) and EATC C2091 Pin 2, circuit VH440 (BU/BN), harness side.
- 

N0148840

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to U9. |
| **No** | REPAIR the circuits. |

**U9 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR AN OPEN**

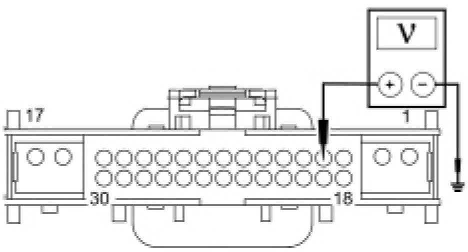
- Measure the resistance between HVAC module — EATC C228A Pin 24, circuit LH111 (BN/WH) and driver side temperature blend door actuator C2091 Pin 1, circuit LH111 (BN/WH), harness side.
- 

N0148798

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to U14. |
| **No** | REPAIR the circuit. |

**U10 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Ignition ON.
- Measure the voltage between ground and HVAC module — EATC C228B Pin 4, circuit VH440 (BU/BN), harness side.
- 

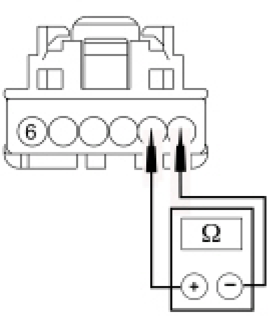
N0148841

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit. |
| **No** | GO to U11. |

**U11 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR A SHORT TO THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Ignition OFF.
- Disconnect: Driver side Temperature Blend Door Actuator C2091 .
- Measure the resistance between HVAC module C2091 Pin 1, circuit LH111 (BN/WH) and HVAC module — EATC C2091 Pin 2, circuit VH440 (BU/BN), harness side.
- 

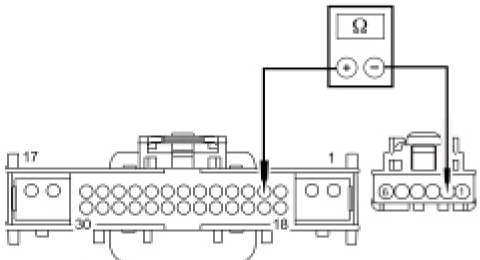
N0148844

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to U12. |

| | |
|---|---|
| **No** | REPAIR the circuits. |

**U12 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT FOR AN OPEN**

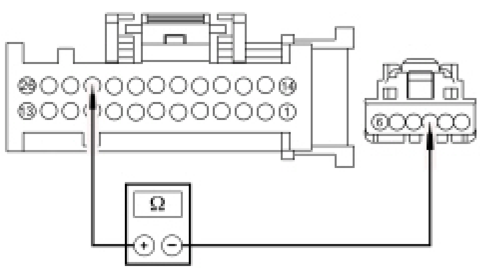
- Measure the resistance between HVAC module — EATC C228B Pin 4, circuit VH440 (BU), harness side and driver side temperature blend door actuator C2091 Pin 2, circuit VH440 (BU/BN), harness side.
- 

  N0148845

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to U13. |
| **No** | REPAIR the circuit. |

**U13 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR RETURN CIRCUIT FOR AN OPEN**

- Measure the resistance between HVAC module — EATC C228A Pin 23, circuit RH111 (GY/BU) and driver side temperature blend door actuator C2091 Pin 3, circuit RH111 (GY/BU), harness side.
- 

  N0148801

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to U14. |
| **No** | REPAIR the circuit. |

**U14 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR CONNECTION**

- Disconnect and inspect the temperature blend door actuator connector.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary

- Reconnect the temperature blend door actuator connector. Make sure it seats and latches correctly.
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Clear the DTCs.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CARRY OUT the Temperature Blend Door Actuator component test in this section. If the actuator tests OK, GO to U15. When installing an actuator, CONNECT the actuator electrical connector before the HVAC module connector(s). This allows the actuator to be calibrated when the HVAC module is reconnected. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

**U15 MODULE ACTUATOR POSITION CALIBRATION**

**NOTE:** *The purpose of the module actuator position calibration is to allow the HVAC module to reinitialize and calibrate the actuator stop points. To carry out the calibration, follow the steps below.*

- Ignition OFF.
- Disconnect and inspect all HVAC module connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Ignition ON.
- Clear the DTCs.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Select any position except OFF.
- **NOTE:** *The HVAC module will now initialize and calibrate the actuators. Calibration of the actuators takes approximately 30 seconds.*

  Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC module. REFER to Section 412-01. |
| **No** | The system is now operating correctly at this time. The concern may have been caused by a foreign object in the HVAC case or temporary binding that restricted actuator door travel. CHECK any |

| | |
|---|---|
| | actuator external linkage. If condition recurs, INSPECT actuator linkage and door for binding and CHECK HVAC case for foreign objects. |

**Pinpoint Test V: Temperature Control is Inoperative/Does Not Operate Correctly — EATC , Dual Zone Passenger Side**

Refer to Wiring Diagrams Cell 55 , Automatic Climate Control System for schematic and connector information.

**Normal Operation**

To rotate the temperature blend door actuator, the HVAC module supplies voltage and ground to the temperature blend door actuators through the door actuator motor circuits. To reverse the temperature blend door actuator rotation, the HVAC module reverses the voltage and ground circuits.

The temperature blend door actuator feedback resistor is supplied a ground from the HVAC module and a 5-volt reference voltage on the temperature blend door actuator reference circuits. The HVAC module reads the voltage on the temperature blend door actuator feedback circuits to determine the temperature blend door actuator position by the position of the actuator feedback resistor wiper arm.

During an actuator calibration cycle, the HVAC module drives the temperature blend door until the door reaches both internal stops in the HVAC case. If the temperature blend door is temporally obstructed or binding during a calibration cycle, the HVAC module may interpret this as the actual end of travel for the door. When this condition occurs and the HVAC module commands the actuator to it's end of travel, the airflow temperature may not be as warm or cold as expected.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B1082:00 | Right Temperature Damper Motor: No Sub Type Information | The HVAC module senses no changes in the feedback circuit when motor movement is commanded and no motor electrical DTCs are present. This indicates either an open in an actuator motor circuit, an actuator internal failure or a binding or stuck linkage or door. |
| B1082:11 | Right Temperature Damper Motor: Circuit Short to Ground | The HVAC module senses lower than expected voltage on an actuator motor circuit when voltage is applied to drive the motor, indicating a short to ground. |
| B1082:12 | Right Temperature Damper Motor: Circuit Short to Battery | The HVAC module senses greater than expected voltage on the actuator motor circuit when ground is applied to drive the motor, indicating a short to voltage. |
| B1082:13 | Right Temperature Damper Motor: Circuit Open | The HVAC module senses no voltage on the actuator motor circuit when ground is applied to drive the motor, indicating an open circuit. |
| B11E6:11 | Right Temperature Damper Position Sensor: Circuit Short to Ground | The HVAC module senses less than 1 volt on the actuator feedback circuit, indicating a short to ground. |
| B11E6:15 | Right Temperature Damper Position Sensor: Circuit Short to Battery or Open | The HVAC module senses greater than 4.88 volts on the actuator feedback circuit, indicating an open circuit or a short to voltage. |

**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- Temperature blend door actuator
- HVAC module
- Front Controls Interface Module (FCIM)
- Stuck or bound linkage or door

**PINPOINT TEST V : TEMPERATURE CONTROL IS INOPERATIVE/DOES NOT OPERATE CORRECTLY — EATC , PASSENGER SIDE**

**V1 CHECK THE HVAC MODULE FOR DTCS**

- Ignition ON.
- Check the HVAC module for DTCs.

**Are any DTCs present?**

| | |
|---|---|
| **Yes** | For DTC B1082:00, INSPECT for binding/damaged linkage or door. REPAIR as necessary. If no condition is found, GO to V5.<br>For DTC B1082:11, GO to V3.<br>For DTC B1082:12, GO to V4.<br>For DTC B1082:13, GO to V5.<br>For DTC B11E6:11, GO to V7.<br>For DTC B11E6:15, GO to V10. |
| **No** | GO to V2. |

**V2 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR OPERATION USING THE HVAC RIGHT BLEND DOOR POSITION (RIGHT_BLEND) ACTIVE COMMAND**

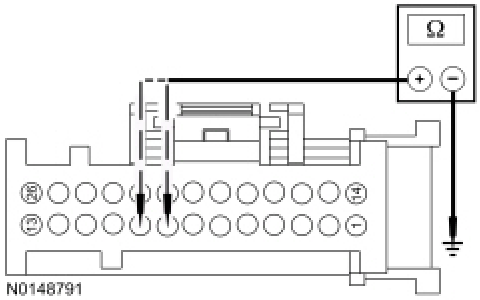
- Start the engine.
- Run the engine until it reaches normal operating temperature.
- Enter the following diagnostic mode on the scan tool: HVAC Module DataLogger .
- Select PANEL mode on the HVAC controls.
- Select the highest blower motor setting.
- Observe airflow while using the scan tool to operate the HVAC module active command RIGHT_BLEND, increasing and decreasing the percentage value.

**Does the airflow temperature change between warm and cold when the active command is changed?**

| | |
|---|---|
| **Yes** | INSTALL a new FCIM . REFER to Refer to the appropriate section in Group 415 for the procedure.. TEST the system for normal operation. |
| **No** | INSPECT for broken linkage or door. REPAIR as necessary. If no condition is found, GO to V14. |

**V3 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TO GROUND**

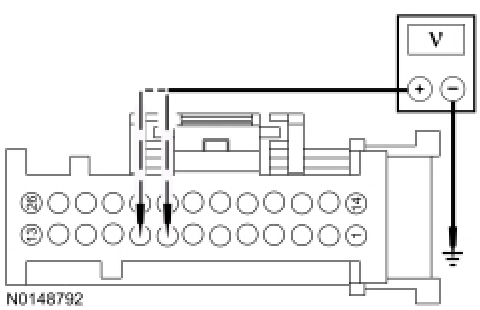
- Ignition OFF.

- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Measure the resistance between ground and:
  - HVAC module — EATC C228A Pin 8, circuit CH213 (BN/GN), harness side.
  - HVAC module — EATC C228A Pin 9, circuit CH212 (BU/OG), harness side.
- 


**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to V6. |
| **No** | REPAIR the circuit in question. |

**V4 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TO VOLTAGE**

- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Ignition ON.
- Measure the voltage between ground and:
  - HVAC module — EATC C228A Pin 8, circuit CH213 (BN/GN), harness side.
  - HVAC module — EATC C228A Pin 9, circuit CH212 (BU/OG), harness side.
- 


**Are any voltages present?**

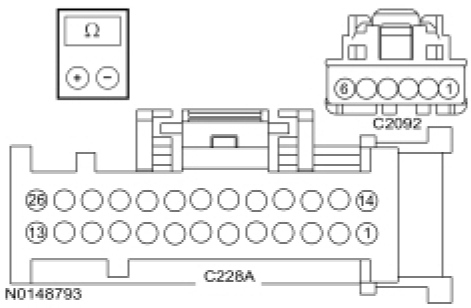
| | |
|---|---|
| **Yes** | REPAIR the circuit in question. |
| **No** | GO to V6. |

**V5 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR MOTOR CIRCUITS FOR AN OPEN**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Disconnect: Passenger side Temperature Blend Door Actuator C2092 .
- Measure the resistance between HVAC module — EATC C228A, harness side and passenger side temperature blend door actuator C2092, harness side using the following chart:

| HVAC Module — EATC | Circuit | Passenger side Temperature Blend Door Actuator |
|---|---|---|
| C228A Pin 9 | CH212 (BU/OG) | C2092 Pin 6 |
| C228A Pin 8 | CH213 (BN/GN) | C2092 Pin 5 |

- 

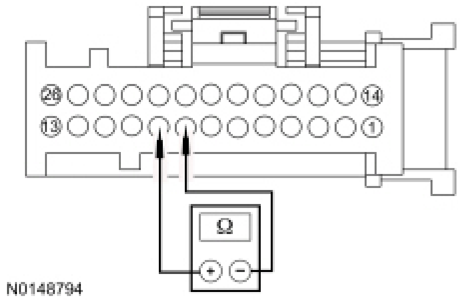

**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to V14. |
| **No** | REPAIR the circuit in question. |

**V6 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR MOTOR CIRCUITS FOR A SHORT TOGETHER**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

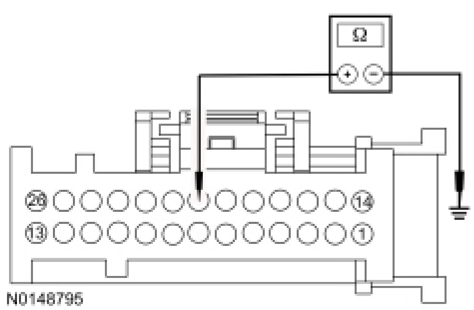
- Ignition OFF.
- Disconnect: Passenger side Temperature Blend Door Actuator C2092 .
- Measure the resistance between HVAC module — EATC C228A Pin 8, circuit CH213 (BN/GN), harness side and HVAC module — EATC C228A Pin 9, circuit CH212 (BU/OG), harness side.

- 


**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to V14. |
| **No** | REPAIR the circuits. |

**V7 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT FOR A SHORT TO GROUND**

- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Measure the resistance between ground and HVAC module — EATC C228A Pin 20, circuit VH441 (WH), harness side.
- 

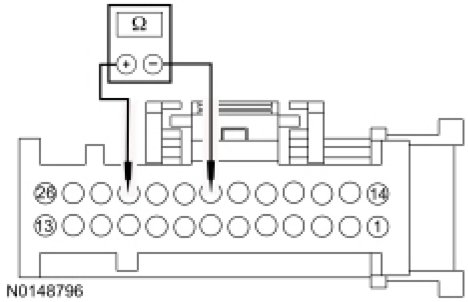

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to V8. |
| **No** | REPAIR the circuit. |

**V8 CHECK THE SIGNAL RETURN CIRCUIT FOR A SHORT TO THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

- Disconnect: Passenger side Temperature Blend Door Actuator C2092 .
- Measure the resistance between HVAC module — EATC C228A Pin 23, circuit RH111 (GY/BU) and HVAC module — EATC C228A Pin 20, circuit VH441 (WH), harness side.

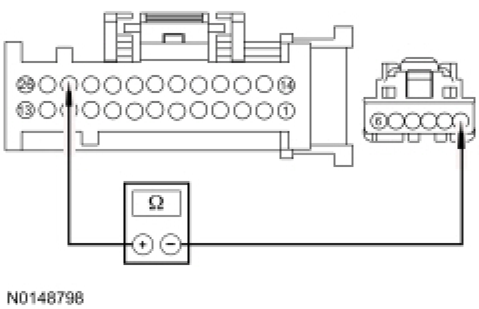
- 


**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to V9. |
| **No** | REPAIR the circuits. |

**V9 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR AN OPEN**

- Measure the resistance between HVAC module — EATC C228A Pin 24, circuit LH111 (BN/WH) and passenger side temperature blend door actuator C2092 Pin 1, circuit LH111 (BN/WH), harness side.

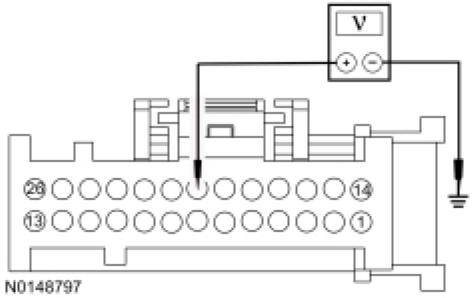
- 


**Are the resistances less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to V14. |
| **No** | REPAIR the circuit. |

**V10 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — EATC C228B .
- Ignition ON.
- Measure the voltage between ground and HVAC module — EATC C228A Pin 20, circuit VH441 (WH), harness side.

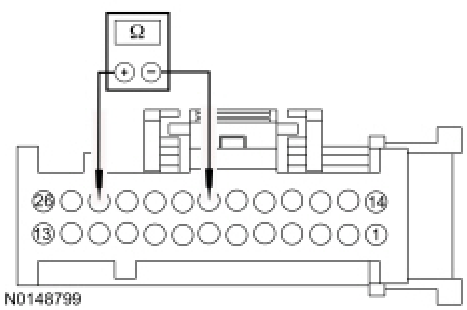
- 


**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit. |
| **No** | GO to V11. |

**V11 CHECK THE REFERENCE VOLTAGE CIRCUIT FOR A SHORT TO THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT**

**NOTE:** *Access to the connector in the following step is difficult. Before performing this step, visually inspect the actuator and its harness for obvious damage. If no damage is evident, proceed with the test.*

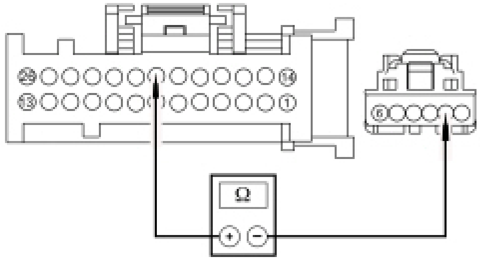
- Ignition OFF.
- Disconnect: Passenger side Temperature Blend Door Actuator C2092 .
- Measure the resistance between HVAC module — C228A Pin 24, circuit LH111 (BN/WH) and HVAC module — EATC C228A Pin 20, circuit VH441 (WH), harness side.
- 


**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to V12. |
| **No** | REPAIR the circuits. |

**V12 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR FEEDBACK CIRCUIT FOR AN OPEN**

- Measure the resistance between HVAC module — EATC C228A Pin 20, circuit VH441 (WH), harness side and passenger side temperature blend door actuator C2092 Pin 2, circuit VH441 (WH), harness side.

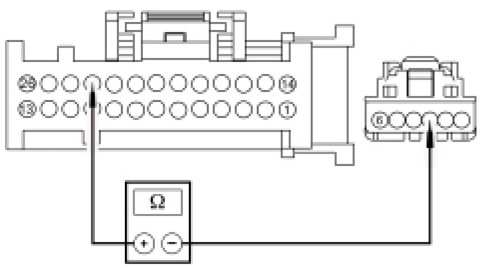



**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to V13. |
| **No** | REPAIR the circuit. |

**V13 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR RETURN CIRCUIT FOR AN OPEN**

- Measure the resistance between HVAC module — EATC C228A Pin 23, circuit RH111 (GY/BU) and passenger side temperature blend door actuator C2092 Pin 3, circuit RH111 (GY/BU), harness side.




**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to V14. |
| **No** | REPAIR the circuit. |

**V14 CHECK THE TEMPERATURE BLEND DOOR ACTUATOR CONNECTION**

- Disconnect and inspect the temperature blend door actuator connector.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Reconnect the temperature blend door actuator connector. Make sure it seats and latches correctly.
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Clear the DTCs.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CARRY OUT the Temperature Blend Door Actuator component test in this section. If the actuator tests OK, GO to V15. When installing an actuator, CONNECT the actuator electrical connector before the HVAC module connector(s). This allows the actuator to be calibrated when the HVAC module is reconnected. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

**V15 MODULE ACTUATOR POSITION CALIBRATION**

**NOTE:** *The purpose of the module actuator position calibration is to allow the HVAC module to reinitialize and calibrate the actuator stop points. To carry out the calibration, follow the steps below.*

- Ignition OFF.
- Disconnect and inspect all HVAC module connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Ignition ON.
- Clear the DTCs.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Select any position except OFF.
- **NOTE:** *The HVAC module will now initialize and calibrate the actuators. Calibration of the actuators takes approximately 30 seconds.*

  Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC module. REFER to Section 412-01. |
| **No** | The system is now operating correctly at this time. The concern may have been caused by a foreign object in the HVAC case or temporary binding that restricted actuator door travel. CHECK any actuator external linkage. If condition recurs, INSPECT actuator linkage and door for binding and CHECK HVAC case for foreign objects. |

**Pinpoint Test W: The Blower Motor Is Inoperative — EMTC Base With 4 Blower Motor Speed Positions**

Refer to Wiring Diagrams Cell 54 , Manual Climate Control System for schematic and connector information.

**Normal Operation**

Voltage is supplied to the relay coil and relay switch contacts from BJB . The coil receives ground from the HVAC module if any function selector position but OFF is selected. When the relay coil is grounded by the HVAC module, the relay is energized and voltage is delivered to the blower motor. Ground for the blower motor is provided from the blower resistor or the blower switch (in HI only). The blower resistor and blower switch are grounded.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B10AF:12 | Blower Fan Relay: Circuit Short to Battery | This DTC sets when the HVAC module senses voltage on the relay coil ground circuit when the relay is grounded by the HVAC module, indicating a short directly to voltage. |
| B10AF:13 | Blower Fan Relay: Circuit Open | This CMDTC sets when the HVAC module senses no voltage on the relay coil ground circuit when the relay is not grounded by the HVAC module, indicating an open. |

**This pinpoint test is intended to diagnose the following:**

- Battery Junction Box (BJB) fuse 51 (40A) and fuse 52 (5A)
- Wiring, terminals or connectors
- Blower motor
- Blower motor relay
- HVAC module

**PINPOINT TEST W : THE BLOWER MOTOR IS INOPERATIVE — EMTC BASE WITH 4 BLOWER MOTOR SPEED POSITIONS**

**W1 CHECK THE HVAC MODULE FOR DTCS**

- Ignition ON.
- Check the HVAC module for DTCs.

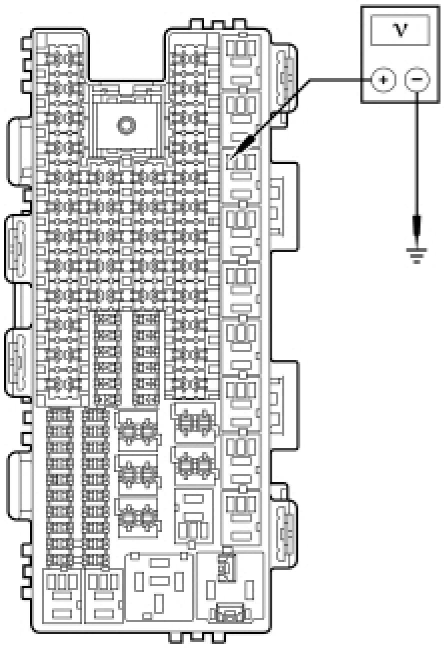
**Are any DTCs present?**

| | |
|---|---|
| **Yes** | For B10AF:12, GO to W2.<br>For B10AF:13, CARRY OUT the blower motor relay component test.<br>Refer to Wiring Diagrams Cell 149 for component testing.<br>If the relay tests OK, GO to W3. |
| **No** | GO to W5. |

**W2 CHECK THE BLOWER MOTOR RELAY COIL SWITCHED GROUND CIRCUIT FOR A SHORT TO VOLTAGE**

- Ignition OFF.

- Disconnect: Blower Motor Relay .
- Disconnect: HVAC Module — Base EMTC C294A .
- Ignition ON.
- Measure the voltage between blower motor relay socket pin 2, circuit CH123 (VT/GN) and ground.

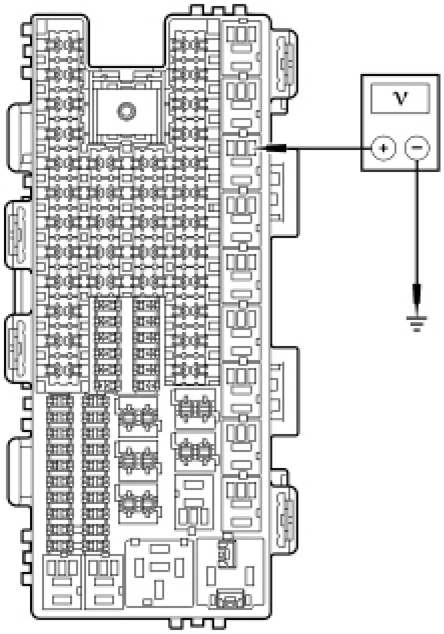
- 
N0122702

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit. |
| **No** | GO to W11. |

**W3 CHECK THE BLOWER MOTOR RELAY COIL SUPPLY VOLTAGE**

- Ignition OFF.
- Disconnect: Blower Motor Relay .
- Ignition ON.
- Measure the voltage between blower motor relay socket pin 1, circuit CBB52 (GN/WH) and ground.



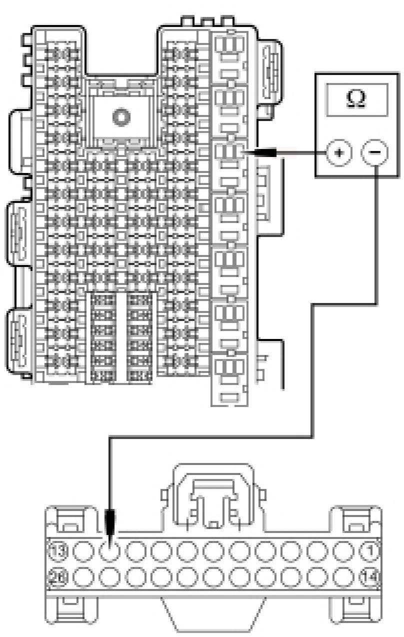

- N0096368

**Is the voltage greater than 11 volts?**

| | |
|---|---|
| **Yes** | GO to W4. |
| **No** | VERIFY Battery Junction Box (BJB) fuse 52 (5A) is OK. If OK, REPAIR circuit CBB52 (GN/WH) for an open. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. |

**W4 CHECK THE BLOWER MOTOR RELAY COIL SWITCHED GROUND CIRCUIT FOR AN OPEN**

- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294A .
- Measure the resistance between blower motor relay socket pin 2, circuit CH123 (VT/GN) and HVAC module — EMTC C294A Pin 11, circuit CH123 (VT), harness side.



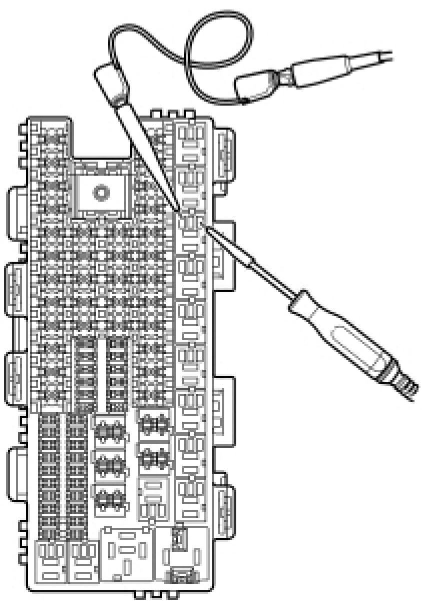

- N0148803

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to W11. |
| **No** | REPAIR the circuit. |

**W5 CHECK THE HVAC MODULE BLOWER MOTOR RELAY OUTPUT**

**NOTICE: The following step uses a test light to simulate normal circuit loads. Use only the test light recommended in the Special Tools table at the beginning of this section or equivalent. To avoid connector terminal damage, use the Flex Probe Kit for the test light probe connection to the vehicle. Do not use the test light probe directly on any connector.**

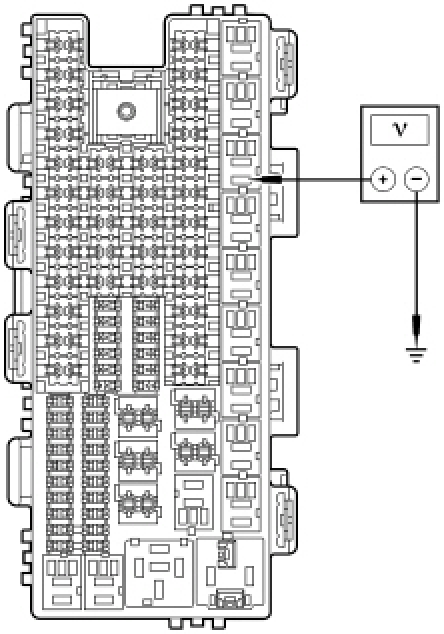
- Ignition OFF.
- Disconnect: Blower Motor Relay .
- Ignition ON.
- Connect the 12-volt test light between blower motor relay socket pin 1, circuit CBB52 (GN/WH) and socket pin 2, circuit CH123 (VT/GN).
- 
  N0096371
- Select PANEL on the HVAC controls.

**Does the test light illuminate?**

| | |
|---|---|
| **Yes** | GO to W6. |
| **No** | GO to W11. |

**W6 CHECK THE BLOWER MOTOR RELAY SWITCH CONTACT VOLTAGE CIRCUIT FOR AN OPEN**

- Measure the voltage between blower motor relay socket pin 3, circuit SBB51 (BU/RD) and ground.



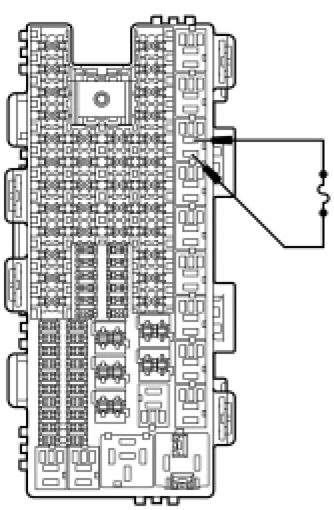

- N0096369

**Is the voltage greater than 11 volts?**

| | |
|---|---|
| **Yes** | GO to W7. |
| **No** | VERIFY Battery Junction Box (BJB) fuse 51 (40A) is OK. If OK, REPAIR circuit SBB51 (BU/RD) for an open. TEST the system for normal operation. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. |

**W7 CHECK THE BLOWER MOTOR RELAY**

- Connect fused jumper lead between blower motor relay socket pin 3, circuit SBB51 (BU/RD) and socket pin 5, circuit CH402 (YE/GN).

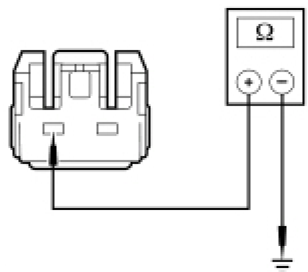


- N0122704

**Does the blower motor operate?**

| | |
|---|---|
| **Yes** | INSTALL a new blower motor relay. TEST the system for normal operation. |
| **No** | GO to W8. |

**W8 CHECK FOR GROUND TO THE BLOWER MOTOR**

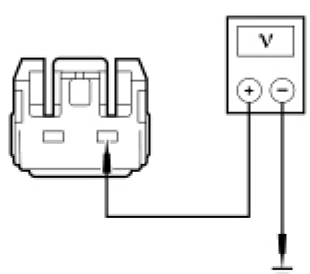
- Ignition OFF.
- Disconnect: Blower Motor C2004 .
- Turn the blower motor switch to the HI position.
- Measure the resistance between blower motor connector C2004 Pin 2, circuit CH426 (YE/OG), harness side and ground.
- 

N0089552

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to W9. |
| **No** | GO to W10. |

**W9 CHECK FOR VOLTAGE TO THE BLOWER MOTOR**

- Connect: Blower Motor Relay .
- Ignition ON.
- Select PANEL on the HVAC controls.
- Measure the voltage between blower motor C2004 Pin 1, circuit CH402 (YE/GN), harness side and ground.
- 

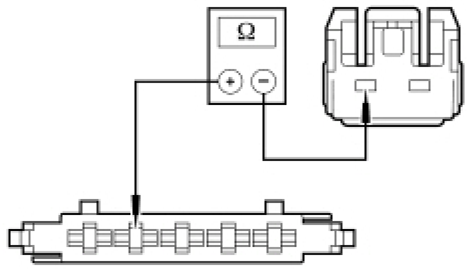
N0089554

**Is the voltage greater than 11 volts?**

| | |
|---|---|
| **Yes** | INSTALL a new blower motor. REFER to Section 412-01. TEST the system for normal operation. |

| | |
|---|---|
| **No** | REPAIR the circuit. |

| **W10 CHECK THE BLOWER MOTOR GROUND CIRCUIT FOR AN OPEN** |
|---|

- Disconnect: HVAC Module — Base EMTC C294B .
- Measure the resistance between HVAC module — EMTC C294B Pin 2, circuit CH426 (YE/OG), harness side and blower motor C2004 Pin 2, circuit CH426 (YE/OG), harness side.
- 
  
  N0089553

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | REPAIR circuit GD138 (BK/WH) for an open. TEST the system for normal operation. |
| **No** | REPAIR circuit CH426 (YE/OG) for an open. TEST the system for normal operation. |

| **W11 CHECK FOR CORRECT HVAC MODULE OPERATION** |
|---|

- Ignition OFF.
- Disconnect and inspect all HVAC module connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Clear the DTCs.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC module. REFER to Section 412-01. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

## Pinpoint Test X: The Blower Motor Does Not Operate Correctly — EMTC Base With 4 Blower Motor Speed Positions

Refer to Wiring Diagrams Cell 54 , Manual Climate Control System for schematic and connector information.

**Normal Operation**

The blower motor is provided a ground through the entire blower resistor in the lowest blower setting. In MED-LO and MED-HI the resistor gets a ground from the blower motor switch. In HI, the blower motor is grounded directly through the blower motor switch. The blower switch is provided a ground.

- DTC B10AF:11 (Blower Fan Relay: Circuit Short to Ground) — the module senses no voltage on the relay coil ground circuit when the module is not grounding the circuit for more than 2 seconds.

**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- Blower motor resistor
- HVAC module

**PINPOINT TEST X : THE BLOWER MOTOR DOES NOT OPERATE CORRECTLY — EMTC BASE WITH 4 BLOWER MOTOR SPEED POSITIONS**

**X1 CHECK THE HVAC MODULE FOR DTCS**

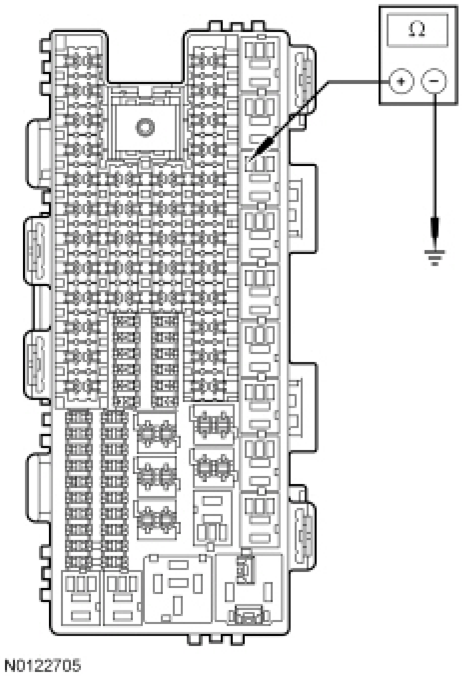
- Ignition ON.
- Check the HVAC module for DTCs.

**Is DTC B10AF:11 present?**

| | |
|---|---|
| **Yes** | GO to X2. |
| **No** | GO to X3. |

**X2 CHECK THE BLOWER MOTOR RELAY COIL SWITCHED GROUND CIRCUIT FOR A SHORT TO GROUND**

- Ignition OFF.
- Disconnect: Blower Motor Relay .
- Disconnect: HVAC Module — Base EMTC C294B .
- Measure the resistance between blower motor relay socket pin 2, circuit CH123 (VT/GN) and ground.




**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to X12. |
| **No** | REPAIR the circuit. |

## X3 CHECK THE BLOWER MOTOR OPERATION

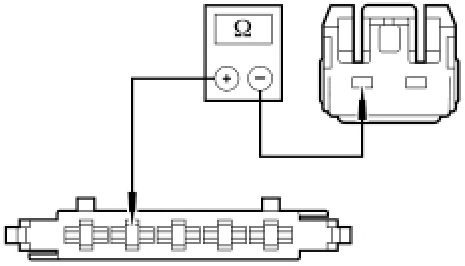
- Ignition ON.
- Select PANEL on the HVAC controls.
- Select all blower speed positions.

**Does the blower motor operate in any position?**

| | |
|---|---|
| **Yes** | If the blower motor does not operate in HI setting only, GO to X4.<br>If the blower motor does not operate in LO setting only, GO to X5.<br>If the blower motor does not operate in MED-LO or MED-HI setting only, GO to X6.<br>If the blower motor operates in HI setting only, GO to X8.<br>If the blower motor operates in LO setting only, GO to X9.<br>For all other symptoms, GO to X10. |
| **No** | GO to Pinpoint Test W. |

## X4 CHECK THE BLOWER MOTOR GROUND CIRCUIT FOR AN OPEN

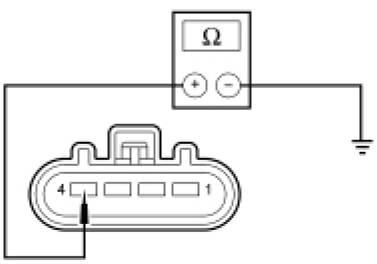
- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294B .
- Disconnect: Blower Motor C2004 .
- Measure the resistance between HVAC module — EMTC C294B Pin 2, circuit CH426 (YE/OG), harness side and blower motor C2004 Pin 2, circuit CH426 (YE/OG), harness side.

- 
  N0089553

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to X12. |
| **No** | REPAIR the circuit. |

**X5 CHECK BLOWER MOTOR RESISTOR GROUND CIRCUIT FOR AN OPEN**

- Ignition OFF.
- Disconnect: Blower Motor Resistor C2185 .
- Measure the resistance between blower motor resistor C2185 Pin 4, circuit GD138 (BK/WH), harness side and ground.
- 
  A0035320

**Is the resistance less than 3 ohms?**

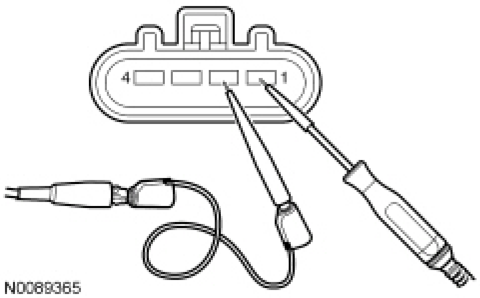
| | |
|---|---|
| **Yes** | INSTALL a new blower motor resistor. REFER to Section 412-01. TEST the system for normal operation. |
| **No** | REPAIR the circuit. |

**X6 CHECK THE BLOWER MOTOR SWITCH**

**NOTICE: The following step uses a test light to simulate normal circuit loads. Use only the test light recommended in the Special Tools table at the beginning of this section or equivalent. To avoid connector terminal damage, use the Flex Probe Kit for the test light probe connection to the vehicle. Do not use the test light probe directly on any connector.**

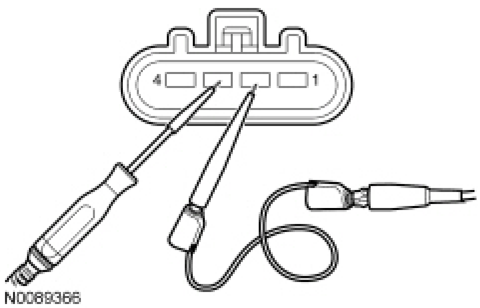
- Ignition OFF.
- Disconnect: Blower Motor Resistor C2185 .
- Ignition ON.

- **For MED-LO inoperative**




For MED-LO inoperative: With the blower motor switch in the MED-LO speed position, connect the 12-volt test light between blower motor resistor C2185 Pin 2, circuit CH426 (YE/OG), harness side and blower motor resistor C2185 Pin 1, circuit CH428 (GN/WH), harness side.

- **For MED-HI inoperative**



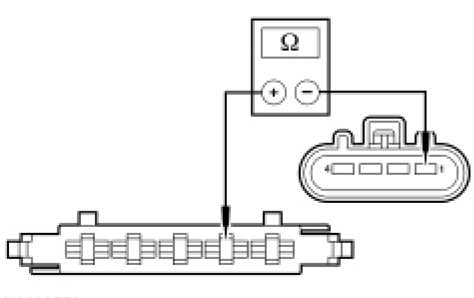

For MED-HI inoperative: With the blower motor switch in the MED-HI speed position, connect the 12-volt test light between blower motor resistor C2185 Pin 2, circuit CH426 (YE/OG), harness side and blower motor resistor C2185 Pin 3, circuit CH429 (GY/BN), harness side.

**Does the test light illuminate?**

| | |
|---|---|
| **Yes** | INSTALL a new blower motor resistor. REFER to Section 412-01. TEST the system for normal operation. |
| **No** | GO to X7. |

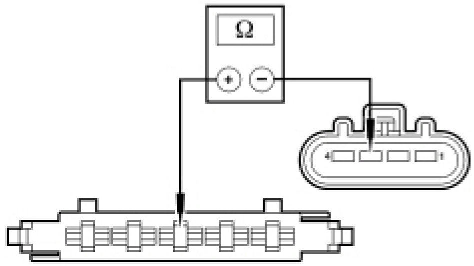
**X7 CHECK THE BLOWER MOTOR SWITCH TO BLOWER MOTOR RESISTOR CIRCUITS FOR AN OPEN**

- Ignition OFF.
- Disconnect: HVAC Module — Base EMTC C294B .
- **For MED-LO inoperative**



For MED-LO inoperative, measure the resistance between HVAC module — EMTC C294B Pin 4, circuit CH428 (GN/WH), harness side and blower motor resistor C2185 Pin 1, circuit CH428 (GN/WH), harness side.

- **For MED-HI inoperative**




N0089557

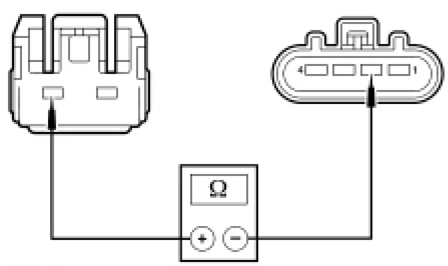
For MED-HI inoperative, measure the resistance between HVAC module — EMTC C294B Pin 3, circuit CH429 (GY/BN), harness side and blower motor resistor C2185 Pin 3, circuit CH429 (GY/BN), harness side.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to X12. |
| **No** | REPAIR the circuit in question. |

**X8 CHECK THE BLOWER MOTOR GROUND CIRCUIT TO THE BLOWER MOTOR RESISTOR FOR AN OPEN**

- Ignition OFF.
- Disconnect: Blower Motor Resistor C2185 .
- Disconnect: Blower Motor C2004 .
- Measure the resistance between blower motor resister C2185 Pin 2, circuit CH426 (YE/OG), harness side and blower motor C2004 Pin 2, circuit CH426 (YE/OG), harness side.

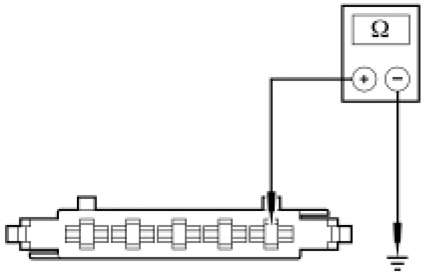



- N0089558

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | INSTALL a new blower motor resistor. REFER to Section 412-01. TEST the system for normal operation. |
| **No** | REPAIR the circuit. |

**X9 CHECK THE BLOWER MOTOR SWITCH GROUND CIRCUIT FOR AN OPEN**

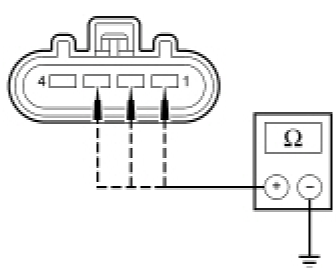
- Disconnect: HVAC Module — Base EMTC C294B .
- Measure the resistance between HVAC module — EMTC C294B Pin 5, circuit GD138 (BK/WH), harness side and ground.
- 

N0089559

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to X12. |
| **No** | REPAIR the circuit. |

## X10 CHECK THE BLOWER MOTOR RESISTOR CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: Blower Motor C2004 .
- Disconnect: Blower Motor Resistor C2185 .
- Disconnect: HVAC Module — Base EMTC C294B .
- Measure the resistance between ground and blower motor resistor:
  - C2185 Pin 1, circuit CH428 (GN/WH), harness side.
  - C2185 Pin 2, circuit CH426 (YE/OG), harness side.
  - C2185 Pin 3, circuit CH429 (GY/BN), harness side.
- 

A0051664

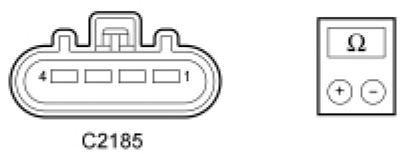
**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to X11. |
| **No** | REPAIR the circuit in question. |

## X11 CHECK THE BLOWER MOTOR RESISTOR CIRCUITS FOR SHORTS TOGETHER

- Measure the resistance between blower motor resistor C2185, harness side using the following chart:

| Blower Motor Resistor | Circuit | Blower Motor Resistor | Circuit |
|---|---|---|---|
| C2185 Pin 1 | CH428 (GN/WH) | C2185 Pin 2 | CH426 (YE/OG) |
| C2185 Pin 1 | CH428 (GN/WH) | C2185 Pin 3 | CH429 (GY/BN) |
| C2185 Pin 2 | CH426 (YE/OG) | C2185 Pin 3 | CH429 (GY/BN) |




- N0122706

**Are the resistances greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | CARRY OUT the Blower Motor Resistor component test in this section. If the resistor tests OK, GO to X12. |
| **No** | REPAIR the circuit in question. |

**X12 CHECK FOR CORRECT HVAC MODULE OPERATION**

- Ignition OFF.
- Disconnect and inspect all HVAC module connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Clear the DTCs.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC module. REFER to Section 412-01. |

| No | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |
|---|---|

**Pinpoint Test Y: The Blower Motor is Inoperative — EATC Dual Zone and EMTC With 4.2 Inch Display System With 7 Blower Motor Speed Positions**

Refer to Wiring Diagrams Cell 54 , Automatic Climate Control System for schematic and connector information.

Refer to Wiring Diagrams Cell 55 , Automatic Climate Control System for schematic and connector information.

**Normal Operation**

Voltage is supplied to the relay coil and relay switch contacts from BJB . When the blower motor relay coil receives a ground from the HVAC module, the relay coil is energized and voltage is delivered to the blower motor and the blower motor speed control from the relay. Ground for the motor is provided from the blower motor speed control. Ground is provided for the blower control module. The HVAC module sends a Pulse Width Modulated (PWM) signal to the blower motor speed control to control the blower speed.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B10AF:12 | Blower Fan Relay: Circuit Short to Battery | This DTC sets when the HVAC module senses voltage on the relay coil ground circuit when the relay is grounded by the HVAC module, indicating a short directly to voltage. |
| B10AF:13 | Blower Fan Relay: Circuit Open | This DTC sets when the HVAC module senses no voltage on the relay coil ground circuit when the relay is not grounded by the HVAC module, indicating an open. |
| B10B9:14 | Blower Control: Circuit Short to Ground or Open | This DTC sets when the HVAC module senses no voltage on the blower motor control target speed signal circuit, indicating an open or a short directly to ground. |

**This pinpoint test is intended to diagnose the following:**

- Battery Junction Box (BJB) fuse 51 (40A) and fuse 52 (5A)
- Wiring, terminals or connectors
- Blower motor relay
- Blower motor speed control
- HVAC module
- Blower motor

**PINPOINT TEST Y : THE BLOWER MOTOR IS INOPERATIVE — EATC DUAL ZONE AND EMTC WITH 4.2 INCH DISPLAY SYSTEM WITH 7 BLOWER MOTOR SPEED POSITIONS**

| Y1 CHECK THE BLOWER MOTOR RELAY FUNCTIONALITY |
|---|
| • Ignition OFF. |

- Install a known good blower motor relay.
- Ignition ON.
- Operate the system and verify the concern is still present.

**Does the blower motor function correctly?**

| | |
|---|---|
| **Yes** | INSTALL a new blower motor relay. |
| **No** | GO to Y2. |

**Y2 CHECK THE HVAC MODULE FOR DTCS**

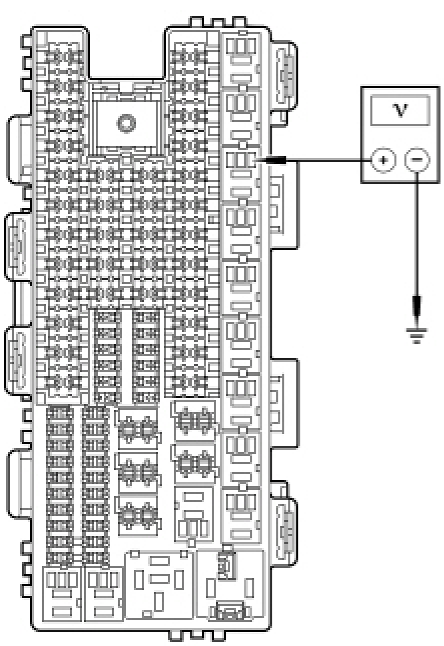
- Ignition ON.
- Check the HVAC module for DTCs.

**Are any DTCs present?**

| | |
|---|---|
| **Yes** | For DTC B10AF:12 and B10AF:13, GO to Y3.<br>For DTC B10B9:14, GO to Y6. |
| **No** | GO to Y6. |

**Y3 CHECK THE BLOWER MOTOR RELAY COIL SUPPLY VOLTAGE**

- Ignition OFF.
- Disconnect: Blower Motor Relay .
- Ignition ON.
- Measure the voltage between ground and BJB blower motor relay socket, pin 1, circuit CBB52 (GN/WH).



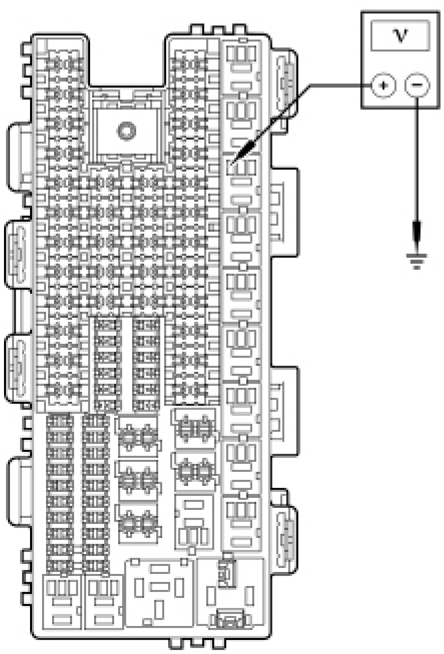

- N0096368

**Is the voltage greater than 11 volts?**

| | |
|---|---|
| **Yes** | GO to Y4. |
| **No** | VERIFY Battery Junction Box (BJB) fuse 52 (5A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. |

**Y4 CHECK THE BLOWER MOTOR RELAY COIL SWITCHED GROUND CIRCUIT FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Ignition ON.
- Measure the voltage between ground and BJB blower motor relay socket, pin 2, circuit CH123 (VY/GN).



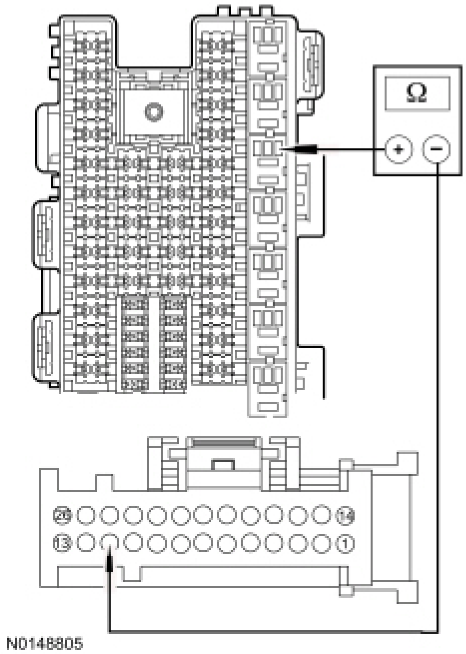

- N0122702

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit. CLEAR the DTCs. RETEST the system.<br>If the blower motor is still inoperative, GO to Y6. |
| **No** | GO to Y5. |

**Y5 CHECK THE BLOWER MOTOR RELAY COIL SWITCHED GROUND CIRCUIT FOR AN OPEN**

- Ignition OFF.



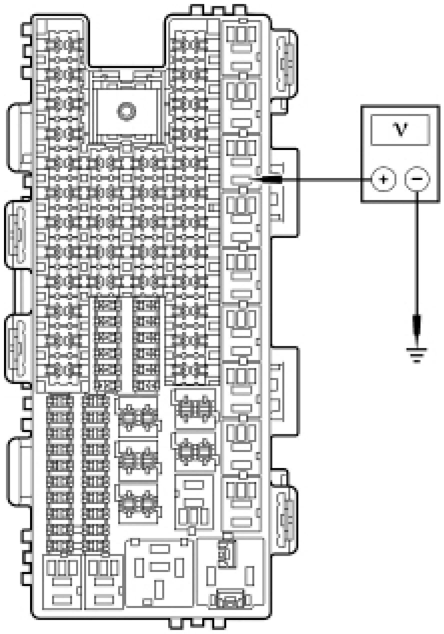

- Measure resistance between BJB blower motor relay socket pin 2, circuit CH123 (VT/GN) and:
  - HVAC module — EATC C228A Pin 11, circuit CH123 (VT/GN), harness side.
  - HVAC module — EMTC C2357A Pin 11, circuit CH123 (VT/GN), harness side.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to Y14. |
| **No** | REPAIR the circuit. CLEAR the DTCs. RETEST the system.<br>If the blower motor is still inoperative, GO to Y6. |

**Y6 CHECK THE BLOWER MOTOR RELAY SWITCH SUPPLY VOLTAGE**

- Ignition OFF.
- Disconnect: Blower Motor Relay .
- Ignition ON.
- Measure the voltage between ground and BJB blower motor relay socket, pin 3, circuit SBB51 (/BU/RD).



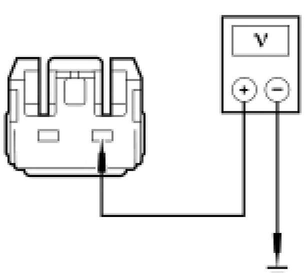
- N0096369

**Is the voltage greater than 11 volts?**

| | |
|---|---|
| **Yes** | GO to Y7. |
| **No** | VERIFY BJB fuse 51 (40A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. |

**Y7 CHECK FOR VOLTAGE TO THE BLOWER MOTOR**

- Ignition OFF.
- Connect: Blower Motor Relay .
- Disconnect: Blower Motor C2004 .
- Ignition ON.
- Select PANEL on the HVAC controls.
- Measure the voltage between blower motor C2004 Pin 1, circuit CH402 (YE/GN), harness side and ground.

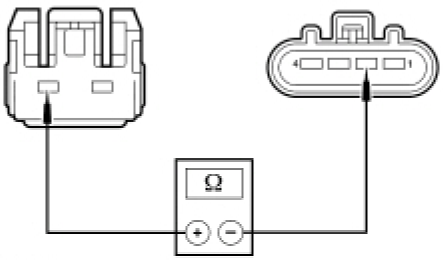


- N0089554

**Is the voltage greater than 11 volts?**

| | |
|---|---|
| **Yes** | GO to Y8. |
| **No** | REPAIR the circuit. |

| Y8 CHECK THE GROUND CIRCUIT BETWEEN THE BLOWER MOTOR AND BLOWER MOTOR SPEED CONTROL FOR AN OPEN |
|---|

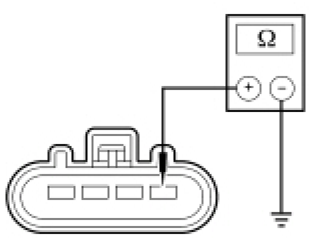
- Ignition OFF.
- Disconnect: Blower Motor Speed Control C271 .
- Measure the resistance between blower motor C2004 Pin 2, circuit VH301 (YE/BU), harness side and blower motor speed control C271 Pin 2, circuit VH301 (YE/BU), harness side.
- 

N0069558

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to Y9. |
| **No** | REPAIR the circuit. |

| Y9 CHECK THE BLOWER MOTOR SPEED CONTROL GROUND CIRCUIT FOR AN OPEN |
|---|

- Measure the resistance between blower motor speed control C271 Pin 1, circuit GD138 (BK/WH), harness side and ground.
- 

N0001799

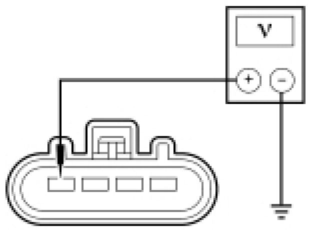
**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to Y10. |
| **No** | REPAIR the circuit. |

| Y10 CHECK FOR VOLTAGE TO THE BLOWER MOTOR SPEED CONTROL |
|---|

- Connect: HVAC Module — EATC C228A .
- Connect: HVAC Module — Remote Mount EMTC C2357A .

- Ignition ON.
- Select PANEL on the HVAC controls.
- Measure the voltage between blower motor speed control C271 Pin 4, circuit CH402 (YE/GN), harness side and ground.



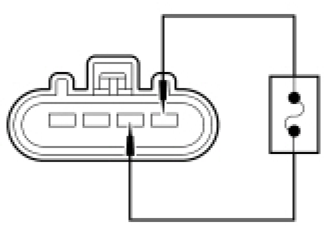

- N0001798

**Is the voltage greater than 11 volts?**

| | |
|---|---|
| **Yes** | GO to Y11. |
| **No** | REPAIR the circuit. |

**Y11  CHECK THE BLOWER MOTOR**

- Ignition OFF.
- Connect: Blower Motor C2004 .
- Connect a fused jumper lead between blower motor speed control C271 Pin 2, circuit VH301 (YE/BU), harness side and C271 Pin 1, circuit GD138 (BK/WH), harness side.



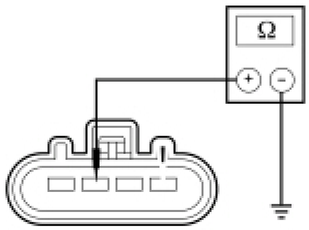
- N0001800
- Ignition ON.
- Select PANEL on the HVAC controls.

**Does the blower motor operate?**

| | |
|---|---|
| **Yes** | GO to Y12. |
| **No** | INSTALL a new blower motor. REFER to Section  412-01. TEST the system for normal operation. |

**Y12  CHECK THE BLOWER MOTOR SPEED CONTROL PULSE WIDTH CIRCUIT FOR A SHORT TO GROUND**

- Ignition OFF.
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Measure the resistance between blower motor speed control C271 Pin 3, circuit VH101 (WH/VT), harness side and ground.




N0148253

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to Y13. |
| **No** | REPAIR the circuit. |

**Y13 CHECK THE BLOWER MOTOR SPEED CONTROL CONNECTION**

- Disconnect and inspect the blower motor speed control connector.
  - Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect the blower motor speed control connector. Make sure it seats and latches correctly.
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Clear the DTCs.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | INSTALL a new blower motor speed control. REFER to Section 412-01.<br>RETEST the system. If concern Is still present, GO to Y14. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

**Y14 CHECK THE HVAC MODULE OPERATION**

- Ignition OFF.
- Disconnect and inspect all HVAC module connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Clear the DTCs.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC module. REFER to Section 412-01. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test Z: The Blower Motor Does Not Operate Correctly — EATC Dual Zone and EMTC With 4.2 Inch Display System With 7 Blower Motor Speed Positions**

Refer to Wiring Diagrams Cell 54 , Automatic Climate Control System for schematic and connector information.

Refer to Wiring Diagrams Cell 55 , Automatic Climate Control System for schematic and connector information.

**Normal Operation**

Voltage is supplied to the relay coil from Battery Junction Box (BJB) fuse 52 (5A) and relay switch contacts from BJB fuse 51 (40A). When the blower motor relay coil receives a ground from the HVAC module, the relay coil is energized and voltage is delivered to the blower motor and the blower motor speed control from the relay. Ground for the motor is provided from the blower motor speed control. Ground is provided for the blower control module. The HVAC module sends a Pulse Width Modulated (PWM) signal to the blower motor speed control to control the blower speed.

**DTC Fault Trigger Conditions**

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B10AF:11 | Blower Fan Relay: Circuit Short to Battery | This DTC sets when the HVAC module senses no voltage on the relay coil ground circuit when the HVAC module is not grounding the circuit, indicating a short directly to voltage. |

| DTC | Description | Fault Trigger Conditions |
|---|---|---|
| B10B9:12 | Blower Control: Circuit Short to Battery | This DTC sets when the HVAC module senses excessive voltage on the blower motor control PWM circuit for more than 1 second, indicating a short to voltage. |
| B10B9:14 | Blower Control: Circuit Short to Ground or Open | This DTC sets when the HVAC module senses no voltage on the blower motor control blower motor control PWM circuit for more than 1 second, indicating an open or a short directly to ground. |

**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- Blower motor speed control
- HVAC module
- Front Controls Interface Module (FCIM)

## PINPOINT TEST Z : THE BLOWER MOTOR DOES NOT OPERATE CORRECTLY — EATC DUAL ZONE AND EMTC WITH 4.2 INCH DISPLAY SYSTEM WITH 7 BLOWER MOTOR SPEED POSITIONS

### Z1 CHECK THE HVAC MODULE FOR DTCS

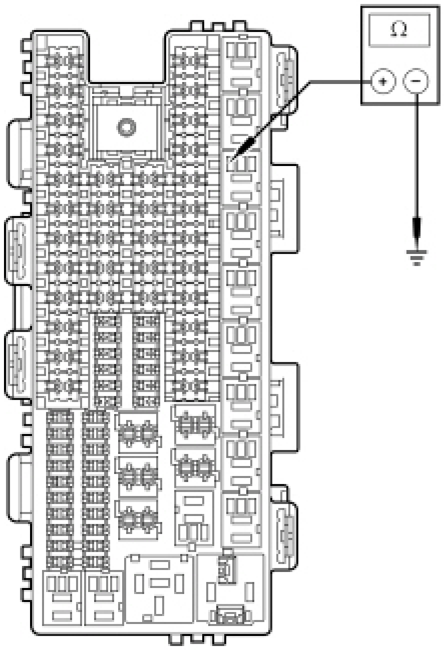
- Ignition ON.
- Check the HVAC module for DTCs.

**Are any DTCs present?**

| | |
|---|---|
| **Yes** | For DTC B10AF:11, GO to Z2.<br>For DTC B10B9:12, GO to Z3.<br>For DTC B10B9:14, GO to Z4. |
| **No** | GO to Z5. |

### Z2 CHECK THE BLOWER MOTOR RELAY COIL SWITCHED GROUND CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: Blower Motor Relay .
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Ignition ON.
- Measure the resistance between blower motor relay socket pin 2, circuit CH123 (VT/GN) and ground.



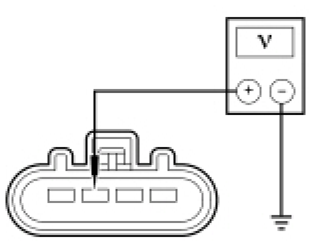

- N0122705

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to Z11. |
| **No** | REPAIR the circuit. |

**Z3 CHECK THE BLOWER MOTOR CONTROL CIRCUIT FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: Blower Motor Speed Control C271 .
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- Ignition ON.
- Measure the voltage between blower motor speed control C271 Pin 3, circuit VH101 (WH/VT), harness side and ground.

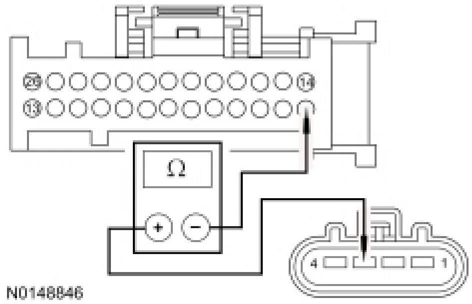



- N0001804

**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit. |
| **No** | GO to Z10. |

## Z4 CHECK THE BLOWER MOTOR CONTROL CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Blower Motor Speed Control C271 .
- Disconnect: HVAC Module — EATC C228A .
- Disconnect: HVAC Module — Remote Mount EMTC C2357A .
- 

N0148846

  Measure the resistance between HVAC module — EATC C228A Pin 1, circuit VH101 (WH/VT), harness side and blower motor speed control C271 Pin 3, circuit VH101 (WH/VT), harness side.

**Is the resistance less than 3 ohms?**

| | |
|---|---|
| **Yes** | GO to Z10. |
| **No** | REPAIR the circuit. |

## Z5 CHECK THE BLOWER MOTOR OPERATION USING THE BLOWER MOTOR SPEED (BLOWRSPD) ACTIVE COMMAND

- Enter the following diagnostic mode on the scan tool: HVAC Module DataLogger .
- Select PANEL mode on the HVAC controls.
- Select the lowest blower motor setting.
- Observe blower motor speed while using the scan tool to operate the HVAC module active command BLOWRSPD, increasing and decreasing the percentage value.

**Does the blower motor speed change between low and high when the active command is changed?**

| | |
|---|---|
| **Yes** | INSTALL a new FCIM . REFER to Refer to the appropriate section in Group 415 for the procedure.. TEST the system for normal operation. |
| **No** | GO to Z6. |

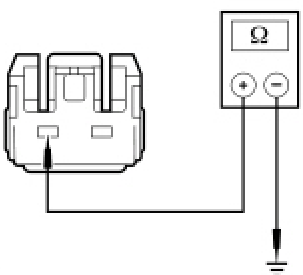
## Z6 VERIFY THE BLOWER MOTOR OPERATION

- Select PANEL on the HVAC controls. Adjust the blower motor setting to LO and then to HI then press the POWER button on the HVAC module (OFF).

**Does the blower motor operate at any setting and/or turn OFF?**

| | |
|---|---|
| **Yes** | If the blower motor operates always in HI, GO to Z7.<br>If the blower motor is always ON, GO to Z8.<br>All other symptoms, INSTALL a new blower motor speed control. REFER to Section 412-01. TEST the system for normal operation. If the blower motor still does not operate correctly, GO to Z11. |
| **No** | GO to Pinpoint Test Y. |

**Z7 CHECK THE BLOWER MOTOR PWM GROUND CIRCUIT FOR A SHORT TO GROUND**

- Ignition OFF.
- Disconnect: Blower Motor Speed Control C271 .
- Disconnect: Blower Motor C2004 .
- Measure the resistance between blower motor connector C2004 Pin 2, circuit VH301 (YE/BU), harness side and ground.
- 

N0069552

**Is the resistance greater than 10,000 ohms?**

| | |
|---|---|
| **Yes** | GO to Z10. |
| **No** | REPAIR the circuit. |

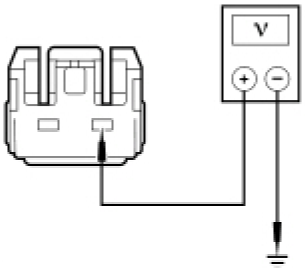
**Z8 CHECK THE BLOWER MOTOR RELAY**

- Ignition OFF.
- Disconnect: Blower Motor Relay .
- Turn OFF the HVAC system.

**Does the blower motor operate?**

| | |
|---|---|
| **Yes** | GO to Z9. |
| **No** | INSTALL a new blower motor relay. TEST the system for normal operation. |

**Z9 CHECK THE BLOWER MOTOR VOLTAGE CIRCUIT FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: Blower Motor C2004 .
- Disconnect: Blower Motor Speed Control C271 .
- Ignition ON.
- Measure the voltage between blower motor C2004 Pin 1, circuit CH402 (YE/GN), harness side and ground.
- 


**Is any voltage present?**

| | |
|---|---|
| **Yes** | REPAIR the circuit. |
| **No** | GO to Z10. |

## Z10 CHECK THE BLOWER MOTOR SPEED CONTROL CONNECTION

- Disconnect and inspect the blower motor speed control connector.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect the blower motor speed control connector. Make sure it seats and latches correctly.
- Clear the DTCs.
- Enter the following diagnostic mode on the scan tool: Self-Test — HVAC Module .
- Operate the system and verify the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | INSTALL a new blower motor speed control. REFER to Section 412-01. TEST the system for normal operation. If the blower motor is still inoperative, GO to Z11. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

## Z11 CHECK FOR CORRECT HVAC MODULE OPERATION

- Ignition OFF.

- Disconnect and inspect all HVAC module connectors.
- Repair:
  - corrosion (install new connector or terminals - clean module pins)
  - damaged or bent pins - install new terminals/pins
  - pushed-out pins - install new pins as necessary
- Connect all HVAC module connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

| | |
|---|---|
| **Yes** | CHECK OASIS for any applicable TSBs . If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSB addresses this concern, INSTALL a new HVAC module. REFER to Section 412-01. |
| **No** | The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. ADDRESS the root cause of any connector or pin issues. |

**Pinpoint Test AA: DTC U0140:00**

**Normal Operation**

- DTC U0140:00 (Lost Communication With Body Control Module: No Sub-Type Information) — sets in continuous memory in the HVAC module if data messages received from the Body Control Module (BCM) over the Medium Speed Controller Area Network (MS-CAN) are missing for 3 seconds or more. For a complete list of all network messages, refer to Section 418-00.

This pinpoint test is intended to diagnose the following:

- BCM
- HVAC module

**PINPOINT TEST AA : DTC U0140:00**

**AA1 VERIFY THE CUSTOMER CONCERN**

- Ignition ON.
- Verify there is an observable symptom present.

**Is an observable symptom present?**

| | |
|---|---|
| **Yes** | GO to AA2. |
| **No** | The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AA2 CHECK THE COMMUNICATION NETWORK**

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test .

- Carry out the network test.

**Does the BCM pass the network test?**

| | |
|---|---|
| **Yes** | GO to AA3. |
| **No** | REFER to Section 418-00. |

**AA3 RETRIEVE THE RECORDED DTCS FROM THE BCM SELF-TEST**

- Check for recorded DTCs from the BCM self-test.

**Is DTC U3003:16 or DTC U3003:17 recorded?**

| | |
|---|---|
| **Yes** | Section 419-10. |
| **No** | GO to AA4. |

**AA4 RETRIEVE THE RECORDED DTCS FROM THE HVAC MODULE SELF-TEST**

- Check for recorded DTCs from the HVAC module self-test.

**Is DTC U3003:16 or DTC U3003:17 recorded?**

| | |
|---|---|
| **Yes** | GO to Pinpoint Test H. |
| **No** | GO to AA5. |

**AA5 RECHECK THE HVAC MODULE DTCS**

**NOTE:** *If new modules were installed prior to the DTC being set, the module configuration may be incorrectly set during the Programmable Module Installation (PMI) or the PMI may not have been carried out.*

- Clear the DTCs.
- Wait 10 seconds.
- Repeat the HVAC module self-test.

**Is DTC U0140:00 still present?**

| | |
|---|---|
| **Yes** | GO to AA6. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AA6 CHECK FOR DTC U0140:00 SET IN OTHER MODULES**

- Clear all DTCs.
- Ignition OFF.

- Ignition ON.
- Wait 10 seconds.
- Enter the following diagnostic mode on the scan tool: Self-Test .
- Retrieve the continuous memory DTCs from all modules.

**Is DTC U0140:00 set in the Driver Seat Module (DSM) ?**

| | |
|---|---|
| **Yes** | INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . CLEAR the DTCs. REPEAT the self-test. |
| **No** | INSTALL a new HVAC module. REFER to Section 412-01. TEST the system for normal operation. |

**Pinpoint Test AB: DTC U0163:00**

**Normal Operation**

- DTC U0163:00 (Lost Communication With navigation Control Module: No Sub-Type Information) — sets in continuous memory in the HVAC module if data messages received from the Accessory Protocol Interface Module (APIM) over the Medium Speed Controller Area Network (MS-CAN) are missing for 3 seconds or more. For a complete list of all network messages, refer to Section 418-00.

This pinpoint test is intended to diagnose the following:

- APIM
- HVAC module

**PINPOINT TEST AB : DTC U0163:00**

**AB1 VERIFY THE CUSTOMER CONCERN**

- Ignition ON.
- Verify there is an observable symptom present.

**Is an observable symptom present?**

| | |
|---|---|
| **Yes** | GO to AB2. |
| **No** | The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AB2 CHECK THE COMMUNICATION NETWORK**

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test .
- Carry out the network test.

**Does the APIM pass the network test?**

| | |
|---|---|
| **Yes** | GO to AB3. |
| **No** | REFER to Section 418-00. |

**AB3 RETRIEVE THE RECORDED DTCS FROM THE APIM SELF-TEST**

- Check for recorded DTCs from the APIM self-test.

**Is DTC U3003:16 or DTC U3003:17 recorded?**

| | |
|---|---|
| **Yes** | Refer to the appropriate section in Group 415 for the procedure.. |
| **No** | GO to AB4. |

**AB4 RETRIEVE THE RECORDED DTCS FROM THE HVAC MODULE SELF-TEST**

- Check for recorded DTCs from the HVAC module self-test.

**Is DTC U3003:16 or DTC U3003:17 recorded?**

| | |
|---|---|
| **Yes** | GO to Pinpoint Test H. |
| **No** | GO to AB5. |

**AB5 RECHECK THE HVAC MODULE DTCS**

**NOTE:** *If new modules were installed prior to the DTC being set, the module configuration may be incorrectly set during the Programmable Module Installation (PMI) or the PMI may not have been carried out.*

- Clear the DTCs.
- Wait 10 seconds.
- Repeat the HVAC module self-test.

**Is DTC U0163:00 still present?**

| | |
|---|---|
| **Yes** | INSTALL a new APIM . Refer to the appropriate section in Group 415 for the procedure.. CLEAR the DTCs. REPEAT the HVAC module self-test. If HVAC DTC U0163:00 returns, INSTALL a new HVAC module. REFER to Section 412-01. TEST the system for normal operation. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**Pinpoint Test AC: DTC U0256:00**

**Normal Operation**

- DTC U0256:00 (Lost Communication With FCIM "A") — sets in continuous memory in the HVAC module if data messages received from the FCIM over the MS-CAN are missing for 3 seconds or more. For a complete list of all network messages, refer to Section 418-00.

This pinpoint test is intended to diagnose the following:

- FCIM
- HVAC module

**PINPOINT TEST AC : DTC U0256:00**

**AC1 VERIFY THE CUSTOMER CONCERN**

- Ignition ON.
- Verify there is an observable symptom present.

**Is an observable symptom present?**

| | |
|---|---|
| **Yes** | GO to AC2. |
| **No** | The system is operating normally at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

**AC2 CHECK THE COMMUNICATION NETWORK**

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test .
- Carry out the network test.

**Does the FCIM pass the network test?**

| | |
|---|---|
| **Yes** | GO to AC3. |
| **No** | REFER to Section 418-00. |

**AC3 RETRIEVE THE RECORDED DTCS FROM THE FCIM SELF-TEST**

- Check for recorded DTCs from the FCIM self-test.

**Is DTC U3003:16 or DTC U3003:17 recorded?**

| | |
|---|---|
| **Yes** | Refer to the appropriate section in Group 415 for the procedure.. |
| **No** | GO to AC4. |

**AC4 RETRIEVE THE RECORDED DTCS FROM THE HVAC MODULE SELF-TEST**

- Check for recorded DTCs from the HVAC module self-test.

**Is DTC U3003:16 or DTC U3003:17 recorded?**

| | |
|---|---|
| **Yes** | GO to Pinpoint Test H. |
| **No** | GO to AC5. |

| **AC5 RECHECK THE HVAC MODULE DTCS** |
|---|

**NOTE:** *If new modules were installed prior to the DTC being set, the module configuration may be incorrectly set during the PMI or the PMI may not have been carried out.*

- Clear the DTCs.
- Wait 10 seconds.
- Repeat the HVAC module self-test.

**Is DTC U0256:00 still present?**

| | |
|---|---|
| **Yes** | GO to AC6. |
| **No** | The system is operating correctly at this time. The DTC may have been set due to high network traffic or an intermittent fault condition. |

| **AC6 CHECK FOR DTC U0256:00 SET IN OTHER MODULES** |
|---|

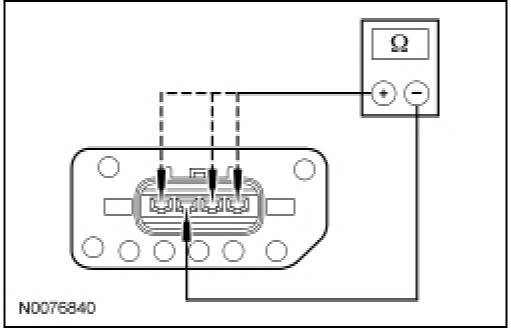
- Clear all DTCs.
- Ignition OFF.
- Wait 10 seconds.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test .
- Retrieve the continuous memory DTCs from all modules.

**Is DTC U0256:00 set in the APIM ?**

| | |
|---|---|
| **Yes** | INSTALL a new FCIM . Refer to the appropriate section in Group 415 for the procedure.. |
| **No** | INSTALL a new HVAC module. REFER to Section 412-01. TEST the system for normal operation. |

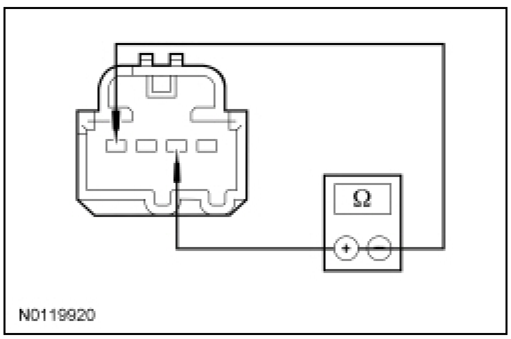
### Component Tests

### Blower Motor Resistor



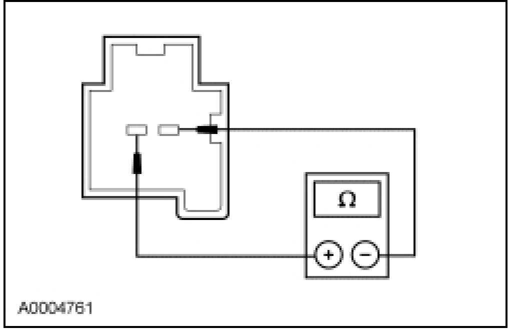

| Blower Motor Resistor Pins | Resistance (Ohms) |
|---|---|
| 4 and 2 | 2.0-2.6 |
| 2 and 3 | 0.2-0.4 |
| 2 and 1 | 0.8-1.1 |

### Temperature Sensor — In-Vehicle



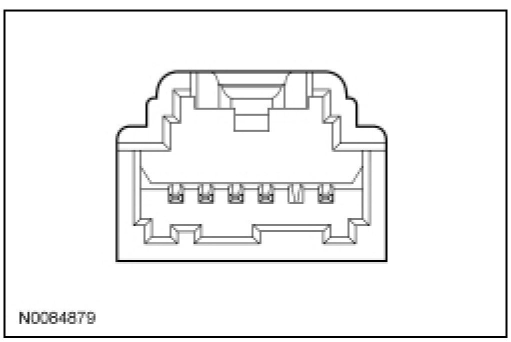

| Ambient Temperature | Resistance (Ohms) |
|---|---|
| 0-10°C (32-50°F) | 57,950-97,640 |
| 10-20°C (50-68°F) | 36,880-59,920 |
| 20-25°C (68-77°F) | 29,700-37,790 |
| 25-30°C (77-86°F) | 23,960-30,300 |
| 30-35°C (86-95°F) | 19,440-24,550 |
| 35-40°C (95-104°F) | 15,860-20,000 |
| 40-50°C (104-122°F) | 10,740-16,390 |

### Temperature Sensor — Evaporator Discharge



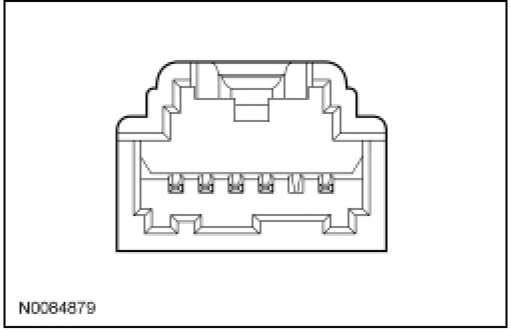

| Ambient Temperature | Resistance (Ohms) |
|---|---|
| -40°C (-40°F) | 1,005,00-1,104,000 |
| -20°C (-4°F) | 287,811-307,800 |
| 0°C (32°F) | 96,890-101,300 |
| 20°C (68°F) | 37,080-38,000 |
| 25°C (77°F) | 29,700-30,300 |
| 30°C (86°F) | 23,840-24,430 |
| 35°C (95°F) | 19,260-19,820 |
| 40°C (104°F) | 15,660-16,180 |
| 60°C (140°F) | 7,239-7,594 |
| 100°C (212°F) | 1,943-2,091 |

**Air Inlet Mode Door Actuator**



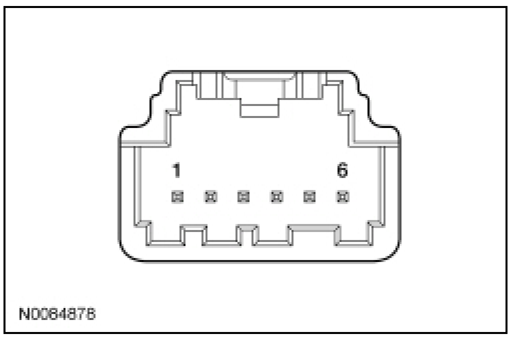

| Actuator Pins | Approx. Resistance (Ohms) |
|---|---|
| 2 and 3 | 2,775-4,625 |
| 2 and 4 | 187-4,313 |
| 3 and 4 | 187-4,313 |
| 1 and 6 | 43-53 |

**Defrost/Panel/Floor Door Actuator**




| Actuator Pins | Approx. Resistance (Ohms) |
|---|---|
| 2 and 3 | 2,775-4,625 |
| 2 and 4 | 187-4,313 |
| 3 and 4 | 187-4,313 |
| 1 and 6 | 43-53 |

**Temperature Blend Door Actuator**




| Actuator Pins | Approx. Resistance (Ohms) |
|---|---|
| 1 and 3 | 2,850-4,750 |
| 1 and 2 | 113-4,563 |
| 2 and 3 | 113-4,563 |
| 5 and 6 | 32-44 |

**Heater Core**

1. **NOTE:** *If a heater core leak is suspected, the heater core must be tested by following the plugged heater core component test before the heater core pressure test. Carry out a system inspection by checking the heater system thoroughly as follows:*

   Inspect for evidence of coolant leakage at the heater water hose to heater core attachments. A coolant leak in the heater water hose could follow the heater core tube to the heater core and appear as a leak in the heater core.

2. **NOTE:** *Spring-type clamps are installed as original equipment. Installation and overtightening of nonspecified clamps can cause leakage at the heater hose connection and damage the heater core.*

   Check the integrity of the heater water hose clamps.

**Heater Core — Plugged**

1. Check to see that the engine coolant is at the correct level.

2. Start the engine and turn on the heater.

3. When the engine coolant reaches operating temperature, feel the heater core inlet and outlet hose to see if they are hot.
   - If the inlet hose is not hot:
     - the thermostat is not working correctly.
   - If the outlet hose is not hot:
     - the heater core may have an air pocket.
     - the heater core may be restricted or plugged.

**Heater Core — Pressure Test**

Use the Pressure Test Kit to carry out the pressure test.

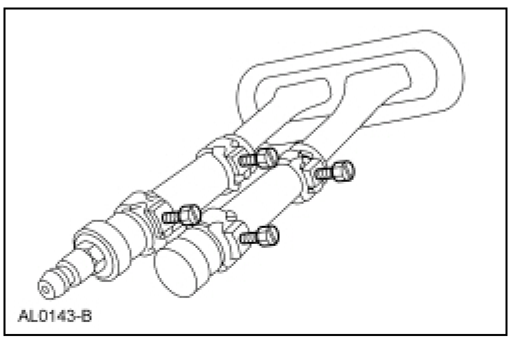
1. **NOTE:** *Due to space limitations, a bench test may be necessary for pressure testing*.

   Drain the coolant from the cooling system. Refer to Section 303-03.

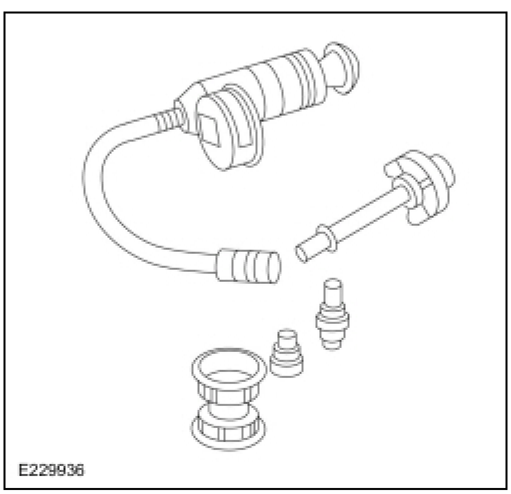
2. Disconnect the heater hoses from the heater core.

3. Install a short piece of heater hose, approximately 101 mm (4 in) long on each heater core tube.

4. Fill the heater core and heater hoses with water and install the plug (221373) and the adapter (221374) from the Pressure Test Kit. Secure the heater hoses, plug and adapter with hose clamps.




5. Attach the pump and gauge assembly from the Pressure Test Kit to the adapter.
   - 014-R1072 Radiator Pressure Test Set




6. Close the bleed valve at the base of the gauge. Pump 138 kPa (20 psi) of air pressure into the heater core.

7. Observe the pressure gauge for a minimum of 3 minutes.

8. If the pressure drops, check the heater hose connections to the core tubes for leaks. If the heater hoses do not leak, remove the heater core from the vehicle and carry out the bench test.

**Heater Core — Bench Test**

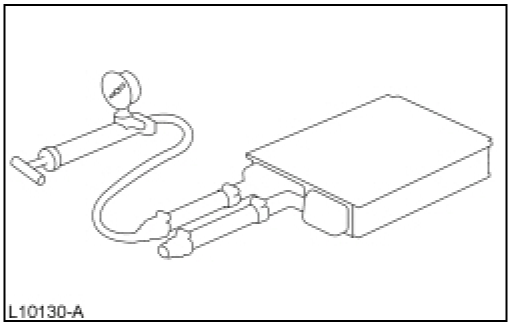
1. Remove the heater core from the vehicle. Refer to Section 412-01.

2. Drain all of the coolant from the heater core.

3. Connect the 101 mm (4 in) test heater water hoses with plug and adapter to the core tubes. Then connect the radiator/heater core pressure tester to the adapter.

4. Apply 138 kPa (20 psi) of air pressure to the heater core. Submerge the heater core in water.

5. If a leak is observed, install a new heater core.




**Evaporator/Condenser Core — On-Vehicle Leak Test**

1. Recover the refrigerant. Refer to Air Conditioning (A/C) System Recovery, Evacuation and Charging.

2. **NOTE:** *DO NOT leak test an evaporator core with the suction accumulator attached to the evaporator core tubes.*

   Disconnect the suspect evaporator core or condenser core from the A/C system. Refer to Section 412-01.

3. Use the correct adapters with a Rotunda approved A/C Service Unit to test the evaporator or condenser.
   - 219-00082 ACF-3000 33PC Adapter Kit - Revised 1st released 44pc kit.
   - 219-00083 A/C Flushing Adapter Kit 2 of 3 (Previously "Supplement A", 2nd released kit).
   - 219-00084 A/C Flushing Adapter Kit 3 of 3.

4. **NOTE:** *The automatic shut-off valves on some hoses do not open when connected to the fittings. If available, use hoses without shut-off valves. If hoses with shut-off valves are used, make sure the valve opens when attached to the adapter fittings. The test is not valid if the shut-off valve does not open.*

   Connect the hoses from the A/C Service Unit to the adapter fittings on the evaporator or condenser.
   - 265-37887 Ritchie R134A A/C Refrigerant Hybrid Management System

5. Open both valves and start the vacuum. Allow the A/C Service Unit to vacuum for a minimum of 45 minutes after the low pressure gauge indicates 101 kPa (30 in-Hg). The 45-minute evacuation is necessary to remove any refrigerant from oil left in the evaporator or condenser. If the refrigerant is not completely removed from the oil, outgassing will degrade the vacuum and appear as a refrigerant leak.

6. If the low pressure gauge reading will not drop to 101 kPa (30 in-Hg) when the valves are open and the A/C Service Unit is operating, close the valves and observe the low pressure reading. If the pressure rises rapidly to zero, a large leak is indicated. Recheck the adapter fitting connections before installing a new evaporator or condenser.

7. After evacuating for 45 minutes, close the valves and stop the service unit. Observe the low pressure gauge; it should remain at the 101 kPa (30 in-Hg) mark.
   - If the low pressure gauge reading rises 34 or more kPa (10 or more in-Hg) of vacuum from the 101 kPa (30 in-Hg) position in 10 minutes, a leak is indicated.
   - If a very small leak is suspected, wait 30 minutes and observe the vacuum gauge.

- If a small amount of vacuum is lost, operate the vacuum pump with the valves open for an additional 30 minutes to remove any remaining refrigerant from the oil in the evaporator core or condenser core. Then recheck for loss of vacuum.
- If a very small leak is suspected, allow the system to set overnight with vacuum applied and check for vacuum loss.

8. If the evaporator core or condenser core leaks, as verified by the above procedure, install a new evaporator core or condenser core. Refer to Section 412-01.

**AIR CONDITIONING (A/C) Compressor — External Leak Test**

1. Recover the refrigerant. Refer to Air Conditioning (A/C) System Recovery, Evacuation and Charging.

2. Disconnect the refrigerant lines from the A/C compressor. Refer to Section 412-01.

3. Install the adapters from the A/C Flush Adapter Kit on the ports of the A/C compressor, using the existing retaining bolts.
   - 219-00082 ACF-3000 33PC Adapter Kit - Revised 1st released 44pc kit.
   - 219-00083 A/C Flushing Adapter Kit 2 of 3 (Previously "Supplement A", 2nd released kit).
   - 219-00084 A/C Flushing Adapter Kit 3 of 3.

4. Connect the high and low pressure lines of the A/C Service Unit to the corresponding fittings on the adapter.
   - 265-37887 Ritchie R134A A/C Refrigerant Hybrid Management System

5. Charge the A/C compressor following the air conditioning service unit instructions. Open the low pressure valve, the high pressure valve and set the refrigerant charge amount to 0.23 kg (8 oz).

6. **NOTE:** *Use a Rotunda-approved Electronic Leak Detector for R-134a refrigerant SAE Certified to J2791.*

   Using the Refrigerant Leak Detector, check for leaks at the compressor shaft.
   - 023-22791 Robinair Infrared A/C Refrigerant Leak Detector w/Case

7. When the leak test is complete, recover the refrigerant from the compressor.

8. If an external leak is found, install a new A/C compressor. Refer to Section 412-01.